

The Alspach–Zhang conjecture: every cubic Cayley graph is 3-edge-colourable

A survey

September 2026

Abstract

In 1992 Brian Alspach and Cun-Quan Zhang conjectured that every Cayley graph of valency at least two admits a nowhere-zero 4-flow. By theorems of Mader and Jaeger this is equivalent to the statement that every connected cubic Cayley graph is 3-edge-colourable, or, in the language introduced by Nedela and Škoviera, that there are no *Cayley snarks*. The conjecture is implied by the folklore conjecture that every Cayley graph is hamiltonian, and it is known to hold for solvable groups (Alspach, Liu and Zhang), for all cubic graphs admitting a solvable vertex-transitive group of automorphisms other than the Petersen graph (Potočník), and for all cubic Cayley graphs on at most 1280 vertices (Potočník, Spiga and Verret). A minimal counterexample, if one exists, is a Cayley graph on a non-abelian simple group or on a group with a unique non-trivial proper normal subgroup, which has index two (Nedela and Škoviera). We survey these results, give short self-contained proofs of the elementary reductions, record several reformulations (in terms of arc-regular cubic graphs, of bi-Cayley graphs, and of small girth), and prove a quotient lemma for arbitrary subgroups (with semi-edges). Its matching-monodromy refinement, together with a known Hamiltonicity result in one boundary case, proves the conjecture whenever $G = S_N$ and the involutory generator is a transposition; point- and two-set-stabiliser arguments with finite certificates also prove it when N is odd, x is an N -cycle and the involution is a product of at most seven transpositions, or more generally has r transpositions with $N \geq 4r + 1$. We also describe the 3-edge-colourings of $\text{Cay}(G, \{a, x, x^{-1}\})$ as a -invariant functions $G \rightarrow \mathbb{Z}_3$ with a local admissibility condition along the x -cycles, and prove that the index-two case is always colourable when $\text{ord}(x) = 3$. We analyse the new $\mathbb{F}_2^3$ construction in the 2026 proof of the cycle double cover theorem, extracting an exact eight-vertex palette reformulation, constructing generator-separated rank-three flows, and reducing the potentially non-circular part of the method to twelve explicit augmented affine systems and an exact interlacement obstruction for their layer potentials. For generators of order five we further identify the quotient as an orientably regular map and decompose the exact constrained 4-flow target into face potentials and surface homology. Finally, we report a computer verification of the conjecture for all cubic Cayley graphs on 85 non-solvable groups of order up to 300 696, including every group permitted by the Nedela–Škoviera reduction of order less than 352 440; consequently a smallest Cayley snark, if one exists, has more than 352 439 vertices.

Contents

1	Introduction	2
2	Preliminaries	4
2.1	Cayley graphs	4
2.2	Cubic Cayley graphs	4
2.3	Tait colourings, even 2-factors and flows	5
2.4	Snarks and cyclic connectivity	5

2.5	Coverings and quotients	6
3	The easy cases and some reformulations	6
3.1	Trivial colourings	6
3.2	Normal subgroups containing the involution	7
3.3	Small girth	7
3.4	Generators of order three: arc-regular cubic graphs	8
3.5	Normal subgroups of index two: bi-Cayley graphs	9
4	Quotients by arbitrary subgroups, and a reformulation over $\mathbb{Z}_3$	10
4.1	Pregraph quotients	10
4.2	Hamiltonian and cyclewise quotients	11
4.3	Monodromy without invariant colours	14
4.4	The colourings as functions $G \rightarrow \mathbb{Z}_3$	19
5	Solvable groups	20
5.1	Solvable vertex-transitive groups	21
6	The structure of a minimal counterexample	21
7	The hamiltonicity route and Cayley maps	23
7.1	$(2, s, 3)$ -Cayley graphs	23
7.2	Cayley maps and independent sets	23
7.3	Other hamiltonicity results	23
8	Computational evidence	24
8.1	The census of cubic vertex-transitive graphs	24
8.2	Beyond the census	24
9	Related problems on flows in Cayley graphs	26
9.1	The cycle-double-cover proof and an eight-point palette	26
10	Open problems	55

1 Introduction

A *Cayley graph* $\text{Cay}(G, S)$ of a finite group G with respect to a subset $S \subseteq G \setminus \{1\}$ with $S = S^{-1}$ has vertex set G , with g adjacent to gs for every $s \in S$. Cayley graphs are the most symmetric graphs one can build from a group: G itself acts regularly on the vertices by left multiplication. Many of the classical “hard” cubic graphs of graph theory—the Petersen graph, the Coxeter graph, the snarks—are highly symmetric, yet none of them is a Cayley graph. The conjecture surveyed here asserts that this is not an accident.

Conjecture 1.1 (Alspach–Zhang, 1992). *Every connected cubic Cayley graph is 3-edge-colourable.*

The conjecture was formulated by Alspach and Zhang at the Louisville workshop on Hamilton cycles in 1992, in the language of flows: *every Cayley graph of valency at least two admits a nowhere-zero 4-flow* (see [29, §1], [53, §1]). It first appeared in print in the paper of Alspach, Liu and Zhang [4], who proved it for solvable groups. Since a connected vertex-transitive graph of valency k is k -edge-connected (Mader [31]) and every 4-edge-connected graph has a nowhere-zero 4-flow (Jaeger [21]), the flow version reduces at once to the cubic case, where a nowhere-zero 4-flow is the same thing as a 3-edge-colouring (Tutte [48]); see Proposition 2.5

below. Nedela and Škoviera [37] coined the term *Cayley snark* for a cubic Cayley graph that is not 3-edge-colourable; Conjecture 1.1 says that Cayley snarks do not exist.

The conjecture sits between two much older problems.

- *Snarks and the Four Colour Theorem.* Tait observed in 1880 that the Four Colour Theorem is equivalent to the statement that every bridgeless planar cubic graph is 3-edge-colourable. Non-planar bridgeless cubic graphs need not be: the Petersen graph is the smallest example, and the non-trivial examples are called snarks. A large part of the theory of snarks consists in showing that highly structured cubic graphs are not snarks; the Alspach–Zhang conjecture is the statement of this kind for the strongest possible symmetry assumption compatible with the Petersen graph not being an exception.
- *Hamilton cycles in vertex-transitive graphs.* Lovász asked in 1969 whether every connected vertex-transitive graph has a Hamilton path, and a folklore conjecture asserts that every connected Cayley graph on at least three vertices has a Hamilton cycle [11, 28]. A cubic Cayley graph has even order (Lemma 2.1), so a Hamilton cycle in it is an even cycle whose edges can be coloured alternately, the remaining perfect matching receiving the third colour. Thus Conjecture 1.1 is a consequence of the hamiltonicity conjecture for cubic Cayley graphs, and every hamiltonicity result for cubic Cayley graphs is a result about Conjecture 1.1. Only four connected vertex-transitive graphs without a Hamilton cycle are known—the Petersen graph, the Coxeter graph, and their truncations—and none of them is a Cayley graph.

What is known

Table 1 summarises the state of the conjecture. The three main theorems are those of Alspach, Liu and Zhang (solvable groups), Nedela and Škoviera (structure of a minimal counterexample) and Potočník (solvable vertex-transitive groups); the strongest computational evidence comes from the census of cubic vertex-transitive graphs of Potočník, Spiga and Verret. We add to this a computation of our own (Section 8) which pushes the verification well beyond the range of the census for the groups singled out by the Nedela–Škoviera reduction: a smallest Cayley snark would have more than 352 439 vertices.

Organisation

Section 2 fixes terminology and recalls the classical facts about 3-edge-colourings, flows, snarks and coverings that are used throughout. Section 3 disposes of the trivial cases and reduces the conjecture to Cayley graphs $\text{Cay}(G, \{a, x, x^{-1}\})$ with a an involution and x of odd order; it also contains several elementary reformulations that we have not seen stated explicitly in the literature (arc-regular cubic graphs, bi-Cayley graphs, small girth), and settles the order-three case when G has a normal subgroup of index two. Section 4 extends the quotient lemma to arbitrary subgroups (the quotient is then a pregraph with semi-edges), gives a cyclewise colouring criterion and a matching-monodromy refinement, proves the transposition case on every symmetric group and a sparse-involution case paired with an odd full cycle, describes the 3-edge-colourings of $\text{Cay}(G, \{a, x, x^{-1}\})$ as a -invariant functions $G \rightarrow \mathbb{Z}_3$ satisfying a local condition along the x -cycles, and derives a consecutive-2-factor criterion for the index-two case. Section 5 treats solvable groups and Potočník’s generalisation to solvable vertex-transitive groups. Section 6 presents the Nedela–Škoviera reduction with a self-contained proof. Section 7 surveys the hamiltonicity route and the Cayley-map approach of Glover, Kutnar, Malnič, Marušič and Hujdurović. Section 8 describes the computational evidence, including our own. Section 9 discusses related flow problems and reduces the potentially useful part of the 2026 cycle-double-cover method to augmented affine systems; Section 10 lists open questions.

Class of cubic Cayley graphs $\text{Cay}(G, S)$	Status / reference
S consists of three involutions	trivial (colour by generator), §3
$S = \{a, x, x^{-1}\}$ with $\text{ord}(x)$ even	trivial, §3
G abelian, dihedral, nilpotent, supersolvable, of odd-order-times-two, ...	all solvable: [4]
G solvable	Alspach–Liu–Zhang [4]
any cubic graph with a solvable vertex-transitive group of automorphisms, other than the Petersen graph	Potočník [42]
$ G \leq 1280$	census [44] (all are hamiltonian)
$G = \langle a, x \rangle$ with $a^2 = x^s = (ax)^3 = 1$ (“(2, s, 3)-Cayley graphs”)	hamiltonian for s odd [17, 16]; trivial for s even
G has a normal subgroup containing a , or a normal subgroup avoiding a and x whose quotient graph is colourable	Lemmas 2.8, 3.3
minimal counterexample	G non-abelian simple, or G has a unique non-trivial proper normal subgroup T , $ G : T = 2$, T simple or $T \cong S \times S$ [37]
girth 4; girth 3 or 5 unless $\text{ord}(x) = 3$ or 5	Proposition 3.4
G has a subgroup H avoiding the conjugacy class of x such that the pregraph $H \setminus X$ is 3-edge-colourable	Lemma 4.1
$\text{ord}(x) = 3$ and G has a normal subgroup of index two	Proposition 3.9
$G = S_N$ and a is a transposition	Theorem 4.13
$G = S_N$, N odd, x is an N -cycle and a has $r \leq 7$ transpositions, or $N \geq 4r + 1$	Proposition 4.7
$\text{ord}(x) = 3$ and $ G \leq 30\,000$	Conder’s census [9], Corollary 3.8
every group admissible for a smallest Cayley snark of order $< 352\,440$, and further non-solvable groups of order $\leq 300\,696$	Section 8

Table 1: Known cases of Conjecture 1.1. Throughout, a denotes an involution and x an element of odd order ≥ 3 generating G together with a ; by §3 this is the only case that matters.

2 Preliminaries

2.1 Cayley graphs

Let G be a finite group and $S \subseteq G \setminus \{1\}$ a subset closed under inversion. The Cayley graph $\text{Cay}(G, S)$ has vertex set G and edge set $\{\{g, gs\} : g \in G, s \in S\}$. It is a simple $|S|$ -regular graph; it is connected if and only if S generates G . Every edge $\{g, gs\}$ carries the label $\{s, s^{-1}\}$, and left multiplication by any $h \in G$ is a label-preserving automorphism, so G acts regularly on the vertices. Conversely, a graph admitting a group of automorphisms acting regularly on its vertices is a Cayley graph on that group. All graphs in this survey are finite; “Cayley graph” will always mean a connected one.

2.2 Cubic Cayley graphs

Let $\text{Cay}(G, S)$ be cubic, so $|S| = 3$. Since $S = S^{-1}$ and $|S|$ is odd, S contains an involution. Either all three elements of S are involutions, or $S = \{a, x, x^{-1}\}$ with $a^2 = 1$ and $\text{ord}(x) \geq 3$. We use the notation

$$X(G; a, x) := \text{Cay}(G, \{a, x, x^{-1}\}), \quad a^2 = 1, \quad \text{ord}(x) \geq 3, \quad G = \langle a, x \rangle.$$

In $X(G; a, x)$ the a -edges form a perfect matching M_a , and the x -edges form a 2-factor F_x consisting of $|G|/\text{ord}(x)$ cycles of length $\text{ord}(x)$ (the left cosets of $\langle x \rangle$).

Lemma 2.1. *A connected cubic Cayley graph has even order and is either $\text{Cay}(G, \{a, b, c\})$ with three involutions a, b, c , or $X(G; a, x)$ as above.*

Proof. The generating set contains an involution, so $|G|$ is even; the dichotomy was explained above. $\square$

2.3 Tait colourings, even 2-factors and flows

A *3-edge-colouring* (or Tait colouring) of a cubic graph X is a colouring of the edges with three colours such that the three edges at each vertex receive distinct colours; equivalently, a partition of $E(X)$ into three perfect matchings. The following reformulation is used constantly.

Lemma 2.2. *A cubic graph is 3-edge-colourable if and only if it has a 2-factor all of whose cycles are even.*

Proof. Two colour classes of a 3-edge-colouring form a 2-factor of even cycles; conversely, colour the cycles of an even 2-factor alternately and give the complementary perfect matching the third colour. $\square$

In particular a cubic graph with a Hamilton cycle is 3-edge-colourable if and only if its order is even, and a bipartite cubic graph is 3-edge-colourable (König).

A *nowhere-zero k -flow* on a graph is an orientation together with an assignment of integers from $\{\pm 1, \dots, \pm(k-1)\}$ to the edges satisfying Kirchhoff's law at every vertex. Tutte [48, 49] showed that a cubic graph admits a nowhere-zero 4-flow if and only if it is 3-edge-colourable, and that a planar graph admits a nowhere-zero 4-flow if and only if its dual is 4-vertex-colourable; his 5-flow conjecture (every bridgeless graph has a nowhere-zero 5-flow) and 3-flow conjecture (every 4-edge-connected graph has a nowhere-zero 3-flow) remain open. Two theorems are needed to relate the flow form of the Alspach–Zhang conjecture to its colouring form.

Theorem 2.3 (Jaeger [21]). *Every 4-edge-connected graph admits a nowhere-zero 4-flow.*

Theorem 2.4 (Mader [31]). *A connected vertex-transitive graph of valency k is k -edge-connected.*

Proposition 2.5. *The following statements are equivalent.*

- (i) *Every connected Cayley graph of valency at least 2 admits a nowhere-zero 4-flow.*
- (ii) *Every connected cubic Cayley graph admits a nowhere-zero 4-flow.*
- (iii) *Every connected cubic Cayley graph is 3-edge-colourable (Conjecture 1.1).*

Proof. A connected 2-regular graph is a cycle and has a nowhere-zero 2-flow. A Cayley graph of valency $k \geq 4$ is k -edge-connected by Theorem 2.4 and hence has a nowhere-zero 4-flow by Theorem 2.3. So (ii) $\Rightarrow$ (i), the converse is trivial, and (ii) $\Leftrightarrow$ (iii) is Tutte's theorem. $\square$

We also recall that a 3-edge-colourable cubic graph has a cycle double cover (take the three 2-factors formed by pairs of colour classes). Thus Conjecture 1.1 implied the cycle double cover conjecture for cubic Cayley graphs; the latter was proved in full generality in 2026 [38, 18, 39]. We analyse whether that proof can conversely help with Conjecture 1.1 in §9.1.

2.4 Snarks and cyclic connectivity

A *snark* is usually defined as a cyclically 4-edge-connected cubic graph of girth at least 5 which is not 3-edge-colourable; the connectivity and girth conditions exclude examples obtained trivially from smaller ones. Following [37] we drop these conditions and call *any* cubic Cayley graph that is not 3-edge-colourable a *Cayley snark*. For vertex-transitive cubic graphs the distinction is small:

Theorem 2.6 (Nedela–Škoviera [35]). *The cyclic edge-connectivity of a connected cubic vertex-transitive graph equals its girth.*

Thus a Cayley snark of girth at least 5 would be a snark in the classical sense, and Jaeger's conjecture [22, 23] that there are no cyclically 7-edge-connected snarks (still open; the analogous conjecture for girth was refuted by Kochol [24]) would imply Conjecture 1.1 for all cubic Cayley graphs of girth at least 7. We return to this in §3.3.

2.5 Coverings and quotients

A surjective graph homomorphism $\pi: \tilde{X} \rightarrow X$ between (multi)graphs is a *covering projection* if it maps the set of edges incident with each vertex $\tilde{v}$ bijectively onto the set of edges incident with $\pi(\tilde{v})$. If a group N acts semiregularly on $\tilde{X}$ (every non-identity element acts without fixed vertices and without inverting edges) and the quotient $\tilde{X}/N$ has no loops or semi-edges, then $\tilde{X} \rightarrow \tilde{X}/N$ is a covering projection, called a *regular N -covering*.

Lemma 2.7. *If $\pi: \tilde{X} \rightarrow X$ is a covering projection of cubic multigraphs and X is 3-edge-colourable, then so is $\tilde{X}$.*

Proof. Colour each edge $\tilde{e}$ of $\tilde{X}$ with the colour of $\pi(\tilde{e})$. At each vertex $\tilde{v}$ the three edges map bijectively onto the three (distinctly coloured) edges at $\pi(\tilde{v})$. $\square$

For Cayley graphs the relevant quotients come from normal subgroups. Let $N \trianglelefteq G$ and write $\bar{g} = gN$.

Lemma 2.8 (quotient lemma). *Let $X = X(G; a, x)$ with $\text{ord}(x)$ odd and let $1 \neq N \trianglelefteq G$ be a proper normal subgroup with $a \notin N$ and $x \notin N$. Then $\bar{G} = G/N$ is generated by the involution $\bar{a}$ and the element $\bar{x}$ of odd order at least 3, $\bar{X} = X(\bar{G}; \bar{a}, \bar{x})$ is a connected cubic Cayley graph on a group of order $|G|/|N|$, and $g \mapsto \bar{g}$ is a regular N -covering $X \rightarrow \bar{X}$. Consequently, if $\bar{X}$ is 3-edge-colourable then so is X .*

Proof. $\bar{a} \neq 1$ since $a \notin N$, and $\text{ord}(\bar{x})$ divides the odd number $\text{ord}(x)$ and is > 1 since $x \notin N$; hence $\text{ord}(\bar{x}) \geq 3$, $\bar{x} \neq \bar{x}^{-1}$ and $\bar{x} \neq \bar{a}$. So $\{\bar{a}, \bar{x}, \bar{x}^{-1}\}$ is an inverse-closed 3-subset of $\bar{G} \setminus \{1\}$ generating $\bar{G}$. The map $g \mapsto \bar{g}$ sends the three edges $\{g, ga\}, \{g, gx\}, \{g, gx^{-1}\}$ at g to the three distinct edges $\{\bar{g}, \bar{g}\bar{a}\}, \{\bar{g}, \bar{g}\bar{x}\}, \{\bar{g}, \bar{g}\bar{x}^{-1}\}$ at $\bar{g}$, and N acts semiregularly by left multiplication with quotient $\bar{X}$. Apply Lemma 2.7. $\square$

When N contains a or x the quotient degenerates (the a -edges become semi-edges, or the x -edges become loops). These two cases behave very differently: the first is harmless (Lemma 3.3), the second is the heart of the problem (§3.5).

3 The easy cases and some reformulations

3.1 Trivial colourings

Proposition 3.1. *Let X be a connected cubic Cayley graph.*

- (a) *If $X = \text{Cay}(G, \{a, b, c\})$ with a, b, c involutions, then colouring each edge by its generator is a 3-edge-colouring.*
- (b) *If $X = X(G; a, x)$ with $\text{ord}(x)$ even, then F_x is an even 2-factor and X is 3-edge-colourable.*
- (c) *If G is abelian, then X is 3-edge-colourable.*

Proof. (a) and (b) are immediate from Lemma 2.2. For (c), by (a) and (b) we may assume $X = X(G; a, x)$ with $\text{ord}(x)$ odd; as G is abelian, $G = \langle x \rangle \times \langle a \rangle$ and X is the prism $C_{\text{ord}(x)} \square K_2$, which has a Hamilton cycle (use two a -edges and all but one edge of each of the two x -cycles). $\square$

Corollary 3.2. *Conjecture 1.1 is equivalent to the following statement: for every finite group $G = \langle a, x \rangle$ generated by an involution a and an element x of odd order $s \geq 3$, the graph $X(G; a, x)$ is 3-edge-colourable.*

From now on we therefore assume that $X = X(G; a, x)$ with $s = \text{ord}(x)$ odd, $s \geq 3$. Note that X then contains the odd cycles of F_x , so X is never bipartite, and a 3-edge-colouring must use a 2-factor different from the “obvious” one. In terms of the group, X is 3-edge-colourable if and only if there is an a -invariant subset $M \subseteq G$ and a set $E' \subseteq E(F_x)$ such that every vertex of M is incident with two edges of E' , every vertex of $G \setminus M$ with one, and the 2-factor consisting of E' together with the a -edges at $G \setminus M$ has only even cycles. Here M is the set of vertices whose a -edge is not used by the even 2-factor; there seems to be no useful algebraic characterisation of this condition.

3.2 Normal subgroups containing the involution

Lemma 3.3. *Let $X = X(G; a, x)$ with $\text{ord}(x)$ odd. If G has a proper normal subgroup N with $a \in N$ (equivalently, if there is a surjective homomorphism $\varphi: G \rightarrow \mathbb{Z}_m$, $m \geq 3$, with $\varphi(a) = 0$), then X is 3-edge-colourable.*

Proof. $G/N = \langle xN \rangle$ is cyclic of order $m \mid \text{ord}(x)$, and $m \geq 3$ because $N \neq G$; let $\varphi: G \rightarrow \mathbb{Z}_m$ be the quotient map, so $\varphi(a) = 0$ and $\varphi(x) = 1$. Choose a proper vertex 3-colouring $c: \mathbb{Z}_m \rightarrow \{1, 2, 3\}$ of the cycle C_m (it exists for every $m \geq 3$). Colour the x -edge $\{g, gx\}$ with $c(\varphi(g))$ and the a -edge $\{g, ga\}$ with the colour in $\{1, 2, 3\} \setminus \{c(\varphi(g) - 1), c(\varphi(g))\}$; the latter is well defined because $\varphi(ga) = \varphi(g)$. At a vertex g with $\varphi(g) = i$ the two x -edges $\{gx^{-1}, g\}$ and $\{g, gx\}$ have the distinct colours $c(i - 1)$ and $c(i)$, and the a -edge has the third colour. $\square$

Note that m is automatically odd. The lemma says that the “degenerate quotient” in which the a -edges collapse to semi-edges is never an obstacle.

3.3 Small girth

A cycle of length ℓ through the identity of $X = X(G; a, x)$ corresponds to a cyclically reduced word of length ℓ in a, x, x^{-1} (no factor aa , xx^{-1} or $x^{-1}x$, also cyclically) representing 1 in G .

Proposition 3.4. *Let $X = X(G; a, x)$ with $s = \text{ord}(x)$ odd, $s \geq 3$, and G not cyclic.*

- (a) *X has girth 3 if and only if $s = 3$.*
- (b) *If X has girth 4, then $\langle x \rangle \trianglelefteq G$ and $G \cong D_{2s}$ is dihedral.*
- (c) *If X has girth 5, then either $s = 5$ or $\langle x \rangle \trianglelefteq G$, so that $G = \langle x \rangle \rtimes \langle a \rangle$ is metacyclic.*

In particular Conjecture 1.1 holds for all cubic Cayley graphs of girth 4, and for those of girth 3 (resp. 5) unless $s = 3$ (resp. $s = 5$).

Proof. A cyclically reduced word in $a, x^{\pm 1}$ of length $\ell \leq 5$ has one of the forms $x^{\pm \ell}$; ax^k with $|k| = \ell - 1$; or (for $\ell \in \{4, 5\}$) $ax^i ax^j$ with $i, j \neq 0$ and $|i| + |j| = \ell - 2$. A relation $x^{\pm \ell} = 1$ with $\ell \leq 5$ and s odd forces $s = \ell \in \{3, 5\}$. A relation $ax^k = 1$ gives $a \in \langle x \rangle$ and G cyclic, which is excluded. A relation $ax^i ax^j = 1$ gives $ax^i a = x^{-j}$ with $|i|, |j| \in \{1, 2, 3\}$ and $|i| + |j| \leq 3$; since s is odd, x^i and x^{-j} generate $\langle x \rangle$ unless $s = 3$ and $3 \mid i$ or $3 \mid j$, which would give girth 3. Hence a normalises $\langle x \rangle$, $\langle x \rangle \trianglelefteq G$, and $G = \langle x \rangle \rtimes \langle a \rangle$. For $\ell = 4$ the only such relations are $(ax)^2 = 1$ (giving D_{2s}) and $axax^{-1} = 1$ (giving G abelian, hence cyclic). The final assertion follows from Theorem 5.1 below, since metacyclic groups are solvable. $\square$

Combined with Theorem 2.6, this shows that in a proof of Conjecture 1.1 by “girth” the cases requiring an idea are girth 3 with $s = 3$, girth 5 with $s = 5$, girth 6, and girth at least 7; a proof of Jaeger’s conjecture on cyclically 7-edge-connected snarks would settle the last case. This parallels the observation of Aboomahigir and Nedela [1] that, modulo Thomassen’s conjecture that cyclically 7-connected cubic graphs other than the Coxeter graph are hamiltonian, the hamiltonicity of cubic Cayley graphs reduces to girth at most 6.

3.4 Generators of order three: arc-regular cubic graphs

The case $s = 3$ has a pleasant reformulation. Recall that the *truncation* $T(Y)$ of a cubic graph Y is obtained by replacing every vertex by a triangle; its vertices are the arcs (ordered pairs of adjacent vertices) of Y , two arcs being adjacent if they have the same initial vertex or are inverse to each other.

Lemma 3.5. *A cubic multigraph Y is 3-edge-colourable if and only if $T(Y)$ is.*

Proof. In a 3-edge-colouring of $T(Y)$ the three edges of a triangle receive three distinct colours, hence so do the three edges leaving the triangle; restricting the colouring to the edges of $T(Y)$ that come from edges of Y gives a 3-edge-colouring of Y . Conversely, given a 3-edge-colouring of Y , colour each triangle edge of $T(Y)$ with the colour of the edge of Y opposite to it in the triangle. $\square$

A group $G \leq \text{Aut}(Y)$ is *arc-regular* if it acts regularly (transitively with trivial stabilisers) on the arcs of Y .

Proposition 3.6. *Let $G = \langle a, y \rangle$ with $a^2 = y^3 = 1$ and $|G| > 6$. Let Y be the coset graph with vertex set $\{\langle y \rangle g : g \in G\}$ in which $\langle y \rangle g$ is adjacent to $\langle y \rangle ay^i g$ for $i = 0, 1, 2$. Then Y is a connected simple cubic graph on $|G|/3$ vertices, G acts on it by right multiplication as an arc-regular group of automorphisms, and $X(G; a, y) \cong T(Y)$. Conversely, if Y is a connected cubic graph with an arc-regular group G of automorphisms, then $G = \langle a, y \rangle$ for an involution a and an element y of order 3, and $T(Y) \cong X(G; a, y)$.*

Proof. Put $H = \langle y \rangle$. The three neighbours Ha, Hay, Hay^2 of the vertex H are distinct: $Ha = Hay^i$ would give $ay^i a^{-1} \in H$, so that a normalises H and $|G| \leq 6$. Right multiplication by G preserves adjacency and is transitive on vertices; the stabiliser H of the vertex H permutes its three neighbours cyclically ($Hay^i \cdot y = Hay^{i+1}$), so G is transitive on arcs, and $|G| = 3|V(Y)|$ equals the number of arcs, so the action on arcs is regular. Identify the arc (Hg, Hag) with $g \in G$; arcs with the same initial vertex correspond to g and $y^{\pm 1}g$, and the inverse of the arc g is the arc ag . So $T(Y)$ is the Cayley graph of G with respect to $\{a, y, y^{-1}\}$ and left multiplication, which is isomorphic to $X(G; a, y)$ via $g \mapsto g^{-1}$.

Conversely, if G is arc-regular on Y , the stabiliser G_v of a vertex v acts regularly on the three arcs leaving v , so $G_v = \langle y \rangle \cong \mathbb{Z}_3$; some $a \in G$ maps the arc (v, w) to (w, v) , and a^2 fixes (v, w) , so a is an involution. Connectedness of Y gives $G = \langle G_v, a \rangle$, and identifying arcs with group elements as above gives $T(Y) \cong X(G; a, y)$. $\square$

Corollary 3.7. *Conjecture 1.1 for generating pairs (a, x) with $\text{ord}(x) = 3$ is equivalent to the statement that every connected cubic graph admitting an arc-regular group of automorphisms is 3-edge-colourable.*

Conder’s census [9] of all connected cubic arc-transitive graphs on at most 10 000 vertices records that every one of them other than the Petersen graph and the Coxeter graph has a Hamilton cycle. Since the Coxeter graph is 3-edge-colourable, Lemma 3.5 and Proposition 3.6 give:

Corollary 3.8. *Every connected cubic graph on at most 10 000 vertices that admits an arc-regular group of automorphisms is 3-edge-colourable. Equivalently, Conjecture 1.1 holds for every $X(G; a, x)$ with $\text{ord}(x) = 3$ and $|G| \leq 30\,000$.*

The Petersen graph is arc-transitive with automorphism group S_5 , but has no arc-regular subgroup (S_5 has no subgroup of order 30), in accordance with the conjecture; the same applies to its truncation, which is a vertex-transitive cubic graph that is not 3-edge-colourable but not a Cayley graph. Cubic arc-transitive graphs have been classified into seventeen “action types” by Conder and Nedela [10] according to which of the Djoković–Miller groups [12] occur as subgroups of the automorphism group; the ones containing an arc-regular (“1-regular”) subgroup are exactly the graphs in Corollary 3.7. Since every finite non-abelian simple group, apart from the symplectic groups $\text{PSp}(4, 2^f)$, $\text{PSp}(4, 3^f)$ and finitely many others (among them A_6 , A_7 , A_8 , $\text{PSL}(3, 4)$, $\text{PSU}(3, 3)$, $\text{PSU}(3, 5)$, $\text{PSU}(4, 3)$, $\text{PSU}(5, 2)$ and the sporadic groups M_{11} , M_{22} , M_{23} , McL), is generated by an involution and an element of order 3 (Liebeck and Shalev [30], Woldar [50]; see [41] for the classical groups of small dimension), the case $s = 3$ alone already concerns a cubic Cayley graph on almost every simple group. (In our computations, Table 2, the groups A_6 , A_7 , A_8 , $\text{PSL}(3, 4)$, $\text{PSU}(3, 3)$ and M_{11} indeed have no generating pair with $\text{ord}(x) = 3$.)

3.5 Normal subgroups of index two: bi-Cayley graphs

Suppose that $X = X(G; a, x)$ with $s = \text{ord}(x)$ odd and that $T \trianglelefteq G$ has index 2. Then $x \in T$ (its image in $G/T \cong \mathbb{Z}_2$ has odd order) and $a \notin T$, so $G = T \rtimes \langle a \rangle$ and conjugation by a induces an involutory automorphism α of T with $T = \langle x, \alpha(x) \rangle$. The vertex set of X is $T \sqcup Ta$, the x -edges inside T are the cycles of x , the x -edges inside Ta are, after the identification $ta \leftrightarrow t$, the cycles of $\alpha(x)$ (because $ta \cdot x = t\alpha(x)a$), and the a -edges form the identity matching $t \leftrightarrow ta$. Thus

$$X \cong B(T; x, \alpha(x)), \quad (1)$$

where, for a group T and elements $u, v \in T$, the graph $B(T; u, v)$ has vertex set $T \times \{0, 1\}$ and edges $(t, 0) \sim (tu, 0)$, $(t, 1) \sim (tv, 1)$ and $(t, 0) \sim (t, 1)$ for all $t \in T$. This is a *bi-Cayley graph* of T , a natural generalisation of the generalised Petersen graphs $GP(n, k) = B(\mathbb{Z}_n; 1, k)$. Conversely, for any group T , any $x \in T$ of odd order and any involutory automorphism α of T with $\langle x, \alpha(x) \rangle = T$, the graph $B(T; x, \alpha(x))$ is the cubic Cayley graph $X(T \rtimes \langle \alpha \rangle; \alpha, x)$.

The order-three case of this construction is always colourable, without any assumption on T .

Proposition 3.9. *Let $G = \langle a, x \rangle$ with $a^2 = x^3 = 1$. If G has a normal subgroup of index two, then $X(G; a, x)$ is 3-edge-colourable.*

Proof. Let $T \trianglelefteq G$ have index two. The image of x in $G/T \cong \mathbb{Z}_2$ has odd order, so $x \in T$; since a and x generate G , we have $a \notin T$. Contract every x -triangle of $X = X(G; a, x)$ to a vertex, retaining the a -edges. The resulting graph Y is a cubic multigraph and $X \cong T(Y)$. Every contracted triangle lies wholly in either T or Ta , while every a -edge joins these two sets. Thus Y is bipartite. By König’s edge-colouring theorem Y is 3-edge-colourable, and Lemma 3.5 now gives a 3-edge-colouring of X . $\square$

Corollary 3.10. *If a Cayley snark $X(G; a, x)$ with $\text{ord}(x) = 3$ exists, then every one of smallest order among such graphs has G non-abelian simple.*

Proof. Choose such a graph with $|G|$ minimum and suppose that $1 \neq N \trianglelefteq G$ is proper. If $a \in N$, Lemma 3.3 colours X . If $a, x \notin N$, then xN still has order three, and Lemma 2.8 makes X a cover of a smaller Cayley graph; that quotient must also be a snark, contrary to minimality. The only remaining possibility is $x \in N$ and $a \notin N$, in which case $G/N = \langle aN \rangle \cong \mathbb{Z}_2$ and Proposition 3.9 again colours X . Hence G is simple; it is non-abelian by Proposition 3.1. $\square$

The generalised Petersen graph $GP(n, k)$ is a Cayley graph if and only if $k^2 \equiv 1 \pmod{n}$ (Nedela and Škoviera [36]), which is exactly the condition that multiplication by k be an involutory automorphism of $\mathbb{Z}_n$; and Castagna and Prins [5] proved Watkins' conjecture that every generalised Petersen graph other than the Petersen graph itself is 3-edge-colourable. (The Petersen graph is $GP(5, 2)$ with $2^2 \equiv -1$, not $+1$, modulo 5.) So Conjecture 1.1 holds for all generalised Petersen graphs, and the results of §6 show that the whole conjecture reduces to graphs of the form (1) with T simple or $T \cong S \times S$, together with Cayley graphs on simple groups.

4 Quotients by arbitrary subgroups, and a reformulation over $\mathbb{Z}_3$

This section collects elementary quotient and colouring constructions. The first extends the quotient lemma (Lemma 2.8) from normal subgroups to arbitrary subgroups, at the price of allowing semi-edges in the quotient; it is the tool behind the computations of Section 8. A monodromy refinement allows invariant matchings whose complementary colours do not descend, and gives a uniform construction for a symmetric-group family. The final reformulation describes the 3-edge-colourings of $X(G; a, x)$ as the a -invariant functions $G \rightarrow \mathbb{Z}_3$ satisfying a local condition along the x -cycles.

4.1 Pregraph quotients

A *pregraph* is a graph in which, besides ordinary edges, *semi-edges* (edges with only one end) and loops are allowed; each vertex of a cubic pregraph is incident with three *darts*, an ordinary edge contributing a dart at each of its ends and a semi-edge one dart. A 3-edge-colouring of a pregraph colours the ordinary edges and the semi-edges so that the three darts at every vertex carry three different colours.

Let $X = X(G; a, x)$ with $s = \text{ord}(x)$ odd and let $H \leq G$ be *any* subgroup. The left action of H on $V(X) = G$ is free, and so is its action on the darts (g, gs) , $s \in \{a, x, x^{-1}\}$. The quotient pregraph $H \backslash X$ has vertex set $H \backslash G = \{Hg\}$ and one dart for each H -orbit of darts; the darts (g, gs) and (gs, g) lie in the same orbit if and only if $hg = gs$ and $hgs = g$ for some $h \in H$, i.e. $s^2 = 1$ and $gs g^{-1} \in H$, which happens only for $s = a$. Hence the a -edge at Hg is a semi-edge exactly when $gag^{-1} \in H$, and the x -edge at Hg is a loop exactly when $g x g^{-1} \in H$.

Lemma 4.1 (generalised quotient lemma). *Let $X = X(G; a, x)$ with $\text{ord}(x)$ odd and let $H \leq G$ be a subgroup containing no conjugate of x . Then $\bar{X} = H \backslash X$ is a loopless cubic pregraph on $|G : H|$ vertices with exactly $|H \cap a^G| \cdot |C_G(a)|/|H|$ semi-edges, and $X \rightarrow \bar{X}$, $g \mapsto Hg$, is a covering of pregraphs. If $\bar{X}$ is 3-edge-colourable, then so is X ; more precisely, the 3-edge-colourings of $\bar{X}$ correspond bijectively to the 3-edge-colourings of X that are invariant under left multiplication by H .*

Proof. The three darts at g are mapped to the three darts at Hg , which are distinct because $\bar{X}$ has no loops; so the projection is a covering. Given a colouring of $\bar{X}$, colour every dart of X with the colour of its image; the two darts of an ordinary edge $\{g, ga\}$ with $gag^{-1} \notin H$ map to the two darts of an ordinary edge of $\bar{X}$ and receive the same colour, and if $gag^{-1} \in H$ they map to the same semi-edge. Thus the colouring is a well defined 3-edge-colouring of X , obviously H -invariant, and conversely an H -invariant colouring of X is constant on H -orbits of darts and descends to $\bar{X}$. The count of semi-edges is the number of cosets Hg with $gag^{-1} \in H$, i.e. the number of elements g with $gag^{-1} \in H \cap a^G$, which is $|H \cap a^G| |C_G(a)|$, divided by $|H|$. $\square$

Lemma 2.8 is the case $H \trianglelefteq G$ with $a \notin H$, and Lemma 2.7 in the form used by the program of §8 is the case of odd $|H|$ (no semi-edges). The point of Lemma 4.1 is that H may now be

large: any subgroup whose order is not divisible by $\text{ord}(x)$, the centraliser $C_G(a)$ whenever no conjugate of x centralises a (the quotient is then the pregraph on the conjugacy class a^G in which b is joined to b^a , b^x and $b^{x^{-1}}$), a point stabiliser of any permutation representation in which x has no fixed points, and so on. Semi-edges carry a parity constraint which, in practice, is the main reason why the largest admissible subgroup need not work.

Lemma 4.2 (parity). *Let $\bar{X}$ be a cubic pregraph with n vertices and σ semi-edges. If $\bar{X}$ is 3-edge-colourable then the semi-edges fall into three colour classes each of size $\equiv n \pmod{2}$; in particular $\sigma \equiv n \pmod{2}$, and $\sigma \geq 3$ if n is odd.*

Proof. Each colour class contains e ordinary edges and σ' semi-edges with $2e + \sigma' = n$. $\square$

Example 4.3. Let $G = \text{PSL}(2, q)$ act on the projective line $\Omega = \text{GF}(q) \cup \{\infty\}$, and let x lie in a non-split torus, so that x has no fixed point on Ω and $H = G_\infty$ (a Borel subgroup) is admissible in Lemma 4.1; the quotient is the Schreier graph of $\langle a, x \rangle$ on the $q + 1$ points. If q is even, an involution a fixes exactly one point, so $\bar{X}$ has $q + 1$ (odd) vertices and one semi-edge, and Lemma 4.2 shows that $\bar{X}$ is never 3-edge-colourable—although X is, for $q \in \{4, 8, 16, 32, 64\}$ (Table 2). If $q \equiv 3 \pmod{4}$ the involutions are fixed-point-free on Ω and if $q \equiv 1 \pmod{4}$ they have two fixed points, so the parity condition holds; for these q we found by computer that the $(q + 1)$ -vertex quotient is 3-edge-colourable for most but not all generating pairs (for instance for all 192 generating pairs with $\text{ord}(x) = 13$ in $\text{PSL}(2, 25)$ or $\text{ord}(x) = 15$ in $\text{PSL}(2, 29)$, but for only 11 of the 13 generating pairs with $\text{ord}(x) = \text{ord}(ax) = 7$ in $\text{PSL}(2, 13)$).

4.2 Hamiltonian and cyclewise quotients

The quotient $H \backslash X$ is itself “hamiltonian” when $\langle x \rangle$ acts transitively on $H \backslash G$, i.e. when $G = H \langle x \rangle$ is an exact factorisation; the x -cycle through H then passes through all $|G : H| = \text{ord}(x)$ vertices. The following sufficient condition also applies when the x -edges form several cycles.

Lemma 4.4 (transition criterion). *Let Y be a loopless cubic pregraph with a spanning 2-factor F all of whose cycles are odd. Contract the cycles of F to obtain a pregraph Q , retaining the complementary darts in their cyclic order around each contracted vertex. Suppose that Q has a spanning 2-regular subpregraph R which uses two consecutive darts at every vertex and whose circuit components all have even length. Then Y is 3-edge-colourable.*

Proof. At each vertex of Q , the two darts selected by R correspond to complementary darts of Y at adjacent vertices u, u' of a cycle C of F . Delete the edge uu' from C and retain the two selected darts. Doing this on every cycle of F produces a spanning 2-regular subpregraph Z of Y . A circuit component of R expands to a circuit of Z whose length has the same parity: every visit to C replaces a vertex by the path $C - uu'$ of even length. Hence every circuit of Z is even. Every other component of R , and therefore of Z , is a path capped by two semi-edges. Colour the circuits of Z alternately 0, 1, do the same along each capped path (including its two semi-edges), and colour every dart outside Z with 2. Every vertex sees all three colours. $\square$

Corollary 4.5. *Let $X = X(G; a, x)$ with $\text{ord}(x)$ odd, and let $H \leq G$ contain no conjugate of x . Suppose that every cycle of the permutation $Hg \mapsto Hgx$ of $H \backslash G$ contains adjacent cosets Hg, Hgx for which $gag^{-1}, gxa x^{-1}g^{-1} \in H$. Then X is 3-edge-colourable.*

Proof. By Lemma 4.1, $H \backslash X$ is a loopless cubic pregraph. Its x -edges form a spanning 2-factor of odd cycles, and the displayed membership conditions say precisely that the complementary darts at Hg and Hgx are semi-edges. In the contracted pregraph Q of Lemma 4.4, choose those two adjacent semi-edges at every vertex; their union is a spanning 2-regular subpregraph with no circuit components. Lemma 4.4 colours the quotient, and Lemma 4.1 lifts the colouring to X . $\square$

Proposition 4.6. *Let $X = X(G; a, x)$ with $\text{ord}(x)$ odd. Suppose $G = H\langle x \rangle$ with $H \cap \langle x \rangle = 1$ for a subgroup H containing a and xax^{-1} . Then X is 3-edge-colourable.*

Proof. Since $\langle x \rangle$ acts regularly on $H \setminus G$, x has no fixed point there, so H contains no conjugate of x and Lemma 4.1 applies; the pregraph $H \setminus X$ has the Hamilton cycle $H, Hx, Hx^2, \dots$, and its adjacent vertices H and Hx carry semi-edges because $a \in H$ and $xax^{-1} \in H$. Apply Lemma 4.4. $\square$

For instance, if n is odd, x is an n -cycle in S_n and a is an involution fixing two cyclically consecutive points of x (say $a = (1\ 2)$, or any involution with at least $(n+1)/2$ fixed points), then Proposition 4.6 with $H = S_{n-1}$ shows that $X(\langle a, x \rangle; a, x)$ is 3-edge-colourable; for $a = (1\ 2)$ this is a two-line proof of the colouring consequence of the Compton–Williamson theorem quoted in §7. The adjacency hypothesis cannot simply be replaced by “at least three semi-edges” (the minimum allowed by Lemma 4.2): the Petersen graph minus a vertex is a 9-cycle with three chords and three vertices of valency two, i.e. the pregraph $S_8 \setminus X$ for $x = (0\ 1 \dots 8)$ and $a = (1\ 5)(2\ 7)(4\ 8)$, and it is not 3-edge-colourable, although $\langle a, x \rangle$ (a solvable group of order 162) and X are perfectly harmless. This example illustrates the main difficulty of the quotient method: a small “natural” quotient of a colourable Cayley graph can be a Petersen-like snark, so a proof of Conjecture 1.1 along these lines must be able to choose the subgroup H adaptively.

The generation hypothesis rules out precisely that obstruction in the first families beyond transpositions. The small degrees in the following proposition are certified by the reproducible exhaustive calculation described in its proof.

Proposition 4.7 (an odd cycle and a sparse involution). *Let N be odd, let x be an N -cycle in S_N , and let a be a product of r disjoint transpositions. Suppose that $\langle a, x \rangle = S_N$. If $N \geq 4r + 1$, then $X(S_N; a, x)$ is 3-edge-colourable. The same conclusion holds, with no extra bound on N , whenever $r \leq 7$.*

Proof. Conjugate so that $x = (0\ 1 \dots N-1)$. The involution a fixes $N - 2r$ points. If $N \geq 4r + 1$, then $N - 2r \geq (N + 1)/2$, so two fixed points are consecutive on the x -cycle. Proposition 4.6, with the stabiliser of either one, gives the result. Also x is even, whereas $\langle a, x \rangle = S_N$, so r is odd. The case $r = 1$ is Theorem 4.13. It remains to treat $r = 3, 5, 7$ below the displayed bound.

First let $r = 3$, so the remaining degrees are $N = 7, 9, 11$. Conjugation by a power of x rotates the six endpoints of a , and replacing x by x^{-1} reverses them without changing the Cayley graph. Enumerating the matchings of size three modulo these dihedral operations gives respectively 11, 85 and 350 orbit types. For $N = 9$ and 11, take $H = S_{N-1}$. The pregraph $H \setminus X$ is the N -cycle with the three transposition chords of a and a semi-edge at each of its $N - 6$ fixed points. Directly verified edge-colouring certificates exist for all 350 types when $N = 11$ and for 84 of the 85 types when $N = 9$. The remaining type has

$$a = (0\ 4)(1\ 6)(3\ 7);$$

this is the Petersen-minus-a-vertex quotient displayed above, after a rotation, and exact Schreier recursion gives $|\langle a, x \rangle| = 162$. It is therefore excluded by the hypothesis.

When $(r, N) = (3, 7)$, the point quotient has one semi-edge and cannot be coloured by Lemma 4.2. Instead take for H the stabiliser of a two-set. The quotient pregraph $H \setminus X$ has $\binom{7}{2} = 21$ vertices and three semi-edges. It has a directly verified edge-colouring for ten of the eleven dihedral types. The remaining type is

$$a = (0\ 1)(2\ 6)(3\ 5),$$

the reflection $i \mapsto 1 - i$ of the 7-cycle; hence $\langle a, x \rangle$ is the dihedral group of order 14, and is excluded.

Now let $r = 5$. The remaining degrees are $N = 11, 13, 15, 17, 19$. Configurations with consecutive fixed points are already done, so it suffices to enumerate those whose fixed points are independent on the N -cycle. Up to the same dihedral operations there are respectively

$$513, \quad 5832, \quad 12152, \quad 5832, \quad 513$$

such types. The point-stabiliser quotient colours all of them for $N = 15, 17, 19$, and all but 46 for $N = 13$; the two-set-stabiliser quotient colours those 46. For $N = 11$ the point quotient again has one semi-edge. The two-set quotient colours 512 of its 513 types. Its sole remaining type is

$$a = (01)(210)(39)(48)(57),$$

the reflection $i \mapsto 1 - i$ of the 11-cycle, which together with x generates the dihedral group of order 22. Thus it too is excluded.

Finally let $(r, N) = (7, 15)$. The unique fixed point may be rotated to 0; the remaining dihedral equivalence is the reflection $i \mapsto -i$. This leaves 68 219 matching types. Their two-set quotients are colourable in 68 144 cases. Exact Schreier recursion gives the following orders for the groups generated by the remaining 75 pairs, with multiplicities shown as exponents:

$$30^1, \quad 360^2, \quad 750^2, \quad 2430^4, \quad 3000^5, \quad 29160^{10}, \quad 38880^{21}, \quad 466560^{30}.$$

None is S_{15} , so the generation hypothesis excludes every failure.

For $(r, N) = (7, 17)$, configurations with two consecutive fixed points are again covered by Proposition 4.6. There are 442 possible independent triples of fixed points. Modulo the dihedral normaliser of $\langle x \rangle$ they form 16 orbits: ten have trivial stabiliser and six have a reflection stabiliser. Enumerating the perfect matchings of the other fourteen points modulo these residual stabilisers gives 1 760 664 types. The point-stabiliser quotient colours 1 755 100 of them. For each of the remaining 5 564, the two-set-stabiliser quotient, with $\binom{17}{2} = 136$ vertices and ten semi-edges, is colourable. Thus in this degree the quotient certificates cover every pair, even without the generation hypothesis.

For $r = 7$, five degrees remain below the consecutive-fixed-point bound:

$$N = 19, 21, 23, 25, 27.$$

The numbers of independent fixed sets are respectively

$$2717, \quad 5148, \quad 3289, \quad 650, \quad 27;$$

they give 79, 133, 79, 16, 1 fixed-set orbits and, after reduction by each residual dihedral stabiliser,

$$9\,671\,925, \quad 16\,576\,723, \quad 9\,671\,925, \quad 1\,760\,664, \quad 68\,219$$

matching types. The point-stabiliser quotient colours every type in degrees 21, 23, 25, 27. In degree 19 it colours 9 671 885 types; the two-set quotient, with 171 vertices and 17 semi-edges, colours the other 40.

Here the point calculation has a small transparent search space. Colour the edges of the Hamilton N -cycle properly and record, at each of the fourteen chord endpoints, the missing colour. The chord matching extends this to a Tait colouring exactly when its two endpoints always have equal missing colours. It suffices to enumerate cycle colourings whose fourteen recorded values use at most two colours, so after global complementation a proposed chord matching has only 2^6 assignments to test. Directly checked witnesses of this restricted kind cover 9 671 473 types for $N = 19$, all but two types for $N = 21$, and every type for $N = 23, 25, 27$. The complete three-colour point calculation covers another 412 types for $N = 19$ and the two exceptional types for $N = 21$, leaving precisely the 40 degree-19 two-set cases above.

All the stabilisers used above are admissible: an N -cycle fixes neither a point nor a two-set. Lemma 4.1 lifts every positive quotient colouring to X .

For completeness, `code/sparse_involution_cycle_experiment.py` generates the matchings rather than assuming the orbit counts, reduces by the relevant dihedral operations, constructs both kinds of quotient, obtains each positive certificate with a SAT solver, and then checks directly that the three incident darts at every quotient vertex have distinct colours. Its independent implementation of Schreier's lemma verifies every exceptional group order above. The separate parallel certificate `code/septuple_n17_experiment.py` performs the degree-17 enumeration. The script `code/septuple_later_experiment.py` performs the five later enumerations, directly verifies every projected cycle-colouring pattern, checks each positive chord matching against its pattern, and uses the independently verified SAT routine for the complete point and two-set fallbacks. Thus the positive certificates are checked item by item rather than trusted merely because a solver reports success. $\square$

4.3 Monodromy without invariant colours

Requiring the whole colouring to descend to a subgroup quotient is stronger than requiring its colour- k matching to descend. The distinction has an exact cycle-lifting description.

Lemma 4.8 (the matching-monodromy test). *Let $H \leq G$ contain no conjugate of x , let $Y = H \backslash X(G; a, x)$, and let $\bar{M}$ be a perfect matching of the pregraph Y , allowing semi-edges in the matching. Lift it to a perfect matching M of X . Each component of $Y - \bar{M}$ is a circuit or a path capped by two semi-edges. For a circuit C of length ℓ , let w_C be the product of its directed generator labels, in traversal order. Then the preimage of C consists of $|H|/\text{ord}(w_C)$ circuits of length $\ell \text{ord}(w_C)$. Every capped path lifts to even circuits. Consequently $X - M$ is even if and only if every odd circuit of $Y - \bar{M}$ has monodromy w_C of even order.*

Proof. Starting over a vertex Hg , a traversal of C takes g to gw_C , where $gw_C g^{-1} \in H$. Repeating the traversal closes for the first time after $\text{ord}(w_C)$ repetitions. This proves both the length and the number of lifted circuits; changing the starting lift conjugates the corresponding element of H and does not change its order.

For a capped path, choose a semi-edge and one of its lifts e . An involution of H interchanges the endpoints of e . Since M is H -invariant, this involution preserves the complementary circuit through e and acts on it without fixed vertices. An odd circuit has no fixed-point-free involutory automorphism, so that circuit is even. Every component over the capped path contains a lift of that semi-edge. The final assertion follows from the circuit formula. $\square$

Here is a family in which the monodromy orders can be controlled explicitly, without a bound on the number of lifted circuits.

Proposition 4.9 (the transposition-odd-cycle family). *Let $s \geq 5$ be odd, let $G = S_{s+1}$ on $\{\infty\} \cup \mathbb{Z}_s$, and put*

$$a = (\infty 0), \quad x = (0 1 \cdots s-1), \quad H = C_G(a).$$

Let $\mathcal{F}$ be the perfect matchings of $X = X(G; a, x)$ selecting exactly one a -edge at each x -cycle, as in Proposition 9.25. There are exactly

$$N_s = (s-2)^{(s-1)/2} + (s-1)(s-2)^{(s-3)/2}$$

H -invariant members of $\mathcal{F}$, and every one has an even complement. In particular this supplies minimum-face Tait colourings throughout this infinite family.

Proof. Conjugating a by powers of x gives all the transpositions (∞i) , so a, x generate S_{s+1} . The subgroup $H \cong S_2 \times S_{s-1}$ is the stabiliser of the two-set $\{\infty, 0\}$. Using right actions for permutations, the quotient vertices are the two-subsets, with generators acting on their entries. Write $k = (s-1)/2$. Its x -cycles are

$$A = (A_i)_{i \in \mathbb{Z}_s}, \quad A_i = \{\infty, i\}, \quad L_d = (B_{d,j})_{j \in \mathbb{Z}_s}, \quad B_{d,j} = \{j, j+d\}, \quad 1 \leq d \leq k.$$

All these cycles have length s , including when s is composite. The only ordinary a -edges at L_d join A_d to $B_{d,0}$ and A_{-d} to $B_{d,-d}$; the other $s-2$ a -darts there are semi-edges. At A , the dart at A_0 is a semi-edge and all the others are ordinary edges to the leaves L_d .

An invariant matching in $\mathcal{F}$ is therefore specified by its selected position p on A . If $p = 0$, each L_d independently selects one of its $s-2$ semi-edges. If $p \neq 0$, the ordinary edge selected at A_p also specifies the selected position on its leaf; each other leaf selects a semi-edge. On every x -cycle, deleting the selected position leaves an even path whose perfect matching is unique. This proves both existence and the formula for N_s .

The retained x -edges in the complementary factor consist of isolated edges and one two-edge path, centred at the selected position. A complementary circuit in the quotient must consequently enter a leaf at one of its ordinary a -edges, traverse at most two x -edges there, and exit by the other ordinary a -edge. Only L_1 and L_2 can support such a passage: the two attachment positions on L_d have cyclic distance d . Their passages have lengths one and two, respectively.

First suppose $s \geq 7$. A circuit using only L_2 is impossible, since A_{-2} and A_2 have cyclic distance at least three. A circuit using both leaves would have to use the central edges $A_{-2}A_{-1}$ and A_1A_2 : the other pairing of their four endpoints has distances at least three. At A_{-1} , its leaf edge and $A_{-2}A_{-1}$ would both be complementary, forcing $A_{-1}A_0 \in \bar{M}$; similarly $A_0A_1 \in \bar{M}$, a contradiction. Thus the only possible circuit uses L_1 and the central path A_{-1}, A_0, A_1 , so $p = 0$ and it is a pentagon. Starting at A_{-1} , its word is

$$w = x^2ax^{-1}a = (0\,1 \cdots s-2)(s-1\,\infty)$$

in right-action notation. It has order $s-1$, which is even. Every such quotient pentagon therefore lifts to circuits of length $5(s-1)$.

For $s = 5$ the additional possibility is a circuit using L_2 and the central edge A_2A_{-2} . It is again a pentagon, with word

$$xax^2a = (0\,3\,1\,4)(2\,\infty)$$

of order four. A circuit using both leaves is still impossible: one pairing of central endpoints is excluded by the same matching contradiction, while the other would require two distinct two-edge central paths, of which there is only one. The L_1 pentagon also has monodromy of order four. In all cases every closed quotient component has even-order monodromy, and every other component is a capped path. Apply Lemma 4.8. $\square$

For $s \geq 7$, precisely

$$K_s = \frac{s-3}{2}(s-2)^{(s-3)/2}$$

of these matchings have a pentagon in their quotient complement. Indeed, choose $p = 0$ and on L_1 choose the selected position j from $2, 4, \dots, s-3$; these are exactly the choices retaining the edge from $B_{1,-1}$ to $B_{1,0}$. The other leaves are arbitrary. Such a matching has an even complement upstairs but cannot be the colour- k matching of an H -invariant Tait colouring, since that would alternately colour an odd quotient circuit. This is why the monodromy test genuinely allows more than colouring the pregraph itself. Choosing $p \neq 0$ instead gives an explicit nonempty subfamily with only capped paths downstairs.

Proposition 4.9 concerns one specified generating pair on each S_{s+1} , rather than all involution-odd-order generating pairs. Its stronger assertion is the evenness of every matching in the

displayed invariant minimum-face family. The constructions below settle every symmetric-group pair whose involution is a transposition, while Proposition 4.7 settles sparse involutions paired with a full odd cycle. Beyond these families, the unresolved task is still to choose a subgroup and a matching for which all odd quotient-circuit words have even order.

The same two-subset quotient admits a simpler construction when the transposition joins two nontrivial cycles. Here only one carefully chosen matching is asserted to work.

Proposition 4.10 (two coprime odd cycles). *Let $m, n \geq 3$ be odd and coprime. On the disjoint union*

$$U = \{u_i : i \in \mathbb{Z}_m\}, \quad V = \{v_j : j \in \mathbb{Z}_n\},$$

put

$$x = (u_0 u_1 \cdots u_{m-1})(v_0 v_1 \cdots v_{n-1}), \quad a = (u_0 v_0).$$

Then $\langle a, x \rangle = S_{m+n}$ and $X(S_{m+n}; a, x)$ has a $C_{S_{m+n}}(a)$ -invariant perfect matching with even complement.

Proof. The conjugates of a by powers of x are the transpositions $(u_i v_j)$ for all (i, j) , by the Chinese remainder theorem. Their transposition graph is connected, so they generate S_{m+n} .

Put $H = C_{S_{m+n}}(a)$ and $Y = H \backslash X$. Again the quotient vertices are the two-subsets. No conjugate of x belongs to H : an odd-order permutation stabilising $\{u_0, v_0\}$ fixes its two points, whereas x has no fixed point. The x -orbits on the quotient vertices are the central orbit

$$A_t = \{u_{t \bmod m}, v_{t \bmod n}\} \quad (t \in \mathbb{Z}_{mn})$$

and the leaf orbits

$$P_{d,i} = \{u_i, u_{i+d}\} \quad (1 \leq d \leq (m-1)/2), \quad Q_{e,j} = \{v_j, v_{j+e}\} \quad (1 \leq e \leq (n-1)/2).$$

They have odd lengths mn, m, n , respectively. Every ordinary a -edge joins A to a leaf. On P_d its two ends occur at positions $i = 0, -d$; on Q_e they occur at $j = 0, -e$. Every other a -dart on a leaf is a semi-edge.

Select the ordinary a -edge from $P_{1,0}$ to A_p , where $p \equiv 1 \pmod{m}$ and $p \equiv 0 \pmod{n}$. On every P_d with $d \geq 2$, select the semi-edge at $P_{d,1}$, and on every Q_e , select the semi-edge at $Q_{e,1}$. Thus every x -orbit is incident with exactly one selected a -dart: the first ordinary edge covers both P_1 and A , and each remaining leaf supplies its own semi-edge. On each odd x -orbit, match the vertices left over by that dart along the unique alternating matching of the remaining even path. The result is a perfect matching $\bar{M}$ of Y , which lifts to an H -invariant perfect matching M of X .

It remains to inspect $Y - \bar{M}$. On an odd cycle whose selected a -dart is at position one, the retained x -edges consist of the two-edge path through positions 0, 1, 2 and isolated edges. On P_d with $d \geq 2$, the two ordinary attachment positions 0, $-d$ therefore lie in distinct components. The same holds on every Q_e : for $e = 1$ the edge joining -1 to 0 belongs to the matching, and for $e \geq 2$ the preceding observation applies. On P_1 one of the two ordinary a -edges was itself selected, so only one attachment remains in the complement.

Consequently no component can pass through a leaf from one central attachment to the other. Every component entering a leaf ends instead at a semi-edge; a component contained in A also ends at semi-edges unless it enters a leaf. Hence $Y - \bar{M}$ has no circuit components at all: it is a union of paths capped by semi-edges. Lemma 4.8 says that every component of $X - M$ is even. $\square$

Corollary 4.11 (the transposition case in even degree). *Let $N \geq 4$ be even, let a be a transposition and let $x \in S_N$ have odd order. If $\langle a, x \rangle = S_N$, then $X(S_N; a, x)$ is 3-edge-colourable.*

Proof. All cycles of x have odd length. Since a meets at most two of them, transitivity forces x to have exactly two cycles, of odd lengths m, n , with the two points of a in different cycles. If $d = \gcd(m, n) > 1$, the sets

$$\{u_i : i \equiv r \pmod{d}\} \cup \{v_j : j \equiv r \pmod{d}\}, \quad r \in \mathbb{Z}_d,$$

form a nontrivial block system preserved by a and cyclically permuted by x , contrary to $\langle a, x \rangle = S_N$. Thus m, n are coprime. If both exceed one, apply Proposition 4.10. If one equals one, Proposition 3.9 handles the other length three (using the normal subgroup $A_4 \trianglelefteq S_4$), and Proposition 4.9 handles every larger odd length. $\square$

The restriction to a chosen matching in Proposition 4.10 is real. For $(m, n) = (3, 5)$, the quotient has 111 matchings obtained by selecting one a -dart per x -orbit. Eight have a complementary quotient circuit of length nine and monodromy order five, so their lifts have odd circuits; the construction in the proof avoids them. Thus the monodromy method has changed the missing step from eliminating every odd word to selecting a matching which avoids the bad words.

The remaining transposition case has one odd cycle and a chord of arbitrary generating distance. The same quotient admits an explicit construction, although three small distances require separate choices.

Proposition 4.12 (an odd cycle with a transposition chord). *Let $N \geq 3$ be odd, put $x = (0\ 1 \cdots N-1)$, and let $a = (0\ k)$, where $1 \leq k \leq (N-1)/2$ and $\gcd(k, N) = 1$. Then $X(S_N; a, x)$ is 3-edge-colourable.*

Proof. The conjugates $x^{-i}ax^i$ are the transpositions $(i\ i+k)$. Their transposition graph is connected because $\gcd(k, N) = 1$, so a, x generate S_N .

For $k = 1$ this follows from the Hamilton cycle of Compton and Williamson [6]. Suppose $k \geq 2$, so $N \geq 5$, and write $h = (N-1)/2$. Put $H = C_{S_N}(a)$ and $Y = H \setminus X$. The quotient vertices are the two-subsets, and H contains no conjugate of the fixed-point-free odd-order permutation x . Its x -orbits are

$$B_d = (B_{d,i})_{i \in \mathbb{Z}_N}, \quad B_{d,i} = \{i, i+d\}, \quad 1 \leq d \leq h.$$

On B_d the positions of the ordinary a -darts are

$$O_d = \{0, -d, k, k-d\} \quad (d \neq k), \quad O_k = \{-k, k\};$$

the missing position 0 in the second formula is the semi-edge at the set $\{0, k\}$.

If the selected a -dart on B_d is at position p , the retained x -fragments in the complement of the forced matching are

$$\{p-1, p, p+1\}, \quad \{p+2r, p+2r+1\} \quad (1 \leq r \leq h-1). \quad (*)$$

Here the middle vertex p of the three-vertex fragment has its a -dart in the matching. Call a selected semi-edge *blocking* if every fragment in $(*)$ contains at most one position from O_d . Then each fragment has a complementary semi-edge at one end, so no complementary component can pass through that orbit.

Each selection below covers every x -orbit with exactly one selected a -dart; in the $k = 3$ construction one ordinary edge covers two orbits at once. On every orbit, add the unique alternating matching of the even path left after its selected position. This produces a perfect matching $\bar{M}$ of Y , and all references below to the quotient complement mean $Y - \bar{M}$.

For $k \geq 4$, select a semi-edge at position p_d on every B_d , using the following mutually exclusive cases in their displayed order:

condition	p_d
$d = k < h$, or $d = k = h$ with k odd	0
$d = k = h$ with k even	1
$d = k - 1$	2
$(k, d) = (4, 2)$	5
$d = k - 2$ and $k \geq 5$	3
$d = 1$ and k even	$k + 1$
otherwise	1

Every displayed choice is blocking. Here is the short verification. At $p = 1$ the fragments are $\{0, 1, 2\}, \{3, 4\}, \{5, 6\}, \dots$; substitution of O_d shows that the only collisions not already covered by $d = k$ occur when $k - d = 1$ or 2 , or when $d = 1$ and the adjacent positions $k - 1, k$ form one of the displayed pairs. These are precisely the exceptional rows. At $p = 2, 3, k + 1$ the exceptional positions are separated respectively by the shifted pairings in $(*)$; $(k, d) = (4, 2)$ is separated at $p = 5$. Finally, on B_k the only ordinary positions are $k, -k$: $p = 0$ separates them unless $k = h$, and in that boundary case the parity choice $p = 0$ for odd k and $p = 1$ for even k separates the adjacent positions $h, h + 1$. Thus every complementary component is capped by semi-edges, and Lemma 4.8 applies.

For $k = 3$ and $N \geq 11$, select the ordinary a -edge

$$B_{1,3} = \{3, 4\} \longleftrightarrow \{0, 4\} = B_{4,0},$$

and select semi-edges at positions 4 on B_2 , 0 on B_3 , and 1 on every B_d with $d \geq 5$. Substitution in O_d and $(*)$ shows that every retained x -fragment contains at most one unselected ordinary a -dart, including the fragments on B_1 and B_4 . Hence the quotient complement is again a union of capped paths.

For $k = 2$ and $N \geq 9$, select semi-edges at positions

$$p_1 = 4, \quad p_2 = 0, \quad p_d = 3 \quad (3 \leq d \leq h).$$

Every orbit except B_1 is blocking. On B_1 , the retained pairs $\{-1, 0\}$ and $\{1, 2\}$ are joined by the ordinary a -edge between positions 0 and 1, forming a path between its remaining ordinary positions -1 and 2 . Those two a -edges lead to positions -1 and 0 on B_3 , which lie in distinct capped fragments. Thus no circuit is created.

It remains only to record the small choices. Listing the selected semi-edge positions in the order $B_1, B_2, \dots$, use

$$(4, 0, 1) \quad \text{for } (N, k) = (7, 2), \quad (1, 2, 0) \quad \text{for } (7, 3), \quad (3, 1) \quad \text{for } (5, 2).$$

Direct substitution in O_d and $(*)$ gives only capped paths in the two degree-seven cases. In the degree-five case it gives capped paths and one quotient circuit of length two, whose lifts are even. Lemma 4.8 completes all three cases. $\square$

Combining the one-cycle and two-cycle constructions settles the transposition case completely.

Theorem 4.13 (the transposition case for symmetric groups). *Let $N \geq 3$, let a be a transposition and let $x \in S_N$ have odd order. If $\langle a, x \rangle = S_N$, then $X(S_N; a, x)$ is 3-edge-colourable.*

Proof. For even N this is Corollary 4.11. Suppose N is odd. The transposition a meets at most two cycles of x , while transitivity of $\langle a, x \rangle$ forces it to connect all of them. There cannot be two, since every cycle of x has odd length and their sum would be even. Thus x is an N -cycle. After conjugating and, if necessary, replacing x by x^{-1} , write $x = (0 \ 1 \ \dots \ N - 1)$ and $a = (0 \ k)$ with $1 \leq k \leq (N - 1)/2$. The transposition graph of the conjugates $x^{-i} a x^i$ has edges $\{i, i + k\}$, so transitivity forces $\gcd(k, N) = 1$. Proposition 4.12 applies. $\square$

4.4 The colourings as functions $G \rightarrow \mathbb{Z}_3$

Let $X = X(G; a, x)$ with $s = \text{ord}(x)$ odd, and identify the three colours with the elements of $\mathbb{Z}_3$, so that the three colours at a vertex sum to 0. Given a 3-edge-colouring κ of X , let $t(g) \in \mathbb{Z}_3$ be the colour of the a -edge at g , i.e. the colour missing from the two x -edges at g . Then $t(ga) = t(g)$, and κ is determined by t : on the x -cycle through g , the colour of the edge $\{gx^i, gx^{i+1}\}$ is $-(t(gx^i) + \kappa(\{gx^{i-1}, gx^i\}))$, and the following lemma shows that t determines κ on the cycle without any initial choice. We say that a function $t: \mathbb{Z}_s \rightarrow \mathbb{Z}_3$ is *admissible* if it is the sequence of colours of the pendant edges in some proper 3-edge-colouring of the cycle C_s (with a pendant edge attached at each vertex), equivalently the sequence of colours missing at the vertices of C_s .

Lemma 4.14. *Let $s \geq 3$ be odd and $t: \mathbb{Z}_s \rightarrow \mathbb{Z}_3$. Call $i \in \mathbb{Z}_s$ a change if $t(i+1) \neq t(i)$, and put $\delta_i = t(i+1) - t(i) \in \{\pm 1\}$ for a change i . Then t is admissible if and only if t is not constant and, for any two cyclically consecutive changes i and j (so that t is constant on $i+1, \dots, j$), $\delta_i = \delta_j$ holds precisely when $j-i$ is odd. Each admissible t comes from exactly one 3-edge-colouring of C_s ; in particular there are $2^s - 2$ admissible functions.*

Proof. Let $f_i \in \mathbb{Z}_3$ be the colour of the edge $\{i, i+1\}$ of C_s in a proper 3-edge-colouring, so $t(i) = -(f_{i-1} + f_i)$, and put $\epsilon_i = f_i - f_{i-1} \in \{\pm 1\}$. Then $t(i+1) - t(i) = -(\epsilon_i + \epsilon_{i+1})$, so i is a change if and only if $\epsilon_i = \epsilon_{i+1}$, in which case $\delta_i = -2\epsilon_i = \epsilon_i$, while $\epsilon_{i+1} = -\epsilon_i$ at a non-change. Between consecutive changes $i < j$ the signs $\epsilon_{i+1}, \dots, \epsilon_j$ alternate, so $\delta_j = \epsilon_j = (-1)^{j-i-1}\epsilon_{i+1} = (-1)^{j-i-1}\delta_i$, which is the stated rule; and t is not constant since s is odd. Conversely, let t be non-constant and satisfy the rule. Define $\epsilon_i = \epsilon_{i+1} = \delta_i$ at every change i and let ϵ alternate between changes; the rule says exactly that this is consistent around the cycle. Put $f_i = f_0 + \epsilon_1 + \dots + \epsilon_i$; this is a proper colouring of C_s provided $\epsilon_1 + \dots + \epsilon_s \equiv 0 \pmod{3}$, and indeed $\sum_i \epsilon_i \equiv -2 \sum_i \epsilon_i = \sum_i (t(i+1) - t(i)) = 0$. By construction $-(f_{i-1} + f_i)$ and $t(i)$ have the same differences, so they differ by a constant c , and replacing f by $f+c$ (which changes $-(f_{i-1} + f_i)$ by $-2c = c$) makes them equal. Since ϵ and c are forced, the colouring is unique; the count follows because C_s has $2^s - 2$ proper 3-edge-colourings for odd s . $\square$

Proposition 4.15. *Let $X = X(G; a, x)$ with $s = \text{ord}(x)$ odd. The 3-edge-colourings of X (with colours in $\mathbb{Z}_3$) are in bijection with the functions $t: G \rightarrow \mathbb{Z}_3$ such that $t(ga) = t(g)$ for all g and, for every g , the sequence $i \mapsto t(gx^i)$ on $\mathbb{Z}_s$ is admissible.*

Proof. The map $\kappa \mapsto t$ described above is injective by Lemma 4.14, and conversely, given t , colouring each x -cycle by the unique colouring of Lemma 4.14 and each a -edge $\{g, ga\}$ with $t(g) = t(ga)$ gives a proper colouring: at g the two x -edges have the colours other than $t(g)$. $\square$

Thus a 3-edge-colouring of X is the same thing as a 3-colouring of the perfect matching M_a (of the $|G|/2$ a -edges) such that, along every x -cycle, the colours of the a -edges met form an admissible sequence: a constraint satisfaction problem with $|G|/2$ variables and $|G|/s$ constraints of arity s , each variable occurring in two constraints. The simplest admissible sequences are those with exactly three changes, all with the same δ : t takes the values $c, c+1, c+2$ on three consecutive arcs of odd lengths. Lemma 3.3 is the case in which t is pulled back from a valid function on the quotient cycle $\mathbb{Z}_m = G/N$. Two small cases deserve to be spelled out.

Corollary 4.16. *Let $X = X(G; a, x)$.*

- (a) *If $\text{ord}(x) = 3$, then X is 3-edge-colourable if and only if there is $t: G \rightarrow \mathbb{Z}_3$ with $t(ga) = t(g)$ which takes three different values on every triangle $\{g, gx, gx^2\}$, i.e. if and only if the 4-valent graph X/M_a obtained by contracting the a -edges has a proper 3-vertex-colouring. (In the notation of §3.4, X/M_a is the line graph of the arc-regular graph Y , so this is Lemma 3.5 again.)*

- (b) If $\text{ord}(x) = 5$, then X is 3-edge-colourable if and only if there is $t: G \rightarrow \mathbb{Z}_3$ with $t(ga) = t(g)$ such that on every x -cycle t is constant on three consecutive vertices and takes the two other values on the remaining two vertices.

Proof. For $s = 3$ a sequence is admissible if and only if its three values are distinct (there are $6 = 2^3 - 2$ of them). For $s = 5$ the $30 = 2^5 - 2$ admissible sequences are exactly the rotations of (c, c, c, c', c'') with $\{c, c', c''\} = \mathbb{Z}_3$: these are admissible by Lemma 4.14 (three changes at mutual distances 3, 1, 1, all with the same δ), and there are $5 \cdot 6 = 30$ of them. $\square$

There is a useful way to isolate the remaining difficulty in the index-two case. Suppose $T \trianglelefteq G$ has index two, and contract each cycle of F_x to a vertex while retaining the a -edges. The resulting graph Q is a bipartite s -regular multigraph: its two parts consist of the x -cycles in T and in Ta . The order of the vertices along each x -cycle gives a cyclic order to the s edges incident with the corresponding vertex of Q . Call a 2-factor of Q *consecutive* if its two edges at every vertex are consecutive in this cyclic order. In the terminology of Kratochvíl and Poljak [25], this is a compatible 2-factor for the transition system in which the allowed pairs at each vertex form the cycle C_s .

Proposition 4.17 (consecutive-2-factor criterion). *Let $X = X(G; a, x)$ with $s = \text{ord}(x)$ odd, and suppose that G has a normal subgroup of index two. If the cyclically ordered bipartite multigraph Q defined above has a consecutive 2-factor, then X is 3-edge-colourable.*

Proof. Every cycle of the 2-factor is even because Q is bipartite. Colour its edges alternately 1, 2, and colour every edge outside the 2-factor with 0. Transfer this colour to the corresponding a -edge of X . Around each x -cycle the colours of the incident a -edges are a rotation of

$$(\underbrace{0, \dots, 0}_{s-2}, 1, 2).$$

This sequence is admissible: its three changes have the same difference in $\mathbb{Z}_3$ and occur at mutual cyclic distances $s - 2, 1, 1$, all odd. Proposition 4.15 therefore supplies a 3-edge-colouring of X . $\square$

For $s = 3$, König's theorem decomposes the cubic bipartite multigraph Q into three perfect matchings; the union of any two is a 2-factor, and every pair among three cyclically ordered incident edges is consecutive. Thus Proposition 4.17 recovers Proposition 3.9. For $s \geq 5$, regularity and bipartiteness still guarantee a 2-factor, but not that it is consecutive. In the notation of (1), Q is the coset-incidence graph whose two parts are the left cosets of $\langle x \rangle$ and $\langle \alpha(x) \rangle$ in T , with one edge labelled by each $t \in T$; the open task is therefore to exploit this group-induced cyclic ordering to find a consecutive 2-factor.

Proposition 4.15 combines with Lemma 4.1: an H -invariant colouring is a function t on $H \backslash G$, constant on the orbits of a , admissible on the cycles of x . For the coset space of a large subgroup this is a very small constraint satisfaction problem, and it is the form in which the SAT solver of Section 8 sees the problem. We have not been able to turn either formulation into a proof of Conjecture 1.1 for any infinite family of simple groups; the difficulty is visible already in Example 4.3.

5 Solvable groups

Theorem 5.1 (Alspach, Liu, Zhang [4]). *Every connected cubic Cayley graph on a solvable group is 3-edge-colourable. Equivalently, every connected Cayley graph of valency at least 2 on a solvable group admits a nowhere-zero 4-flow.*

The proof is an induction along a normal series. With the lemmas of §§2–3 in hand it can be organised as follows. Let $X = X(G; a, x)$ with G solvable and $\text{ord}(x)$ odd, and let N be a minimal normal subgroup of G ; it is elementary abelian. If $a \notin N$ and $x \notin N$, Lemma 2.8 reduces to the smaller solvable group G/N . If $a \in N$, Lemma 3.3 applies (note $N \neq G$ as G is not abelian of prime order). Otherwise $x \in N$, N has index 2 and $G = N \rtimes \langle a \rangle$ with N an elementary abelian p -group (p odd) generated by x and x^a ; thus $N \cong \mathbb{Z}_p$ or $\mathbb{Z}_p^2$, and X is either the prism over C_p (if $x^a = x^{\pm 1}$), or the bi-Cayley graph $B(\mathbb{Z}_p^2; (1, 0), (0, 1))$; the prism is hamiltonian, and the second graph is 3-edge-colourable as well (see [4] for the details in the original formulation; we also confirmed this by computer for $p \leq 13$).

Theorem 5.1 covers, among others, all cubic Cayley graphs on abelian, nilpotent, supersolvable, dihedral and generalised dihedral groups, on p -groups, on groups of order $p^a q^b$ (Burnside) and on all groups of order less than 60.

5.1 Solvable vertex-transitive groups

Potočník extended Theorem 5.1 from regular to transitive actions of solvable groups.

Theorem 5.2 (Potočník [42]). *Let X be a connected cubic simple graph whose automorphism group contains a solvable subgroup acting transitively on the vertices. If X is not 3-edge-colourable, then X is the Petersen graph.*

The Petersen graph is a genuine exception: the subgroup of S_5 of order 20 (the affine group $\text{AGL}(1, 5)$) is solvable and acts transitively on its ten vertices. Theorem 5.2 contains Theorem 5.1, since a Cayley graph on a solvable group admits that group as a solvable regular, hence transitive, group of automorphisms, and the Petersen graph is not a Cayley graph.

The strategy of the proof is again inductive, using the theory of elementary abelian regular covers developed by Malnič, Marušič and Potočník [32, 33]. If $G \leq \text{Aut}(X)$ is solvable and vertex-transitive, a minimal normal subgroup N of G is elementary abelian; if N acts semiregularly on $V(X)$ then X is a regular N -cover of the quotient X/N , which admits the solvable vertex-transitive group G/N , and Lemma 2.7 applies provided X/N is a cubic graph. The remaining cases (non-semiregular N , quotients of smaller valency) are analysed directly, and the base of the induction involves the Petersen graph: as Malnič and Potočník put it, “the fact that all elementary abelian covers of the Petersen graph are 3-edge-colourable was used ... in order to prove, by induction, that every connected cubic graph admitting a vertex-transitive solvable group of automorphisms (with the sole exception of the Petersen graph itself) is 3-edge-colourable” [33, §1].

Theorem 5.2 also yields the conjecture for many non-Cayley vertex-transitive cubic graphs, and for every cubic vertex-transitive graph whose order n is such that all transitive permutation groups of degree n are solvable.

6 The structure of a minimal counterexample

Nedela and Škoviera proved that a smallest Cayley snark, if one exists, must come from a group that is “almost as simple as possible”. A group G is *almost simple* if $T \leq G \leq \text{Aut}(T)$ for some non-abelian simple group T .

Theorem 6.1 (Nedela–Škoviera [37]). *Let $X = X(G; a, x)$ be a Cayley snark of smallest possible order. Then either*

- (i) G is a non-abelian simple group; or
- (ii) G has exactly one non-trivial proper normal subgroup T ; moreover $|G : T| = 2$, $x \in T$, $a \notin T$, and T is either a non-abelian simple group (so that G is almost simple) or the direct

product $S \times S$ of two isomorphic non-abelian simple groups, the involution a interchanging the two factors (so that $G \cong S \wr \mathbb{Z}_2$).

In particular every Cayley snark is a regular cover of a Cayley snark on a group of type (i) or (ii).

The last sentence follows from the first by repeated application of Lemma 2.8 to a Cayley snark that is not minimal (the quotient by a normal subgroup avoiding a and x is again a Cayley snark, by Lemma 2.8, or the normal subgroup contains a , which is impossible by Lemma 3.3, or it has index 2). The reader should consult [37] for the original proof; the following short argument, based on the lemmas above and on Theorem 5.1, recovers the statement.

Proof of Theorem 6.1. Let $X = X(G; a, x)$ be a Cayley snark of smallest order; by Proposition 3.1, $s = \text{ord}(x)$ is odd and G is not abelian. Let $1 \neq N \trianglelefteq G$ be a proper normal subgroup.

- If $a \notin N$ and $x \notin N$, then by Lemma 2.8 X covers the smaller cubic Cayley graph $X(G/N; \bar{a}, \bar{x})$, which is 3-edge-colourable by minimality; hence so is X , a contradiction.
- If $a \in N$, then X is 3-edge-colourable by Lemma 3.3, a contradiction.
- Hence $x \in N$ and $a \notin N$. Then $G/N = \langle aN \rangle$ has order 2.

So every non-trivial proper normal subgroup of G has index 2 and contains x . If $N_1 \neq N_2$ were two of them, $N_1 \cap N_2$ would be a normal subgroup of index 4 containing $x \neq 1$, contradicting what we just proved. Hence G has at most one non-trivial proper normal subgroup.

If it has none, G is simple, and non-abelian because $|G|$ is even and divisible by $s \geq 3$.

Otherwise let T be the unique non-trivial proper normal subgroup. It is a minimal normal subgroup, hence characteristically simple: $T \cong S^k$ for a simple group S . If S is abelian then $G = T \rtimes \langle a \rangle$ is solvable, contradicting Theorem 5.1. So S is non-abelian simple, and the k direct factors of T are the minimal normal subgroups of T , which a permutes by conjugation. The product of the factors in one $\langle a \rangle$ -orbit is normalised by T and by a , hence normal in G , hence equal to T ; so a acts transitively on the k factors and $k \leq 2$. If $k = 1$, T is non-abelian simple and G is almost simple. If $k = 2$, a interchanges the two factors; after replacing the second factor's coordinates by their image under a suitable automorphism of S we may assume that a acts by $(u, v) \mapsto (v, u)$, i.e. $G \cong S \wr \mathbb{Z}_2$. $\square$

Remark 6.2. (1) In case (ii) the graph X is the bi-Cayley graph $B(T; x, \alpha(x))$ of (1), where α is the outer involutory automorphism of T induced by a (it is not inner: if a induced an inner automorphism ι_t , then at^{-1} would be a central involution of G , generating a second normal subgroup). Examples of admissible pairs (G, T) : (S_n, A_n) with a an odd permutation; $(\text{PGL}(2, q), \text{PSL}(2, q))$ for odd q ; $(\text{PSL}(2, q^2), \text{PSL}(2, q^2))$; (M_{10}, A_6) ; $(S \wr \mathbb{Z}_2, S \times S)$ with, for instance, $S = A_5$ and $x = ((123), (12345))$.

(2) Groups of type (i) or (ii) are precisely the groups $G = \langle a, x \rangle$ whose socle is a non-abelian minimal normal subgroup T with G/T of order at most 2; this explains the title of [37]. Note that the theorem does not say that every Cayley snark lives on such a group, only that a *smallest* one does; a Cayley snark on an arbitrary group covers one of these. When $\text{ord}(x) = 3$, Proposition 3.9 eliminates type (ii), so Corollary 3.10 reduces that case to non-abelian simple groups alone.

(3) The argument shows more generally that Conjecture 1.1 holds for every group G that has a non-trivial normal subgroup N with $a \in N$ or with $a, x \notin N$ and the conjecture known for G/N . For instance, it holds for every group of the form $G = N \rtimes \mathbb{Z}_m$ with m odd and N containing a , and for all groups G with a solvable non-trivial normal subgroup N such that G/N satisfies the conjecture (by induction, if $a, x \notin N$; by Lemma 3.3 if $a \in N$; and if $x \in N$ then $|G : N| = 2$ and G is solvable). In particular, the conjecture holds for all groups with a non-trivial solvable radical whose quotient by the radical satisfies it.

7 The hamiltonicity route and Cayley maps

As noted in the introduction, a Hamilton cycle in a cubic Cayley graph yields a 3-edge-colouring. The literature on Hamilton cycles in Cayley graphs is vast (see the surveys [11, 28, 40]); we mention only the results that bear directly on cubic Cayley graphs on non-solvable groups, since for solvable groups Theorem 5.1 is already available.

7.1 $(2, s, 3)$ -Cayley graphs

Glover and Marušič [17] initiated the study of cubic Cayley graphs $X(G; a, x)$ on groups with a $(2, s, 3)$ -presentation

$$G = \langle a, x \mid a^2 = x^s = (ax)^3 = 1, \dots \rangle,$$

i.e. generated by an involution a and an element x of order s whose product has order 3. Such a graph has a natural embedding as a *Cayley map* in an orientable surface whose faces are the s -gons bounded by the x -cycles and the hexagons bounded by the closed walks $(ax)^3$; a Hamilton cycle is sought as the boundary of a tree of faces (a “tree of faces” in the Cayley map), using classical results on cyclic stability of cubic graphs.

Theorem 7.1 (Glover–Marušič [17]; Glover–Kutnar–Marušič [15]; Glover–Kutnar–Malnič–Marušič [16]). *Let $X = X(G; a, x)$ where G has a $(2, s, 3)$ -presentation. Then X has a Hamilton path, and it has a Hamilton cycle whenever $|G| \equiv 2 \pmod{4}$, whenever $s \equiv 0 \pmod{4}$, and whenever s is odd. The only open case is $|G| \equiv 0 \pmod{4}$ with $s \equiv 2 \pmod{4}$.*

Corollary 7.2. *Every $(2, s, 3)$ -Cayley graph is 3-edge-colourable.*

Proof. If s is even, Proposition 3.1(b) applies; if s is odd, Theorem 7.1 provides a Hamilton cycle. $\square$

Note that a $(2, s, 3)$ -generated group is the same thing as a $(2, 3)$ -generated group (put $y = ax$), but the Cayley graph $X(G; a, x)$ of Corollary 7.2 is different from the graph $X(G; a, y)$ of §3.4: in the former the element of order 3 is the *product* of the two generators. The two families overlap exactly when $s = 3$.

7.2 Cayley maps and independent sets

Hujdurović, Kutnar and Marušič [19, 20] pursued the Cayley-map approach with Conjecture 1.1 rather than hamiltonicity as the target. In [19] they show that a connected bicirculant of prime valency admitting a suitable automorphism is *near-bipartite* (its vertex set has an independent subset whose removal leaves a bipartite graph), and combine this with the theory of Cayley maps to obtain partial results on the non-existence of Cayley snarks; [20] develops this further, “combining the theory of Cayley maps with the existence of certain kinds of independent sets of vertices in arc-transitive graphs” to obtain “new partial results suggesting promising future research directions” for the conjecture. The underlying idea is that, in $X(G; a, x)$ with s odd, the x -cycles are the vertices of an s -valent G -arc-transitive quotient graph (the coset graph of $\langle x \rangle$), and a suitable independent set in this quotient allows one to modify the odd 2-factor F_x into an even one.

7.3 Other hamiltonicity results

Kutnar, Marušič, Morris, Morris and Šparl [26] proved that every connected Cayley graph whose order is kp with $k < 32$, $k \neq 24$ (p prime), or kpq with $k \leq 5$, or pqr , or kp^2 with $k \leq 4$, or kp^3 with $k \leq 2$ (p, q, r distinct primes) is hamiltonian, and Morris and Wilk [27] extended the first case to all $k < 48$; in particular every connected Cayley graph of order less than 48 is

hamiltonian, and so are, for instance, the cubic Cayley graphs of order 60 on A_5 and of order 120 on S_5 and $\text{SL}(2, 5)$. All of these are, however, well inside the range of the census discussed next.

A different kind of result concerns a single, very natural family: for $\sigma = (1\ 2\ \cdots\ n)$ and $\tau = (1\ 2)$ the graph $X(S_n; \tau, \sigma)$ is the undirected “sigma-tau graph”. Compton and Williamson [6] showed that it is hamiltonian for every $n \geq 3$, and Sawada and Williams [45] solved the much harder directed version (Nijenhuis–Wilf’s sigma-tau problem): the Cayley *digraph* on $\{\sigma, \tau\}$ has a Hamilton cycle if and only if n is odd. For odd n this is a bi-Cayley graph of the form (1) on $T = A_n$ with α conjugation by a transposition, so Conjecture 1.1 holds for these graphs; the case of a general odd-order $x \in A_n$ is open.

8 Computational evidence

8.1 The census of cubic vertex-transitive graphs

Potočník, Spiga and Verret [44] constructed all connected cubic vertex-transitive graphs on at most 1280 vertices: there are 111 360 of them, of which 109 926 are Cayley graphs (97 687 of these are graphical regular representations, i.e. their full automorphism group acts regularly). They report:

“We tested all the graphs in the census for Hamilton cycles and found at least one such cycle for each graph apart from the four well-known exceptions (the Petersen graph, the Coxeter graph and their truncations).” [44, §2]

Corollary 8.1. *Every connected cubic Cayley graph on at most 1280 vertices is 3-edge-colourable. More generally, every connected cubic vertex-transitive graph on at most 1280 vertices other than the Petersen graph and its truncation is 3-edge-colourable.*

Proof. A hamiltonian cubic graph of even order is 3-edge-colourable. The Coxeter graph has chromatic index 3, hence so does its truncation by Lemma 3.5; the Petersen graph and its truncation are not 3-edge-colourable, by Lemma 3.5 again. None of the four exceptions is a Cayley graph. $\square$

8.2 Beyond the census

Combining Corollary 8.1 with Theorem 6.1 gives a concrete list of the smallest groups on which a Cayley snark could conceivably live: a smallest Cayley snark has more than 1280 vertices, and its group is a non-abelian simple group of order > 1280 , or an almost simple group $T.2$ with $|T| > 640$ and T simple, or $S \wr \mathbb{Z}_2$ with $|S| > \sqrt{320}$, i.e. $|S| \geq 60$. In increasing order the first candidates are $\text{PGL}(2, 11)$ (order 1320), $\text{PGL}(2, 13)$ (2184), $\text{PSL}(2, 17)$ (2448), A_7 (2520), $\text{PSL}(2, 19)$ (3420), $\text{PSL}(2, 16)$ (4080), $\text{PGL}(2, 17)$ (4896), S_7 (5040), $\text{PSL}(3, 3)$ (5616), $\text{PSU}(3, 3)$ (6048), $\text{PSL}(2, 23)$ (6072), $\text{PGL}(2, 19)$ (6840), $A_5 \wr \mathbb{Z}_2$ (7200), $\text{PSL}(2, 25)$ (7800), M_{11} (7920), $\text{PSL}(2, 16)$ (8160), ...

We wrote a program (`code/cayley_snark_check2.py` in the repository accompanying this survey; an earlier version, `cayley_snark_check.py`, is described below) which, for a given permutation group G , enumerates all pairs (a, x) with a an involution, x of odd order and $\langle a, x \rangle = G$, up to conjugation in G and inversion of x , and decides whether $X(G; a, x)$ is 3-edge-colourable. By Proposition 3.1 this decides Conjecture 1.1 for *all* cubic Cayley graphs on G . The decision procedure is Lemma 4.1: for a subgroup $H \leq G$ containing no conjugate of x , the quotient pregraph $H \backslash X$ has $|G : H|$ vertices, and a 3-edge-colouring of it—found by the SAT solver CaDiCaL [7] through the python-sat interface, with one Boolean variable “ e has colour 2” and one variable “ e has colour 1” per edge or semi-edge e and the obvious

clauses at each vertex—lifts to an H -invariant 3-edge-colouring of X . The lifted colouring is then verified independently on X : for every $g \in G$ the program checks that the three edges at g have three different colours and that the colour of $\{g, ga\}$ is the same when computed from g and from ga . Candidate subgroups H are generated once per group—point stabilisers, pointwise and setwise stabilisers of pairs and of small sets, stabilisers of partitions into blocks of equal size, centralisers and normalisers of elements of odd order and of involutions, and abelian subgroups of odd order—(conjugate subgroups give isomorphic quotients, so at most two candidates with the same order and the same multiset of element orders are kept), and for each pair (a, x) those not meeting the conjugacy class of x are tried in decreasing order of size, the full Cayley graph (with a time limit) serving as a last resort; the last resort was never needed. Generation is tested through the socle T of G ($T = G$ or $|G : T| = 2$): $\langle a, x \rangle = G$ if and only if $a \notin T$ and $\langle x, x^a \rangle = T$ (or $\langle a, x \rangle = T$), and a subgroup of T generated by two given elements is all of T as soon as its order exceeds $|T|/m$, where m is the minimal index of a proper subgroup of T (its minimal degree, taken from the Atlas). Groups are given by permutation generators (constructed from matrices over finite fields, or copied from the Atlas of Finite Group Representations [8]), and the program checks every group order against the known value; the Petersen graph is rejected as a control, and the SAT encoding for pregraphs was cross-checked against exhaustive search on a thousand random small pregraphs.

The effect of Lemma 4.1 is dramatic. The earlier program, which only used abelian subgroups H of odd order (no semi-edges), reduced $X(S_8; a, x)$ with x of order 15 to graphs on 4480 vertices on which the solver timed out, leaving 19 pairs on S_8 undecided; with $H = S_4 \wr S_2$ (the stabiliser of a partition into two blocks of four, which contains no element of cycle type $5 \cdot 3$) the quotient has 35 vertices and is coloured instantly. The largest quotient that had to be solved for any group in Table 2 has 9 240 vertices, whereas the groups have order up to 300 696. Table 2 lists, for each group, the number of generating pairs, the orders of x that occur, and the number $\max |\bar{V}|$ of vertices of the largest quotient pregraph that was needed; all 25 954 Cayley graphs were found to be 3-edge-colourable.

Theorem 8.2. *Conjecture 1.1 holds for every cubic Cayley graph on each of the 85 groups listed in Table 2; in particular for every non-abelian simple group of order at most 285 852, for $\text{PSL}(2, q)$ with $q \leq 83$, for $\text{PGL}(2, q)$ with $q \leq 67$, for A_n with $n \leq 9$ and S_n with $n \leq 8$, and for the groups M_{10} , $\text{PSL}(3, 3)$, $\text{PSU}(3, 3)$, $A_5 \wr \mathbb{Z}_2$, M_{11} , $\text{P}\Sigma\text{L}(2, 16)$, $\text{PSL}(3, 3).2$, $G_2(2) = \text{PSU}(3, 3).2$, $\text{P}\Sigma\text{L}(2, 25)$, $\text{PSL}(2, 25).2_3$, $\text{PSL}(3, 4)$, $\text{PSU}(4, 2) = \text{PSp}(4, 3)$, $Sz(8)$, $\text{PSL}(3, 4).2$ (field), $\text{PSL}(3, 4).2$ (graph), $\text{PSL}(3, 4).2$ (graph-field), $\text{PSU}(4, 2).2 = W(E_6)$, $\text{PSL}(2, 7) \wr \mathbb{Z}_2$, $\text{PSU}(3, 4)$, M_{12} , $\text{PSL}(2, 49).2_3$, $\text{P}\Sigma\text{L}(2, 49)$, $\text{PSU}(3, 4).2$, $\text{PSU}(3, 5)$, J_1 , $M_{12}.2$, $\text{PSU}(3, 5).2$, $A_6 \wr \mathbb{Z}_2$.*

Two of the groups in the table, M_{10} and $\text{PSL}(2, 25).2_3$, have all their involutions inside the socle and are therefore not generated by an involution and an element of odd order at all; the only cubic Cayley graphs on them are the trivially colourable ones of Proposition 3.1. The groups permitted by Theorem 6.1 for a smallest Cayley snark are the non-abelian simple groups, their extensions $T.2$ of index two and the wreath products $S \wr \mathbb{Z}_2$. Listing these by order (from the orders of the simple groups; note that $\text{PSL}(2, 8)$, $Sz(8)$ and $\text{PSL}(2, 32)$ have no extension of index two, while $\text{PSL}(2, 25)$, $\text{PSL}(2, 49)$ and $\text{PSL}(3, 4)$ have three each), one finds that every such group of order less than 352 440 appears in Table 2, the first ones missing being $\text{PSL}(2, 89)$ (order 352 440) and S_9 (order 362 880). Combining Theorem 6.1 with Corollary 8.1 and Table 2 therefore gives:

Corollary 8.3. *A Cayley snark of smallest order, if one exists, has more than 352 439 vertices. Its group is a non-abelian simple group, or a group $T.2$ or $S \wr \mathbb{Z}_2$ as in Theorem 6.1(ii), of order at least 352 440 and not listed in Table 2.*

We stress that these computations, like the census, are evidence rather than insight: the even 2-factors found by the solver do not seem to follow any pattern that would suggest a general

construction. One structural remark can be made, though. In every case a colouring invariant under a fairly large subgroup was found—the quotient never had more than 9 240 vertices—and for the groups $\text{PSL}(2, q)$ with x in a non-split torus the $(q + 1)$ -vertex quotient by a Borel subgroup sufficed for the great majority of pairs once $q \geq 25$ (Example 4.3); but for small q , and for q even, it does not, so “the” natural quotient is not the answer. Another instructive family is $\text{PGL}(2, q)$ with $\text{ord}(x) = 3$ and $\text{ord}(ax) = q + 1$: the quotients by the dihedral subgroups of order $2(q + 1)$ (admissible when $3 \nmid q + 1$) were never 3-edge-colourable—for $q = 61$ the solver proved all 30 such quotients, on 1830 and 3660 vertices, uncolourable—whereas the quotient by the unipotent subgroup of order q , or by a Borel subgroup when $3 \nmid q - 1$, always was.

9 Related problems on flows in Cayley graphs

The flow formulation of Conjecture 1.1 has stimulated work on the stronger 3-flow property. By Mader’s theorem (Theorem 2.4), Tutte’s 3-flow conjecture restricted to vertex-transitive graphs asserts that every vertex-transitive graph of valency at least 4 admits a nowhere-zero 3-flow. This is known for Cayley graphs on abelian groups (Potočník, Škoviera and Škrekovski [43]), for Cayley graphs of valency at least 4 on groups with a certain Sylow 2-subgroup structure (Nánásiová and Škoviera [34]), for graphs admitting a solvable arc-transitive group of automorphisms (Li and Zhou [29]), for Cayley graphs on supersolvable groups (Zhang and Zhou [53]), for nilpotently vertex-transitive graphs [52], for various dihedral and generalised dihedral and quaternion groups [51, 2], and for Cayley graphs on solvable groups of twice square-free order [3]. The general case for solvable groups, which would be the 3-flow analogue of Theorem 5.1, is open. These results do not bear on the cubic case (a cubic graph never has a nowhere-zero 3-flow), but their proofs use the same quotient-and-lift technique of §2.5.

9.1 The cycle-double-cover proof and an eight-point palette

In July 2026 OpenAI announced a proof of the cycle double cover conjecture [38]; expositions by Geelen [18] and Oum [39] give independent presentations of the argument. Its starting point for a loopless bridgeless cubic graph X is a nowhere-zero flow $f: E(X) \rightarrow \Gamma$, where $\Gamma = \mathbb{F}_2^3$. It constructs a two-element set $P_e \subset \Gamma$ for every edge such that

$$m_v(s) := |\{e \ni v : s \in P_e\}| \in \{0, 2\} \quad (v \in V(X), s \in \Gamma). \quad (2)$$

The even subgraphs $M_s = \{e : s \in P_e\}$ then form a cycle double cover. More explicitly, after ordering the three edges at v as a, b, c and writing $f(a) = x$, $f(b) = y$, $f(c) = x + y$, the proof chooses local offsets $g_{v,a} = 0$, $g_{v,b} = x$, $g_{v,c} = 0$. It proves by a duality argument that the affine system

$$t_u + t_v + \epsilon_e f(e) = g_{u,e} + g_{v,e} \quad (e = uv) \quad (3)$$

has a solution with $t_v \in \Gamma$ and $\epsilon_e \in \mathbb{F}_2$. The sets

$$P_e = \{t_v + g_{v,e}, t_v + g_{v,e} + f(e)\}$$

are consequently independent of the chosen endpoint and satisfy (2).

There is a precise way to measure how far this relaxed colouring is from a Tait colouring. For a family $P = (P_e)$ satisfying (2), define its *palette graph* $H(P)$ to be the simple graph with vertex set Γ and edge set $\{P_e : e \in E(X)\}$ (repetitions are suppressed).

Proposition 9.1 (palette extraction). *Let X be a loopless cubic graph and let $P_e \subset \mathbb{F}_2^3$ be two-element sets satisfying (2). If $H(P)$ is 4-vertex-colourable, then X is 3-edge-colourable. Conversely, if X is 3-edge-colourable, one can start the cycle-double-cover construction with a flow f for which every resulting palette graph is 4-colourable. Hence X is 3-edge-colourable if and only if some output of the construction has a 4-colourable palette graph.*

Proof. Identify the four vertex colours with the elements of $\mathbb{F}_2^2$, and let $c: \Gamma \rightarrow \mathbb{F}_2^2$ be a proper colouring of $H(P)$. For $P_e = \{p_e, q_e\}$ put

$$h(e) = c(p_e) + c(q_e).$$

This is nonzero because $c(p_e) \neq c(q_e)$. At every vertex v ,

$$\sum_{e \ni v} h(e) = \sum_{s \in \Gamma} m_v(s) c(s) = 0,$$

since every $m_v(s)$ is zero or two. Thus h is a nowhere-zero $\mathbb{F}_2^2$ -flow, whose three nonzero values give a 3-edge-colouring of the cubic graph.

Conversely, encode the three classes of a 3-edge-colouring by the three nonzero elements of a two-dimensional subspace $W < \Gamma$. This gives a nowhere-zero Γ -flow f with image in W . Choose a linear map $\pi: \Gamma \rightarrow \mathbb{F}_2^2$ whose restriction to W is injective. In every output of the construction the two ends of the palette edge P_e differ by $f(e) \in W \setminus \{0\}$, so π is a proper 4-colouring of $H(P)$. $\square$

One particularly concrete sufficient target follows. If the complement of $H(P)$ has a perfect matching, give the two vertices in each matched pair the same one of four colours; then Proposition 9.1 gives a Tait colouring of X . Thus a possible route to Conjecture 1.1 is to use the Cayley structure to choose the initial $\mathbb{F}_2^3$ -flow and the affine solution so that the palette is 4-colourable—or, more strongly, so that its complement has a perfect matching.

The converse half of Proposition 9.1 also exposes the danger of circularity: a rank-two starting flow is already a Tait colouring in disguise. The construction itself guarantees the parity condition (2), not the global chromatic bound on $H(P)$. Nor can one simply choose the starting flow to inherit the full Cayley symmetry.

Lemma 9.2. *Let $X = X(G; a, x)$ with $\text{ord}(x) > 2$. No nowhere-zero $\mathbb{F}_2^k$ -flow on X is invariant under the left regular action of G , for any $k \geq 1$.*

Proof. An invariant assignment is constant, say α , on the orbit of a -edges and constant, say ξ , on the orbit of x -edges. At each vertex there is one edge of the first kind and two of the second. In characteristic two the flow equation is therefore $\alpha + \xi + \xi = \alpha = 0$, contrary to nowhere-zerosness. $\square$

Full rank by itself does not remove the circularity. In fact the exact reformulation in Proposition 9.1 remains exact after full rank is imposed.

Proposition 9.3 (full-rank tetrahedral palette). *Let X be a connected loopless cubic graph on $n \geq 6$ vertices. Then X is 3-edge-colourable if and only if the cycle-double-cover construction has an output whose starting $\mathbb{F}_2^3$ -flow has rank three and whose palette pairs are supported on exactly four affinely independent elements of $\mathbb{F}_2^3$.*

Proof. The reverse implication is Proposition 9.1. For the forward implication, let $h: E(X) \rightarrow \mathbb{F}_2^2 \setminus \{0\}$ be the flow encoded by a 3-edge-colouring and put $m = |E(X)| = 3n/2$. Consider the homogeneous linear system

$$r_u + r_v + \delta_e h(e) = 0 \quad (e = uv),$$

with $r_v \in \mathbb{F}_2^2$ and $\delta_e \in \mathbb{F}_2$. It has $2n + m$ scalar variables and at most $2m$ independent equations, so its solution space has dimension at least $2n - m = n/2 \geq 3$. The constant solutions $r_v = r$, $\delta_e = 0$ form only a two-dimensional subspace; choose a nonconstant solution.

Index four affinely independent points $q_z \in \mathbb{F}_2^3$ by $z \in \mathbb{F}_2^2$. For $e = uv$, the four indices split into two pairs $\{z, z + h(e)\}$. Let P_e be the pair of corresponding q 's which does not contain

q_{r_u} . The equation says $r_v \in \{r_u, r_u + h(e)\}$, so the same pair avoids q_{r_v} and P_e is well-defined. At v the three incident values of h are the three nonzero elements of $\mathbb{F}_2^3$; hence the three P_e are precisely the edges of the triangle on the three points other than q_{r_v} . These local triangles have exactly the form used in the cycle-double-cover construction, and their agreement across edges gives a solution of its compatibility system. Put $f(e) = p_e + q_e$ when $P_e = \{p_e, q_e\}$. The triangle condition makes f a nowhere-zero $\mathbb{F}_2^3$ -flow. Since r is nonconstant, at least two different local triangles occur; their union is connected and uses all four affinely independent points, so their edge differences span $\mathbb{F}_2^3$. Thus f has rank three. $\square$

For a fixed starting flow, the freedom in the paper's affine system has a simple intrinsic description. Define the *weighted cut code*

$$\mathcal{K}_f = \left\{ \delta \in \mathbb{F}_2^{E(X)} : \sum_{e \in C} \delta_e f(e) = 0 \text{ for every cycle } C \text{ of } X \right\}. \quad (4)$$

Equivalently, for each of the three coordinates f_j of f , the edge set with indicator $\delta_e f_j(e)$ must be a cut. This follows from the orthogonality of the binary cycle and cut spaces, and explains the terminology.

Proposition 9.4 (compatibility-space rigidity). *Fix a nowhere-zero $\mathbb{F}_2^3$ -flow f on a connected loopless cubic graph and fix the local data $g_{v,e}$ in the cycle-double-cover construction. The nullity of the homogeneous compatibility system associated with (3) is*

$$3 + \dim \mathcal{K}_f.$$

Modulo a common translation of every palette point by an element of $\mathbb{F}_2^3$, its solution space is therefore parameterised by $\mathcal{K}_f$. In particular, if $\mathcal{K}_f = 0$, all outputs for these fixed local data have isomorphic palette graphs.

Proof. Subtract two solutions of (3). Writing their differences as $s_v \in \mathbb{F}_2^3$ and $\delta_e \in \mathbb{F}_2$ gives

$$s_u + s_v = \delta_e f(e) \quad (e = uv). \quad (5)$$

The right-hand side is the coboundary of a vertex labelling s if and only if its sum around every cycle is zero, which is exactly the condition $\delta \in \mathcal{K}_f$. Indeed, necessity follows by summing (5) around a cycle; conversely, fix s at one vertex and define it by path sums, which are path-independent by the cycle condition. For each $\delta \in \mathcal{K}_f$ the resulting s is unique up to one common element of $\mathbb{F}_2^3$, accounting for the other three dimensions. Such a common translation translates every palette pair and hence preserves the palette graph up to isomorphism. $\square$

The choices of local edge orders do not create additional outputs. At v , let W_v be the plane spanned by its three nonzero flow values. Every local triangle has three vertices in W_v , so is W_v with one point omitted; translations of W_v act transitively on these four triangles and preserve their edge differences. Thus for two choices of offsets there are $d_v \in W_v$ and incidence bits $\eta_{v,e}$ with

$$g'_{v,e} = g_{v,e} + d_v + \eta_{v,e} f(e).$$

The change of variables $t'_v = t_v + d_v$, $\epsilon'_e = \epsilon_e + \eta_{u,e} + \eta_{v,e}$ takes one compatibility system to the other and preserves every pair P_e . In particular, the set of attainable palettes for a fixed flow, and the feasibility of any fixed matching target, are independent of the local orders.

Thus a fixed flow with $\mathcal{K}_f = 0$ offers no effective affine choice beyond global translation: to change its palette up to isomorphism, one must change the flow. Any extra matching equations can only be met through the evaluation of the weighted cut code together with global translations. A large code is not sufficient, but the proposition isolates exactly where the cycle-double-cover construction can supply non-circular flexibility.

There is nevertheless a more structured way in which the Cayley generators can enter the construction. Let Q be the multigraph obtained from $X = X(G; a, x)$ by contracting every x -cycle to a vertex; its edges are the a -edges of X . Thus Q is a connected, vertex-transitive, s -regular loopless multigraph, where $s = \text{ord}(x)$.

Proposition 9.5 (generator-separated 8-flow). *Suppose that s is odd and that Q has a nowhere-zero $\mathbb{F}_2^2$ -flow. Then X has a rank-three nowhere-zero flow $f: E(X) \rightarrow \mathbb{F}_2 \oplus \mathbb{F}_2^2$ such that every a -edge has value $(0, p)$ with $p \neq 0$, while every x -edge has value $(1, y)$.*

Proof. Let $\phi: E(Q) \rightarrow \mathbb{F}_2^2 \setminus \{0\}$ be such a flow. Around an x -cycle write the values of its incident a -edges, in cyclic order, as $p_0, \dots, p_{s-1}$. The flow equation in Q says $\sum_i p_i = 0$, so there are $y_i \in \mathbb{F}_2^2$ satisfying

$$y_{i-1} + y_i = p_i \quad (i \bmod s) :$$

choose one y_{s-1} and continue recursively; the last equation is exactly $\sum_i p_i = 0$. Assign $(0, p_i)$ to the corresponding a -edge and $(1, y_i)$ to the i th x -edge. The displayed equation is precisely the flow equation at the corresponding vertex of X , and every assigned value is nonzero.

It remains to check rank. If the three nonzero elements of $\mathbb{F}_2^2$ occur n_1, n_2, n_3 times at a vertex of Q , the flow equation forces n_1, n_2, n_3 to have the same parity. Their sum is the odd number s , so all three are odd and in particular positive. The a -edge values therefore span $\{0\} \oplus \mathbb{F}_2^2$, and any x -edge value supplies the missing first coordinate. $\square$

The hypothesis of Proposition 9.5 is automatic when $s \geq 5$ and Q is simple: Mader's theorem makes Q s -edge-connected, and Jaeger's theorem and Tutte's group-flow theorem give the required $\mathbb{F}_2^2$ -flow. Writing $H = \langle x \rangle$, the neighbours of the vertex H are the cosets hHa with $h \in H$, so two coincide exactly when their quotient lies in $H \cap aHa$. Thus Q is simple exactly when $H \cap aHa = 1$. In particular this holds when s is prime and G is non-solvable, since otherwise $aHa = H$, so a normalises H and $G = \langle a, x \rangle$ is solvable.

The weighted cut code of a generator-separated flow has a useful decomposition. Define $\mathcal{K}_\phi$ for the quotient flow by the analogue of (4) on Q . Write $f(e) = (0, p_e)$ on an a -edge and $f(e) = (1, y_e)$ on an x -edge. Let $\mathcal{R}_f$ be the vector space, modulo constant functions, of binary functions $r: V(X) \rightarrow \mathbb{F}_2$ which are constant across every a -edge and for which there is a function $w: V(X) \rightarrow \mathbb{F}_2^2$ satisfying

$$w_u + w_v \in \{0, p_e\} \quad (e = uv \text{ an } a\text{-edge}), \quad (6)$$

$$w_u + w_v = (r_u + r_v)y_e \quad (e = uv \text{ an } x\text{-edge}). \quad (7)$$

We call its elements *extendable layer potentials*.

Proposition 9.6 (layer-potential exact sequence). *For the generator-separated flow f constructed from ϕ , there is a short exact sequence*

$$0 \longrightarrow \mathcal{K}_\phi \longrightarrow \mathcal{K}_f \longrightarrow \mathcal{R}_f \longrightarrow 0.$$

Consequently $\dim \mathcal{K}_f = \dim \mathcal{K}_\phi + \dim \mathcal{R}_f$.

Proof. For $\delta \in \mathcal{K}_f$, Proposition 9.4 gives a vertex potential $s_v = (r_v, w_v)$, unique up to a constant, with $s_u + s_v = \delta_e f(e)$. On an a -edge, comparison of first coordinates gives $r_u = r_v$, while the lower coordinates give (6). On an x -edge the first coordinate gives $\delta_e = r_u + r_v$, and the lower coordinates then give (7). Thus projection onto r defines a map to $\mathcal{R}_f$. Conversely, given (r, w) satisfying (6) and (7), set $\delta_e = r_u + r_v$ on an x -edge and set δ_e on an a -edge according as $w_u + w_v$ is 0 or p_e . Then $s_u + s_v = \delta_e f(e)$ on every edge, so $\delta \in \mathcal{K}_f$. The map is therefore onto.

Its kernel consists of the classes for which r is constant. After subtracting a global constant, take $r = 0$. The first-coordinate equation on every x -edge then gives $\delta_e = 0$, and (7) says that w

is constant around each x -cycle. It thus descends to a potential on Q ; on an a -edge its difference is 0 or p_e , exactly the condition defining $\mathcal{K}_\phi$. Conversely, every element of $\mathcal{K}_\phi$ extends in this way by taking $r = 0$ and w constant on each contracted cycle. $\square$

There is a necessary condition on the first-coordinate projection which no longer mentions w . Regard an a -invariant function r as an edge indicator on Q , assigning to an a -edge its common value at the two endpoints. If $r \in \mathcal{R}_f$, then at every vertex C of Q ,

$$\sum_{e \ni C} r_e \phi(e) = 0. \quad (8)$$

Indeed, sum (7) around the corresponding x -cycle. If its successive a -edge values are $p_i = y_{i-1} + y_i$, the right-hand side becomes

$$\sum_i (r_i + r_{i+1}) y_i = \sum_i r_i (y_{i-1} + y_i) = \sum_i r_i p_i.$$

Thus the selected labelled edges must themselves form an $\mathbb{F}_2^2$ -flow, although this condition alone need not make r extendable for a fixed choice of lifts.

There is an exact obstruction to making it extendable by changing the lifts. Let $[p, q]$ denote the nondegenerate alternating form on $A = \mathbb{F}_2^2$; thus $[p, q] = 0$ for nonzero p, q exactly when $p = q$. Fix an edge indicator $r \in \mathbb{F}_2^{E(Q)}$ satisfying (8), and define

$$\mathcal{D}_r = \left\{ \lambda \in \mathbb{F}_2^{E(Q)} : \sum_{e \ni C} \lambda_e p_e = 0 = \sum_{e \ni C} \lambda_e r_e p_e \text{ for every } C \in V(Q) \right\}.$$

Thus both λ and its coordinatewise product λr are subflows of ϕ . At C , choose a starting point in the inherited cyclic order and write the incident data as $p_0, \dots, p_{s-1}$, $r_0, \dots, r_{s-1}$, and $\lambda_0, \dots, \lambda_{s-1}$. Put

$$\omega_C(r, \lambda) = \sum_{0 \leq j < i < s} \lambda_i (r_j + r_i) [p_i, p_j] \in \mathbb{F}_2. \quad (9)$$

For $\lambda \in \mathcal{D}_r$ this value is independent of the chosen starting point, as also follows from the proof below.

Proposition 9.7 (lift-flexible obstruction). *There is a choice of the lifts y_i in Proposition 9.5 for which r is an extendable layer potential if and only if*

$$\sum_{C \in V(Q)} \omega_C(r, \lambda) = 0 \quad \text{for every } \lambda \in \mathcal{D}_r. \quad (10)$$

The constant indicator $\mathbf{1}$ belongs to $\mathcal{D}_r$ and $\omega_C(r, \mathbf{1}) = 0$. Hence, if r is nonconstant and $\mathcal{D}_r = \langle \mathbf{1}, r \rangle$, the criterion reduces to the single parity

$$q(r) := \sum_C \omega_C(r, r) = 0.$$

Proof. At a quotient vertex C , choose y_i with $y_{i-1} + y_i = p_i$. Since (8) holds, one may also choose z_i cyclically so that

$$z_i + z_{i+1} = r_i p_i.$$

Then $h_i = z_i + r_i y_{i-1}$ satisfies

$$h_i + h_{i+1} = (r_i + r_{i+1}) y_i,$$

so it is a particular solution of the x -edge equations (7). Translating all the y_i by $c_C \in A$ and choosing the arbitrary translation of h gives every possible solution in the form

$$w_i = h_i + d_C + r_i c_C, \quad c_C, d_C \in A.$$

Let a quotient edge e join position i at C to position j at D . Since the kernel of the functional $[p_e, \cdot]$ is $\langle p_e \rangle$, the a -edge condition (6) is the single affine equation

$$[p_e, d_C + r_e c_C + d_D + r_e c_D] = [p_e, h_{C,i} + h_{D,j}].$$

A set of these equations, indicated by $\lambda \in \mathbb{F}_2^{E(Q)}$, has zero left-hand side exactly when, at every C , the coefficients of both d_C and c_C vanish. Nondegeneracy of the alternating form says that these two conditions are precisely

$$\sum_{e \ni C} \lambda_e p_e = 0, \quad \sum_{e \ni C} \lambda_e r_e p_e = 0;$$

that is, $\lambda \in \mathcal{D}_r$. Linear-system duality therefore says that the equations are soluble exactly when every such dependency annihilates their right-hand sides.

To identify that right-hand side, take $y_{i-1} = \sum_{j < i} p_j$ and $z_i = \sum_{j < i} r_j p_j$. Then

$$h_i = \sum_{j < i} (r_j + r_i) p_j, \quad [p_i, h_i] = \sum_{j < i} (r_j + r_i) [p_i, p_j].$$

Summing over the two ends of every edge gives exactly (10). This derivation also proves independence from all auxiliary choices. Finally,

$$\sum_i [p_i, h_i] = \sum_i [y_{i-1} + y_i, h_i] = \sum_i [y_i, h_i + h_{i+1}] = 0,$$

so $\omega_C(r, \mathbf{1}) = 0$. The last assertion follows because $0, \mathbf{1}, r, \mathbf{1} + r$ are then all the dependencies and ω_C is linear in its second argument. $\square$

In fact the preceding equations admit a smaller matrix formulation. Let $\mathcal{S}_\phi$ be the space of all binary indicators satisfying (8). Choose a starting point at each quotient vertex, take the reference lifts $y_{i-1} = \sum_{j < i} p_j$, and describe any other lifts by translations $c = (c_C)_{C \in V(Q)}$. On $\mathcal{S}_\phi$ define

$$B_c(r, t) = \sum_C \omega_C(r, t) + \sum_C \left[\sum_{e \ni C} r_e t_e p_e, c_C \right]. \quad (11)$$

Proposition 9.8 (quotient bilinear form). *The form B_c is symmetric, and $\mathbf{1}$ belongs to its radical. For the flow f_c with these lifts,*

$$\mathcal{R}_{f_c} \cong \text{rad}(B_c) / \langle \mathbf{1} \rangle, \quad \dim \mathcal{K}_{f_c} = \dim \mathcal{K}_\phi + \dim \mathcal{S}_\phi - \text{rank } B_c - 1.$$

The diagonal $B_c(r, r) = q(r)$ is independent of c and is linear on $\mathcal{S}_\phi$. Consequently every extendable layer potential lies in $\ker q$.

Proof. For $r, t \in \mathcal{S}_\phi$, expansion of (9) gives

$$\omega_C(r, t) + \omega_C(t, r) = \left[\sum_i r_i p_i, \sum_i t_i p_i \right] = 0.$$

The correction term in (11) is symmetric as well. For fixed c , the proof of Proposition 9.7 leaves only the d_C as unknowns. Its dependencies are therefore all $t \in \mathcal{S}_\phi$, and their right-hand sides are $B_c(r, t)$. Hence r extends exactly when $B_c(r, t) = 0$ for all t , which is the radical condition. The constant indicator extends with $w = 0$; quotienting by it and using Proposition 9.6 proves the dimension formula. Finally, the correction term on the diagonal vanishes because $r_e^2 = r_e$ and $r \in \mathcal{S}_\phi$. Symmetry in characteristic two implies $q(r + t) = q(r) + q(t)$. $\square$

Thus several prescribed indicators extend for the same lifts precisely when their span lies in the radical of one member of the affine family B_c . For a prescribed span this is a linear system in the $2|V(Q)|$ coordinates of c ; finding a large span amounts to making B_c have small rank. The rank measures available directions, while the transvection syndrome still tests whether those directions solve the colouring equations.

The same matrix describes the affine, rather than homogeneous, first-coordinate assignments of the actual palette. Let r_e now denote the layer containing the palette pair of an a -edge e ; this is well-defined at both endpoints. Use the local order consisting of the a -edge, the incoming x -edge, and the outgoing x -edge at every vertex of an oriented x -cycle. The associated offsets are $0, (0, p_i), 0$.

Proposition 9.9 (affine layer equation). *For a fixed generator-separated flow f_c , the attainable a -edge layer assignments are exactly*

$$\mathcal{A}_c = \{r \in \mathcal{S}_\phi : B_c(r, t) = B_c(t, t) \text{ for all } t \in \mathcal{S}_\phi\}. \quad (12)$$

This is a nonempty affine translate of $\text{rad } B_c$. In particular, changing the lifts while preserving B_c also preserves the entire set of attainable layer assignments, not just its homogeneous directions.

Proof. Write the vertex translation at position i on an x -cycle as (r_i, w_i) . The first coordinates of the compatibility system make r constant across a -edges. On the outgoing x -edge they give $\epsilon_i = r_i + r_{i+1}$, so the lower-coordinate equation is

$$w_i + w_{i+1} = (r_i + r_{i+1})y_i + p_{i+1}.$$

Summing around the cycle shows $r \in \mathcal{S}_\phi$. If h_i is the particular homogeneous potential used in Proposition 9.7, every solution on this cycle is $w_i = y_i + h_i + d_C$. Thus the remaining a -edge equations have dependencies $t \in \mathcal{S}_\phi$ and are consistent exactly when

$$B_c(r, t) = \kappa(t), \quad \kappa(t) = \sum_C \sum_i t_i [p_i, y_i].$$

The functional κ is independent of the lift translations. Taking the reference partial sums for y gives

$$q(t) + \kappa(t) = \sum_C \sum_{j < i} t_i t_j [p_i, p_j] = \sum_C \sum_i t_i = 0.$$

For the middle equality, the multiplicities of the three nonzero values among the selected edges at C have the same parity, since their sum is zero. Both the sum over pairs of different values and the number of selected edges therefore have that parity. The last equality counts each selected quotient edge twice. Consequently $\kappa(t) = q(t) = B_c(t, t)$, proving (12) and its sufficiency. The diagonal of a symmetric form vanishes on its radical, so belongs to the image of the form as a linear map to the dual space; hence the displayed affine system is always consistent. $\square$

The same argument gives a simultaneous form which is useful when one wants a high-dimensional space $\mathcal{R}_f$. Let $r^1, \dots, r^k$ satisfy (8). Using one fixed starting point at each quotient vertex, one choice of the lifts makes all of them extendable if and only if

$$\sum_{\ell=1}^k \sum_C \omega_C(r^\ell, \lambda^\ell) = 0$$

for every tuple $(\lambda^1, \dots, \lambda^k)$ such that each λ^ℓ is a subflow of ϕ and $\sum_\ell \lambda^\ell r^\ell$ is also a subflow. Under these conditions the displayed total is independent of the starting points. Indeed, in the proof above the lift translations c_C are shared by the k systems while each layer potential has

its own translations d_C^ℓ ; vanishing of their coefficients gives exactly the stated conditions. Once a basis extends for the same lifts, so does its span.

There is also a useful interpretation of the smallest dependency space. Call a nonzero member of $\mathcal{S}_\phi$ a *circuit* if it contains no proper nonempty member of $\mathcal{S}_\phi$. For nonconstant $r \in \mathcal{S}_\phi$,

$$\mathcal{D}_r = \langle \mathbf{1}, r \rangle$$

exactly when both r and its complement $\mathbf{1} + r$ are circuits. One direction follows because any proper subflow contained in either part belongs to $\mathcal{D}_r$. Conversely, if $\lambda \in \mathcal{D}_r$, the four intersections determined by r and λ all lie in $\mathcal{S}_\phi$; circuit minimality forces λ to be one of $0, \mathbf{1}, r, \mathbf{1} + r$. Thus a complementary pair of circuits with $q(r) = 0$ is another concrete sufficient target for a nonzero layer potential.

When $s = 5$, the local obstruction has a particularly small description. The three nonzero values of A occur with multiplicities 3, 1, 1 at every quotient vertex. Call the value occurring three times the *repeated value*.

Proposition 9.10 (order-five interlacement). *Suppose $s = 5$, and let r satisfy (8). At a quotient vertex where r is nonconstant, either its support or its complement consists of two positions carrying the repeated value. The local term $\omega_C(r, \lambda)$ vanishes for every locally admissible λ unless both*

- (i) *the three positions carrying the repeated value are not cyclically consecutive, and*
- (ii) *the distinguished pair given by r is not consecutive.*

Consequently, if this exceptional pattern occurs at no vertex of Q , lifts can be chosen for which r is an extendable layer potential.

Proof. The first assertion follows by listing the subsets of a multiset with values a, a, a, b, c , where $c = a + b$, whose sum is zero: apart from the empty and full subsets, they are the three pairs of a -positions and their complements. Up to a cyclic shift, reversal, and an automorphism of A , there are two cyclic orders,

$$(a, a, a, b, c) \quad \text{and} \quad (a, a, b, a, c).$$

Both the local admissibility conditions and (9) are unchanged when r is replaced by its complement, so it is enough to consider the three two-element supports. Substitution in (9), subject to

$$\sum_i \lambda_i p_i = 0 = \sum_i \lambda_i r_i p_i,$$

shows that every term vanishes in the first order. In the second order it also vanishes when the distinguished pair is the consecutive pair in positions 0, 1; for either of the other two pairs, taking λ to be its indicator gives value one. This is exactly (i)–(ii). If the exceptional pattern is absent everywhere, every term in (10) vanishes, and Proposition 9.7 applies. $\square$

The finite local calculation in this proof has 480 cases before symmetry reduction and is checked exhaustively by `code/cdc_palette_experiment.py`.

For $s = 5$ the matrix family itself also simplifies. At C , write a_C for the repeated value. There are fixed alternating forms T_C on $\mathcal{S}_\phi$, each of rank at most two, such that

$$B_c = B_0 + \sum_C [a_C, c_C] T_C.$$

Indeed, the space of local subflows has dimension three, with basis the constant indicator and two different pairs of a_C -positions. The local correction form vanishes on the constant indicator

and has value $[a_C, c_C]$ on those two pairs, whose intersection is a single a_C -position. It therefore descends to an alternating form on a two-dimensional space, proving the assertion. In particular only one bit of each c_C affects the layer-potential space; adding a_C to c_C preserves B_c . The remaining bit still changes the starting flow and may affect the transvection equations.

For arbitrary odd s , there is also a direct construction of layer potentials. Recall that the cyclic order of the a -edges incident with a vertex of Q is inherited from the corresponding x -cycle.

Proposition 9.11 (consecutive monochromatic cycle). *Suppose the quotient flow ϕ has a cycle R whose edges all receive the same value, and at every vertex of R its two incident edges are consecutive in the inherited cyclic order. The lifts in Proposition 9.5 can be chosen so that $\mathcal{R}_f$ contains a nonzero element supported on the endpoints of the a -edges in R . More generally, k vertex-disjoint cycles of this kind give a choice of lifts for which $\dim \mathcal{R}_f \geq k$.*

Proof. Let $r_v = 1$ when the a -edge at v belongs to R , and put $r_v = 0$ otherwise. At a quotient vertex on R , let the two selected consecutive positions be $i, i+1$ and let their common flow value be p . The lift equations give

$$y_{i-1} + y_i = p = y_i + y_{i+1},$$

so the two x -edges at the boundary of the selected pair have equal lower values: $y_{i-1} = y_{i+1}$. Translating all the y -values around this x -cycle by their common value makes both boundary values zero without changing ϕ . Make this choice independently at every vertex of R , and take $w = 0$ everywhere. Equation (6) is then automatic. On an x -edge where r changes, its lower value was made zero, while on every other x -edge r is constant; hence (7) also holds. The resulting r is nonconstant because R uses only two of the at least three incident edges at its vertices.

For vertex-disjoint cycles the lift translations occur on disjoint x -cycles and can be made simultaneously. Their edge indicators, and hence their layer potentials, are linearly independent. $\square$

For $s = 5$, Proposition 9.10 strengthens the one-cycle part of Proposition 9.11: it is enough that, at every vertex of the monochromatic cycle, either its two edges are consecutive or all three incident edges carrying their common value are consecutive. The latter alternative permits the cycle to use the two ends of that three-edge block.

The generator-separated flow separates the palette into two four-point layers. In any output P , an a -edge is a pair within one layer and an x -edge is a pair crossing the layers. Let $B_x(P)$ be the bipartite graph on the two layers formed by the palette pairs belonging to x -edges. If the bipartite complement of $B_x(P)$ has a perfect matching, colour the two endpoints of each matched pair alike. All within-layer pairs and all unmatched crossing pairs are then properly coloured, so Proposition 9.1 gives a Tait colouring. The paper's method has therefore reduced this structured case to a four-by-four Hall problem. What it does not yet supply is a choice of ϕ , of the lifts y_i , and of the compatibility-system solution which forces Hall's condition.

The twenty-four possible matchings admit a useful exact classification. Put $A = \mathbb{F}_2^2$, so that the layers are $\{0\} \times A$ and $\{1\} \times A$.

Proposition 9.12 (linearised matching target). *Every perfect matching between the two layers has the form*

$$\mathcal{M}_{L,c} = \{ \{ (0, z), (1, Lz + c) \} : z \in A \}$$

for unique $L \in \text{GL}(2, 2)$ and $c \in A$. Let an x -edge have flow value $(1, y_e)$ and write its palette pair as $\{ (0, z_e), (1, z_e + y_e) \}$. This pair belongs to $\mathcal{M}_{L,c}$ exactly when

$$(I + L)z_e = y_e + c. \tag{13}$$

For each of the twelve choices with L of order two, requiring that no x -edge pair belong to $\mathcal{M}_{L,c}$ amounts to adding ordinary affine linear equations to the compatibility system (3).

Proof. The affine group on A has order $4 \cdot 6 = 24$ and induces all 24 permutations of its four points, proving the first statement. The equality of the upper endpoints of the two displayed pairs gives (13). If L has order two, $N = I + L$ has rank one and $K := \ker N = \text{im } N$. Let $\rho: A \rightarrow \mathbb{F}_2$ be the nonzero functional with kernel K , and let $\lambda: K \rightarrow \mathbb{F}_2$ be its nonzero functional. Then $\lambda(Nz) = \rho(z)$. When $y_e + c \notin K$, collision is impossible. Otherwise (13) is equivalent to

$$\rho(z_e) = \lambda(y_e + c),$$

so avoiding collision is the single affine equation $\rho(z_e) = 1 + \lambda(y_e + c)$.

It remains only to see that this really is linear in the paper's variables. At an endpoint v , write $t_v + g_{v,e} = (b, w) \in \mathbb{F}_2 \oplus A$. Since the other endpoint of the pair is obtained by adding $(1, y_e)$, its layer-zero endpoint has lower coordinate

$$z_e = w + by_e,$$

an affine linear function of t_v . Thus each required non-collision equation can be appended directly to (3). $\square$

Proposition 9.4 turns consistency of this augmented system into a smaller, exact target. Let U_f be the homogeneous solution space of the original compatibility system and let $T < U_f$ be its three-dimensional subspace of global translations.

Proposition 9.13 (transvection syndrome criterion). *Fix f , the local data, and a transvection matching $\mathcal{M}_{L,c}$. Let D be the set of x -edges for which $y_e + c \in \text{im}(I + L)$; these are exactly the edges on which collision is possible. Write the corresponding non-collision equations as*

$$A\mathbf{z} = b \quad \text{in } \mathbb{F}_2^D,$$

where $\mathbf{z} = (t, \epsilon)$ denotes the variables of the compatibility system, and fix any solution $\mathbf{z}_0$ of that original system. Then the augmented system is consistent if and only if

$$[b + A\mathbf{z}_0] \in \text{im} \left(\bar{A} : \mathcal{K}_f \cong U_f/T \longrightarrow \mathbb{F}_2^D/A(T) \right). \quad (14)$$

In particular,

$$\dim \text{im } \bar{A} = \text{rank}(A|_{U_f}) - \text{rank}(A|_T) \leq \dim \mathcal{K}_f.$$

Failure is certified by a linear functional on $\mathbb{F}_2^D$ which annihilates $A(U_f)$ but is nonzero on $b + A\mathbf{z}_0$.

Proof. Every solution of the original system is $\mathbf{z}_0 + u$ with $u \in U_f$. It satisfies the extra equations exactly when

$$Au = b + A\mathbf{z}_0.$$

Thus consistency is equivalent to $b + A\mathbf{z}_0 \in A(U_f)$. Quotienting the target by $A(T)$ and using $U_f/T \cong \mathcal{K}_f$ from Proposition 9.4 gives (14) and the rank formula. The final statement is the standard dual characterisation of membership in a linear subspace. $\square$

The explicit equation in the proof of Proposition 9.12 shows also that A depends on f and the chosen matching but not on the local data $g_{v,e}$. The local change of variables described after Proposition 9.4 preserves every palette pair, so changing local orders cannot alter feasibility: apparent changes of the syndrome are coordinate changes, not a new choice for satisfying the target. The last sentence of Proposition 9.13 identifies precisely how the paper's duality methodology would have to be strengthened: its original argument rules out inconsistency

certificates for the compatibility equations, whereas one must now rule out certificates carrying an additional functional supported on the dangerous edges. Merely counting variables cannot do this, since the target syndrome may lie outside the usually much smaller image of $\bar{A}$.

This splits the Hall strategy into four parallel matchings ($L = I$), twelve transvection matchings (the linear case of Proposition 9.12), and eight matchings for which L has order three. In the last case $I + L$ is invertible, so (13) forbids one point rather than one affine line and the condition is no longer linear. The parallel case can be understood completely, and shows precisely where the cycle-double-cover argument adds no leverage.

Proposition 9.14 (parallel-matching criterion). *Retain the quotient flow ϕ of Proposition 9.5. At a vertex of Q , list its incident values cyclically as $p_0, \dots, p_{s-1}$ and put*

$$S = \{0, p_0, p_0 + p_1, \dots, p_0 + \dots + p_{s-2}\} \subseteq A.$$

For any fixed $c \in A$, the lifts can be chosen so that the parallel matching $\mathcal{M}_{I,c}$ misses every palette pair if and only if $S \neq A$ at every vertex of Q . Whenever this happens, the Tait flow is already given directly by

$$h(e) = p_e \quad (e \text{ an } a\text{-edge}), \quad h(e) = y_e + c \quad (e \text{ an } x\text{-edge}).$$

If $s = 5$, the local condition $S \neq A$ says exactly that the value occurring three times among $p_0, \dots, p_4$ occurs in three cyclically consecutive positions.

Proof. All solutions of $y_{i-1} + y_i = p_i$ around the cycle are obtained by translating the set of y_i by a common element of A , and that set is a translate of S . It can be made to avoid c exactly when $S \neq A$. By (13) with $L = I$, an x -edge pair belongs to $\mathcal{M}_{I,c}$ exactly when $y_e = c$, independently of the compatibility-system solution. The displayed h is nonzero, and at a vertex its sum is $p_i + (y_{i-1} + c) + (y_i + c) = 0$.

For $s = 5$, the flow equation makes the multiplicities of the three nonzero elements of A all odd, hence they are 3, 1, 1. Write the repeated value as r and the other two as u, v , so $r = u + v$. If the three r 's are consecutive, a direct partial-sum calculation omits one element of A . Otherwise the two exceptional entries separate the three r 's into nonempty blocks; up to reversal and cyclic shift the word is u, r, v, r, r , whose partial sums visit $u, v, 0, r$, all of A . $\square$

There is a useful map-theoretic interpretation of this direct criterion. A dart of Q is an incidence between an a -edge and an x -cycle, and can therefore be labelled by the element $g \in G$ at that incidence.

Proposition 9.15 (the regular-map flow formulation). *On the dart set G , put*

$$\rho(g) = gx, \quad \lambda(g) = ga.$$

The cycles of ρ give the rotations at the vertices of an orientably regular map with underlying graph Q , edge reversal λ , and type $\{\text{ord}(ax), s\}$. The group G acts regularly on its darts by left multiplication, and X is the truncation of this map.

If $s = 5$, then X is 3-edge-colourable if and only if Q has a nowhere-zero $\mathbb{F}_2^2$ -flow such that, at every map vertex, the value of multiplicity three occupies three cyclically consecutive darts. Moreover, the perfect matching of one fixed colour k belongs to the minimum- a -edge face of Proposition 9.25 if and only if k has multiplicity one at every map vertex.

Proof. The ρ -cycles are the right cosets of $\langle x \rangle$, and λ pairs the two darts of the quotient edge represented by the a -edge $\{g, ga\}$. Left multiplication commutes with both permutations and is free and transitive on the darts. Since $\langle a, x \rangle = G$, the map is connected, so this is an orientably regular map. With the convention that a face step first applies λ and then ρ , it sends g to gax

and has order $\text{ord}(ax)$. Replacing every ρ -cycle by a cycle of its darts and retaining the λ -edges gives exactly the x - and a -edges of X , proving the truncation assertion.

Identify the three Tait colours with the nonzero elements of $A = \mathbb{F}_2^2$. Around an x -pentagon, let p_i be the colour of the incident a -edge and let y_i be the colour of the x -edge from position i to position $i + 1$. Kirchhoff's equation is

$$p_i + y_{i-1} + y_i = 0.$$

Summing it shows that the p_i sum to zero, so restriction to the a -edges is a nowhere-zero A -flow on Q . Conversely, for a quotient flow the displayed recurrence determines the y_i up to adding a common element of A . A translate avoiding zero exists precisely when the cyclic partial sums of the p_i omit an element of A .

For five nonzero elements summing to zero, the multiplicities of the three possible values are all odd, hence are 3, 1, 1. Proposition 9.14 says that their partial sums omit an element exactly when the three equal values are consecutive. This proves the equivalence. Finally, the colour- k perfect matching uses as many a -edges on an x -pentagon as the multiplicity of k at the corresponding map vertex. Proposition 9.25(i) proves the last assertion. $\square$

Let Q^* denote the dual of this map. Its vertices are the right cosets of $\langle ax \rangle$: the quotient edge with darts g, ga separates the faces $g\langle ax \rangle$ and $ga\langle ax \rangle$. Thus Q^* is another small coset graph on which ordinary vertex-colouring can be used.

Corollary 9.16 (a three-colourable dual criterion). *Suppose $s = 5$. If Q^* is 3-vertex-colourable, then X is 3-edge-colourable. More generally it is enough to 4-colour the faces of Q so that no map vertex is incident with all four colours.*

Proof. Identify the four face colours with $A = \mathbb{F}_2^2$. If an edge of Q separates faces of colours z, z' , give it value $z + z'$. Properness makes this value nonzero. At a map vertex the five edge values are the consecutive differences in the cyclic word of incident face colours, so they sum to zero and define an A -flow.

If the cyclic face word uses at most three colours, it actually uses exactly three because an odd cycle is not 2-colourable. Up to a cyclic shift and relabelling, every proper three-colouring of a 5-cycle has the form

$$u, v, u, v, w.$$

The value $u + v$ then occurs on three consecutive edges. Proposition 9.15 gives a Tait colouring of X . $\square$

For example, when $\text{ord}(ax) = 3$, the loopless dual Q^* is cubic. Brooks' theorem gives a 3-vertex-colouring unless its underlying simple graph is K_4 ; that exception would force $|G| = 12$, which is incompatible with $\text{ord}(x) = 5$. Thus Corollary 9.16 gives a second proof, in the order-five setting, of the corresponding case of Corollary 7.2. For larger $\text{ord}(ax)$ the criterion is not automatic, but it replaces a Tait-colouring problem on $|G|$ vertices by a constrained face-colouring problem on $|G|/\text{ord}(ax)$ vertices.

The missing flows have an exact topological parameter. Let Σ be the orientable surface of the regular map and write ∂z for the A -valued edge function obtained by assigning $z(F) \in A$ to each face and summing the values on the two sides of every edge.

Proposition 9.17 (homology-corrected face potentials). *Choose one A -valued 1-cycle h in each class of $H_1(\Sigma; A)$. Every A -flow on Q has the form*

$$p = h + \partial z, \quad p(e) = h(e) + z(F_e^+) + z(F_e^-), \quad (15)$$

for a unique chosen representative h and a face assignment z unique up to a common translation. Consequently, when $s = 5$, X is Tait-colourable if and only if some pair (h, z) makes every value in (15) nonzero and makes its multiplicity-three value consecutive around every vertex of Q .

If the map has genus g and $m = \text{ord}(ax)$, then

$$2 - 2g = |G| \left(\frac{1}{s} - \frac{1}{2} + \frac{1}{m} \right), \quad |H_1(\Sigma; A)| = 4^{2g}.$$

Proof. In the cellular chain complex of the map with coefficients in A , graph flows are the 1-cycles $\ker \partial_1$, face potentials give the 1-boundaries $\text{im } \partial_2$, and their quotient is $H_1(\Sigma; A)$. This gives (15). Since Σ is connected, the kernel of ∂_2 consists of the constant face assignments, proving uniqueness. Proposition 9.15 gives the colouring criterion. Finally the map has $|G|/s$ vertices, $|G|/2$ edges and $|G|/m$ faces; Euler's formula and $H_1(\Sigma; A) \cong A^{2g}$ give the two displayed identities. $\square$

Corollary 9.16 is the special choice $h = 0$. Proposition 9.17 isolates a possible use of the cycle-double-cover proof's linear-algebraic methodology: use symmetry and duality to control the homology representative h , then solve for the face variables z . It does not by itself force the nonzero or consecutive-triple inequalities. For the first A_5 example the map is spherical, so $h = 0$ is the only class. For the $\text{PSL}(2, 11)$ cases with $m = 11$ the map has genus 70; the computational obstruction below shows that $h = 0$ cannot work, while a suitable nonzero class does. The broader topological transition- and Penrose-polynomial literature packages circuit partitions, face colourings and edge colourings of embedded graphs into polynomial evaluations [13]. Such an evaluation could provide a counting version of the target, but no known non-vanishing theorem for these regular maps supplies the missing existence argument.

Thus Jaeger's theorem supplies a nowhere-zero 4-flow on the 5-edge-connected graph Q , but it does not supply the consecutive-triple condition. The minimum- a -edge strategy asks in addition for a nonzero value that is a singleton at every vertex; this is a global constraint, since either of the two singleton values may otherwise vary from vertex to vertex. Requiring *both* singleton values to be fixed globally would give two disjoint perfect matchings of Q whose incident darts are consecutive at every vertex, and would be a simple sufficient construction. It is too strong already for

$$G = A_5, \quad a = (01)(23), \quad x = (01234) :$$

an exact enumeration of the 125 quotient perfect matchings checks all 125^2 ordered pairs and finds no such pair. Consequently a proof through the stronger face must allow the identity of the second singleton value—equivalently, the identity of the triple value—to change between vertices.

Thus the four parallel targets are direct Tait-flow criteria in different coordinates. The transvection targets can also be tested against a simpler necessary condition on the quotient labels. Call a cyclic word of nonzero values in A *admissible* if it sums to zero and its cyclic partial sums omit an element of A , as in Proposition 9.14.

Proposition 9.18 (quotient switches for transvection targets). *Fix a transvection $L \in \text{GL}(2, 2)$, and for $r \in \mathcal{S}_\phi$ define the switched quotient flow*

$$\psi_r(e) = L^{r_e} p_e.$$

If a palette for f_c is certified by a matching $\mathcal{M}_{L,d}$, its a -edge layer assignment r belongs to $\mathcal{A}_c$, and every cyclic word of incident ψ_r -values is admissible. Conversely, finding any $r \in \mathcal{S}_\phi$ with all these words admissible gives a Tait colouring of X directly, without needing a palette or the original lifts c .

When $s = 5$, the set of such switches r is an affine subspace of $\mathcal{S}_\phi$, possibly empty, and can be found by solving a binary linear system on $E(Q)$.

Proof. Colour a palette point $(0, z)$ by z and $(1, z)$ by $L(z + d)$. These are the four colours identified by the matching, because $L^2 = I$. The extracted Tait flow on an a -edge in layer r_e has value $L^{r_e}p_e$. Its nonzero x -edge values around a cycle have consecutive differences $\psi_r(e)$, so their partial sums omit a value. The assertion $r \in \mathcal{A}_c$ is Proposition 9.9.

For the converse, ψ_r is indeed a quotient flow: writing $N = I + L$, its sum at a quotient vertex is

$$\sum_{e \ni C} \psi_r(e) = \sum_{e \ni C} p_e + N \sum_{e \ni C} r_e p_e = 0.$$

Assign $\psi_r(e)$ to each a -edge and take cyclic partial sums on each x -cycle, translated by an omitted value. Every resulting x -edge value is nonzero and the vertex flow equations hold. This is a nowhere-zero $\mathbb{F}_2^2$ -flow, hence a Tait colouring. This converse asserts a colouring, not necessarily a palette for the prescribed f_c .

For $s = 5$, write the repeated value as a and the other two as $b, c = a + b$. The local subflow condition has eight solutions. Up to cyclic order reversal, starting point, and relabelling of A , the table below gives the extra condition for the switched word to be admissible; positions are indexed from 0 to 4. Its columns specify the nonzero value fixed by L .

Original cyclic word	fixes a	fixes b	fixes c
(a, a, a, b, c)	no extra condition	$r_0 + r_1 = 0$	$r_1 + r_2 = 0$
(a, a, b, a, c)	impossible	$r_1 + r_3 = 1$	$r_0 + r_3 = 1$

To check the table, the eight local subflows are the empty and full sets, the three pairs of repeated-value positions, and their complements. Switch the indicated labels in each case and apply the consecutive-triple criterion of Proposition 9.14. Thus each vertex contributes either no extra equation, a contradiction, or one affine equation. Together with the local subflow equations these form the asserted global linear system. $\square$

In particular, if the value fixed by L is repeated at a quotient vertex in three nonconsecutive positions, no choice of lifts can produce a palette certified by any $\mathcal{M}_{L,d}$. If each of the three values has such a vertex, all twelve transvection targets are excluded for that quotient flow. This obstruction precedes the choice of the form B_c and is unchanged by every lift modification.

For a fixed B_c , the affine switch system may also be intersected with (12); if this intersection is empty, every lift giving that form is excluded. For constructing a Tait colouring, however, the switch system alone suffices. The remaining existence problem is to choose ϕ so that it is consistent for at least one transvection. Merely removing the impossible local pattern does not guarantee global consistency.

There is a further simplification if one no longer requires the changes of quotient labels to come from a subflow in $\mathcal{S}_\phi$. A transvection fixes one nonzero value k , so it preserves the set of quotient edges with that value. Instead of keeping the rest of ϕ , prescribe only this *colour support* $P \subseteq E(Q)$. For $s = 5$, the remaining two labels can be recovered by an exact signed-cycle test.

Proposition 9.19 (fixed-colour support criterion). *Suppose $s = 5$. A set $P \subseteq E(Q)$ can be exactly the set of a -edges receiving a specified Tait colour k if and only if the following conditions hold.*

- (i) *At every quotient vertex, P occupies either one position or three cyclically consecutive positions.*
- (ii) *The signed 2-regular multigraph H_P described below has even sign sum on every circuit.*

The vertices of H_P are the edges of Q outside P . At a quotient vertex with three positions in P , join the other two edges by a sign-1 edge. At a vertex with one position in P , index its cyclic order as e_0, e_1, e_2, e_3, e_4 with $e_0 \in P$; join e_2 to e_3 with sign 0 and e_1 to e_4 with sign 1.

When both conditions hold, there are exactly $2^{c(H_P)}$ labelled Tait colourings with the prescribed support. The number of circuits with odd sign sum is always even. In particular, a connected H_P is sufficient.

Proof. Write the two colours other than k as b and $b + k$, and encode their choices by bits u_e on $E(Q) \setminus P$. In an admissible five-letter word every colour appears once or three times, and the repeated colour occupies consecutive positions; this proves the necessity of (i). If k occurs three times consecutively, the other two labels must differ, giving the sign-1 equation. If only e_0 has value k , admissibility is equivalent to

$$u_{e_2} + u_{e_3} = 0, \quad u_{e_1} + u_{e_4} = 1.$$

Indeed, the three equal labels must occupy either positions 1, 2, 3 or positions 2, 3, 4. These are exactly the two relations defining H_P at this vertex.

Each quotient edge outside P participates in one relation at each endpoint, so H_P is 2-regular. Propagating the bits around a circuit is consistent exactly when the sum of its signs is zero, and then there are two choices on that circuit. Every locally admissible quotient word has a unique extension to nonzero x -edge values: of its four possible partial-sum translations, exactly one avoids zero. This proves sufficiency and the count.

There is exactly one sign-1 relation per quotient vertex. Since $|V(Q)| = |G|/5$ is even, the total sign sum is zero. Thus the number of odd-sign circuits is even, and a single circuit must have even sign sum. $\square$

The support criterion has a useful counting form. It is important here to retain the signs: an unsigned circuit-partition polynomial records the number of circuits but loses exactly the obstruction that remains.

Proposition 9.20 (the signed-loop state sum). *Fix one Tait colour k , let $\mathcal{P}(Q)$ be the set of perfect matchings of Q , and for $P \in \mathcal{P}(Q)$ let $\epsilon_P(C) \in \mathbb{F}_2$ be the sign sum of a circuit C of H_P . The number $Z_k(Q)$ of labelled Tait colourings in which the colour- k matching belongs to the minimum- a -edge face is*

$$\begin{aligned} Z_k(Q) &= \sum_{P \in \mathcal{P}(Q)} \prod_{C \in \mathcal{C}(H_P)} (1 + (-1)^{\epsilon_P(C)}) \\ &= \sum_{\substack{P \in \mathcal{P}(Q) \\ \epsilon_P(C)=0 \text{ for every } C}} 2^{c(H_P)}. \end{aligned} \tag{16}$$

Equivalently, this is a fixed local three-state partition function on the embedded graph Q . Give every quotient edge one of the states $\{*, 0, 1\}$. At every vertex require a unique $*$ -edge, rotate the incident order so that it is $e_0 = *, e_1, e_2, e_3, e_4$, and require

$$e_1 = e_4, \quad e_2 = e_3.$$

Give this local state weight $(-1)^{e_1}$. Then $Z_k(Q)$ is the sum of the products of the local weights over all admissible edge-state assignments.

The regular action gives the exact orbit expansion

$$Z_k(Q) = \sum_{\substack{[P] \in \mathcal{P}(Q)/G \\ \epsilon_P(C)=0 \text{ for every } C}} |G : \text{Stab}_G(P)| 2^{c(H_P)}. \tag{17}$$

In particular, $Z_k(Q) > 0$ is equivalent to the stronger, minimum-face version of the desired Tait-colouring statement.

Proof. For a fixed P , Proposition 9.19 gives zero colourings if some circuit is sign-odd and $2^{c(H_P)}$ otherwise. Each factor in the first line of (16) is respectively zero or two, proving both displayed lines.

For the local formulation, expand the product by choosing an arbitrary subset of the circuits of H_P . Label a non- P edge of Q by one exactly when its corresponding vertex of H_P lies on a chosen circuit. At a quotient vertex the two ends of each local relation lie on the same circuit, giving $e_1 = e_4$ and $e_2 = e_3$. The first relation is the unique sign-one relation, so the product of the local weights is (-1) to the total sign sum of the chosen circuits. Conversely, the equalities propagate a 0 or 1 constantly around every circuit of H_P , and the unique $*$ -edges form P . This is a bijection between the terms of the expanded product and the admissible three-state assignments. Finally G preserves the rotations, circuit counts and circuit signs, so grouping the second sum into orbits proves (17). $\square$

There is also a direct edge-colouring-polynomial interpretation. If

$$E_k(t) = \sum_{\chi} t^{|\{e:e \text{ is an } a\text{-edge and } \chi(e)=k\}|},$$

where χ ranges over labelled Tait colourings of X , then Proposition 9.25 gives

$$Z_k(Q) = [t^{|G|/10}]E_k(t).$$

Thus the target is not merely the usual edge-colouring count: it is its lowest possible a -edge coefficient. The Ω polynomial of Ellis-Monaghan, Kauffman and Moffatt [14] does count edge colourings of arbitrary ribbon graphs, while the embedded Penrose polynomial at 3 has that interpretation for cubic plane graphs but not for all higher-genus embeddings. Formula (16) identifies the additional coefficient and sign filter that a polynomial proof would have to control. The local transition identity alone cannot prove non-vanishing, since the Petersen control has five quotient matchings and $Z_k = 0$. The regular action does, however, control the cancellation modulo a power of two.

Proposition 9.21 (an unsigned-count congruence). *Let X be a cubic Cayley graph on G , let $h = |G|_2$ be the largest power of two dividing $|G|$, and let $\mathcal{B}$ be any G -invariant family of perfect matchings of X . Define*

$$U_{\mathcal{B}} = \sum_{M \in \mathcal{B}} 2^{c(X-M)}, \quad T_{\mathcal{B}} = \sum_{\substack{M \in \mathcal{B} \\ X-M \text{ has only even circuits}}} 2^{c(X-M)}.$$

Then $T_{\mathcal{B}}$ counts the labelled Tait colourings whose fixed colour- k matching belongs to $\mathcal{B}$, and

$$U_{\mathcal{B}} \equiv T_{\mathcal{B}} \pmod{\mu(G)}, \quad \mu(G) = \begin{cases} 2h, & h = 2, \\ 4h, & h \geq 4. \end{cases} \quad (18)$$

In particular, a nonzero residue of $U_{\mathcal{B}}$ modulo $\mu(G)$ proves that $\mathcal{B}$ contains a matching with even complement. If r is its least nonnegative residue, then $T_{\mathcal{B}} \geq r$.

Proof. Consider states (M, B) where $M \in \mathcal{B}$ and B is the union of the vertex sets of any subset of the circuits of $X - M$. Give the state sign $(-1)^{|B|}$. There are $2^{c(X-M)}$ states over each M , and their signed sum is

$$\prod_{C \in \mathcal{C}(X-M)} (1 + (-1)^{|C|}).$$

It is zero if a circuit is odd and $2^{c(X-M)}$ otherwise. Thus, writing N_- for the number of negative states,

$$U_{\mathcal{B}} - T_{\mathcal{B}} = 2N_-.$$

The usual alternating two-colouring of each even complementary circuit gives the asserted Tait-colouring interpretation.

The group G acts on these states by left translation. A negative state has a stabiliser of odd order: if its stabiliser contained an involution t , then $tB = B$; but t acts freely on vertices, so it partitions B into pairs, contradicting odd $|B|$. Every orbit of negative states therefore has size divisible by h , giving the congruence modulo $2h$.

If $h \geq 4$, adjoin the commuting involution κ sending (M, B) to $(M, V(X) \setminus B)$. Since $|V(X)| = |G|$ is even, κ preserves the sign. A negative state again has a stabiliser of odd order under $G \times \langle \kappa \rangle$. An involution in this stabiliser could not be $(1, \kappa)$, which fixes no subset, or $(t, 1)$, which was excluded above. The remaining possibility (t, κ) would imply

$$tB = V(X) \setminus B, \quad |B| = |G|/2.$$

The last number is even when $4 \mid |G|$, another contradiction. Hence every negative orbit under the enlarged group has size divisible by $2h$. It follows that $4h$ divides $2N_-$, proving (18). The final assertions use $T_{\mathcal{B}} \geq 0$. $\square$

This argument uses only that every involution acts without fixed vertices; it also applies to a cubic graph with an even-order automorphism group whose vertex stabilisers all have odd order. In that setting h divides the number of vertices, so the same complement argument applies. In particular, it applies to the cubic arc-regular quotient of the order-three case, whose vertex stabilisers have order three. Neither planarity nor the order-five local rule is required for the congruence itself.

Corollary 9.22 (a circuit-count and stabiliser certificate). *Let M be a perfect matching of a cubic Cayley graph on G , and write $2^r = |\text{Stab}_G(M)|_2$. Its complement is even if*

$$c(X - M) \leq \begin{cases} r, & |G|_2 = 2, \\ r + 1, & |G|_2 \geq 4. \end{cases}$$

Within any fixed invariant matching class, an invariant subfamily with nonzero unsigned residue exists if and only if some matching satisfies this bound.

Proof. For the orbit $\mathcal{O} = GM$, the unsigned count is

$$U_{\mathcal{O}} = |G : \text{Stab}_G(M)| 2^{c(X-M)}.$$

Its power of two is $|G|_2 2^{c(X-M)-r}$, so it is nonzero modulo $\mu(G)$ exactly under the displayed bound. Proposition 9.21 then supplies an even-complement member of the orbit; all its members have the same complementary circuit lengths. Finally every invariant family is a disjoint union of matching orbits, so a nonzero residue for a family requires a nonzero residue for at least one orbit. $\square$

For the order-five minimum face $\mathcal{F}$, put

$$U(Q) = \sum_{P \in \mathcal{P}(Q)} 2^{c(H_P)}.$$

The quotient/lift correspondence gives $U(Q) \equiv Z_k(Q) \pmod{\mu(G)}$. Thus an unsigned circuit count, requiring no individual sign tests, can certify colourability through its residue. More generally, if

$$U_X(t) = \sum_{M \text{ a perfect matching of } X} 2^{c(X-M)} t^{|M \cap E_a|},$$

then $U_X(t) \equiv E_k(t) \pmod{\mu(G)}$ coefficientwise: matchings with any prescribed number of a -edges form an invariant family. This gives a precise modular substitute for the missing colouring coefficient.

The moduli in Proposition 9.21 cannot be doubled uniformly over all invariant families. On S_3 with $a = (01)$ and $x = (012)$, take $\mathcal{B} = \{M_a\}$. Its complement is two triangles, so $U_{\mathcal{B}} - T_{\mathcal{B}} = 4 = \mu(S_3)$. On A_4 with $a = (01)(23)$ and $x = (012)$ the same family gives four triangles and difference $16 = \mu(A_4)$. These examples concern a restricted family, not the colourability of the graphs. Freeness of involutions is essential: for the order-twenty transitive action on the Petersen graph and its five pentagon-face matchings, $U = 20$ and $T = 0$, so even congruence modulo $2|G|_2 = 8$ fails.

The signs can be eliminated by recording where their unique local contributions occur.

Lemma 9.23 (the defect-incidence multigraph). *Let P satisfy condition (i) of Proposition 9.19. Form a multigraph K_P whose vertices are the circuits of H_P . At each quotient vertex, mark the unique local relation of sign one. For every edge $uv \in P$, join in K_P the circuit containing the marked relation at u to the circuit containing the marked relation at v ; loops are allowed and count twice toward degree. Then*

$$o(P) = |\{C \in V(K_P) : \deg_{K_P}(C) \text{ is odd}\}|.$$

In particular, P is the support of a Tait colour if and only if K_P is Eulerian.

Proof. Every quotient vertex contributes exactly one sign-one relation to exactly one circuit of H_P . The corresponding vertex of K_P receives one endpoint for each incident edge of P . There are either one or three of these, so their parity is precisely that single sign contribution. Summing over the quotient vertices on a circuit shows that its degree in K_P has the same parity as its sign sum. Proposition 9.19 completes the proof. $\square$

Thus the order-five face problem can also be stated without signed graphs or matrices: choose a quotient perfect matching P for which the ordinary multigraph K_P has no odd-degree vertex. When H_P has one circuit, K_P has one vertex and only loops, recovering the Hamiltonian sufficient case. When it has two circuits, the obstruction is simply an odd number of P -edges joining the two vertices of K_P . The construction does not yet show how to choose P , but it turns the parity defect into an incidence condition on the matching itself.

This criterion is the ordinary even-2-factor obstruction expressed in quotient coordinates, not an independent existence theorem. Once P satisfies (i), the perfect matching of colour k is forced: on each x -cycle match the vertices not incident with a selected a -edge along the remaining even path or edge. The circuits of H_P record the components of its complementary 2-factor, and their sign sums record the parities of those components. In particular the connected case is a Hamilton-cycle criterion. The benefit over the CDC-restricted switches is that no unnecessary layer-subflow constraint remains. The unresolved task is to choose P so that all these circuit parities vanish; the Cayley action does not force this for every locally valid P .

The starting support, however, can be obtained without assuming any colouring condition.

Proposition 9.24 (a locally valid starting support). *If G is non-solvable and $s = 5$, the quotient Q has a perfect matching. Any such matching, used as P , satisfies condition (i) of Proposition 9.19.*

Proof. As observed after Proposition 9.5, Q is simple, 5-regular and 5-edge-connected. It has even order. For a nonempty vertex set U , each odd component of $Q - U$ sends at least five edges to U , while at most $5|U|$ such edges are available. Thus the number of odd components is at most $|U|$. For $U = \emptyset$ it is zero, since Q is connected and even. Tutte's 1-factor theorem [47] supplies a perfect matching. Its support has exactly one position at each quotient vertex, as required. $\square$

These starts belong to a substantially smaller exchange-closed class. This observation applies to any odd order of x , provided the quotient has a perfect matching.

Proposition 9.25 (the quotient-matching face). *Let $s = \text{ord}(x)$ be odd, let $q = |G|/s$, and suppose Q has a perfect matching. Let $\mathcal{F}$ consist of perfect matchings of X selecting exactly one a -edge at each x -cycle.*

- (i) *The minimum number of a -edges in a perfect matching of X is $q/2$, and $\mathcal{F}$ is exactly the set attaining this minimum. Projection to the selected a -edges gives a bijection with the perfect matchings of Q .*
- (ii) *If $M, N \in \mathcal{F}$, every perfect matching contained in $M \cup N$ belongs to $\mathcal{F}$. In particular every circuit flip against every group translate stays in $\mathcal{F}$.*
- (iii) *The circuits of $M \triangle N$ project bijectively onto those of $P \triangle P'$, where P, P' are the projected quotient matchings. Each quotient circuit lifts by inserting the unique even path between its two incident positions at every visited x -cycle.*

Moreover, projection and lifting give inverse linear maps between the perfect-matching polytope of Q and the minimum- a -edge face of the perfect-matching polytope of X .

Proof. An a -edge cannot join two vertices of the same x -cycle, since that would put the involution a in the odd-order subgroup $\langle x \rangle$. Every perfect matching uses an odd positive number of a -edges at each odd cycle. Counting their endpoints gives the lower bound $q/2$, with equality exactly when each cycle has one selected position. A quotient perfect matching attains the bound: deleting its selected vertex from each x -cycle leaves an even path with a unique perfect matching.

At a fixed x -cycle, $M \cup N$ offers at most two a -edges. Any perfect matching in this union must use an odd number of them, hence exactly one. This proves (ii). If the selected positions for M, N agree, their internal matchings agree; otherwise the internal symmetric difference is one even alternating path between the two positions. It has no additional circuit, since an alternating circuit cannot lie in a single odd cycle. Consequently the quotient difference circuits are expanded one by one, proving (iii).

For linearity, index an x -cycle by $v_0, \dots, v_{s-1}$, with b_i its incident a -edge and $d_i = v_i v_{i+1}$. If y is the characteristic vector of a quotient matching and z that of its lift, then, over the real numbers,

$$z_{b_i} = y_{b_i}, \quad z_{d_i} = \sum_{j=1}^{(s-1)/2} y_{b_{i-(2j-1)}},$$

with indices modulo s . Exactly one y_{b_i} equals one. These formulas give the forced internal matching and extend linearly to convex hulls. Taking a perfect-matching polytope to mean the convex hull of matching vectors, its minimum-cost face is the convex hull of the minimisers in (i), giving the final assertion. $\square$

Thus the local-invalidity barrier disappears completely within $\mathcal{F}$; the implication graph of Lemma 9.29 has no arcs there. The two counterexamples in Proposition 9.32 lie outside this face. It would suffice for the order-five case to prove that some $M \in \mathcal{F}$ has no odd complementary circuits, without requiring improvement from every locally valid matching. This is a stronger existence claim than Tait colourability alone, and remains unproved.

There is also a parity-sensitive way to use circuit-partition linear algebra. Traldi's extended Cohn–Lempel equality [46, Theorem 4] says that a circuit partition T of a 4-regular graph R has $c(T) = \nu(A_T) + c(R)$, where ν is binary nullity. Choose one Euler circuit per component, form its interlacement matrix, delete rows and columns at transitions following that Euler circuit, and retain the other rows and columns, putting a diagonal one precisely at orientation-inconsistent transitions. This defines A_T even when R has parallel edges.

Proposition 9.26 (a parity-refined nullity formula). *Let $s = 5$ and let P be a perfect matching of Q . Put $R = Q - P$ and, at every vertex with cyclic order e_0, e_1, e_2, e_3, e_4 and $e_0 \in P$, prescribe transitions*

$$T : \quad (e_1, e_4), (e_2, e_3).$$

Give the half-edge at position e_1 weight one and the other three half-edges weight zero. Give each edge uv of R voltage equal to the sum of its two half-edge weights in $\mathbb{F}_2$. Let $\tilde{R}$ be the two-sheeted voltage cover, with the same transitions $\tilde{T}$ at each lifted vertex. The number $o(P)$ of odd circuits in $X - M(P)$ is

$$o(P) = 2(\nu(A_T) + c(R)) - (\nu(A_{\tilde{T}}) + c(\tilde{R})). \quad (19)$$

In particular an even complementary 2-factor is equivalent to equality between the second parenthesis and twice the first.

Proof. The transitions of T are exactly the circuits of H_P in Proposition 9.19. Along a circuit, the sum of edge voltages is the sum of all its traversed half-edge weights. The transition (e_1, e_4) contributes one and (e_2, e_3) contributes zero, so this voltage sum equals its sign sum. A sign-even circuit lifts to two circuits and a sign-odd circuit to one. Hence $c(\tilde{T}) = 2c(T) - o(P)$. Applying the extended Cohn–Lempel equality downstairs and upstairs proves the formula. $\square$

The two nullities track parity, not just the total circuit count. Nevertheless R , its voltage cover and their forced transitions all depend on P : arbitrary principal-submatrix choices need not correspond to allowable quotient matchings. Equation (19) therefore gives an exact algebraic target, not an existence proof or a monotonicity theorem.

One tempting use of the group action is a translated single-circuit repair. Let $M(P)$ be the forced matching upstairs and let $g \in G$. The symmetric difference $M(P) \triangle gM(P)$ is a union of even alternating circuits. Flipping any one circuit D gives another perfect matching $M' = M(P) \triangle D$. Its a -edge support P' is locally valid provided it does not select all five positions at any quotient vertex; matching the remaining vertices of a pentagon forces precisely the one-position or consecutive-three-position patterns. If every obstructed support had such a locally valid flip with fewer odd-sign circuits, Proposition 9.24 would give a proof by descent. However, this universal single-circuit statement is *false*: Proposition 9.32 gives a locally valid support on $\text{PSL}(2, 11)$ for which every single-circuit flip is invalid. The complete A_5 checks below remain correct but cannot be extrapolated to a universal rule. Simultaneous flips lead to the more flexible criterion of Lemma 9.29.

A strict improvement necessarily uses a proper component of the symmetric difference: if $M(P) \triangle gM(P)$ has just one circuit, flipping it gives $gM(P)$ itself and leaves the number of odd complementary circuits unchanged. Finding a translate with several difference circuits is therefore necessary, but not sufficient. A partial flip can both join and split complementary circuits, so simply showing that it meets two odd circuits does not prove a decrease.

The regular action does impose an additional arithmetic restriction which is absent for a general vertex-transitive graph. For a positive integer m , write m_2 for the largest power of two dividing m .

Proposition 9.27 (odd-cycle orbit divisibility). *Let M be any perfect matching of a cubic Cayley graph on G , and suppose $H \leq G$ preserves M under left translation. The number of odd circuits in $X - M$ is divisible by $|H|_2$. Consequently, if this number is less than $|H|_2$, then M is the complement of an even 2-factor and gives a Tait colouring.*

Proof. The subgroup H permutes the circuits of $X - M$. If C has odd length ℓ , its setwise stabiliser H_C acts faithfully on C : an element acting trivially fixes a vertex, and the left regular action has no nonidentity element fixing a vertex. Every reflection of an odd circuit fixes a vertex, so H_C consists only of rotations. Thus $|H_C|$ is odd and divides ℓ . The orbit of C has

size $|H|/|H_C|$, which is divisible by $|H|_2$. Summing the sizes of the orbits of odd circuits proves the assertion; the last statement follows from Lemma 2.2. $\square$

In particular, a matching with exactly two odd complementary circuits cannot be preserved by a subgroup whose order is divisible by four. This gives a genuine constraint from the free group action, but does not force the odd-circuit count to vanish: one still needs a suitable matching with a sufficiently small count, or a decreasing move.

There is also a useful distinction between two types of difference circuit for an involution. Here invariance refers to the matching as an edge set, not to a choice of its complementary colours.

Lemma 9.28 (involution exchange). *Let t be a fixed-point-free involutory automorphism of a finite simple cubic graph, and let M be a perfect matching. The circuits of $M \triangle tM$ are either setwise fixed by t or occur in pairs D, tD of distinct circuits.*

- (i) *Every setwise fixed circuit has length congruent to 2 modulo 4.*
- (ii) *There is a t -invariant perfect matching contained in $M \cup tM$ if and only if there is no setwise fixed difference circuit. If there are p pairs and no fixed circuit, there are exactly 2^p such matchings.*
- (iii) *If $D \neq tD$ and $M' = M \triangle D$, then*

$$M' \triangle tM' = (M \triangle tM) \setminus (D \cup tD).$$

$$\text{In particular, } |M' \triangle tM'| = |M \triangle tM| - 2|D|.$$

Proof. On a setwise fixed circuit of length $2k$, the involution interchanges its two alternating matchings. A reflection interchanging these matchings fixes two vertices: writing the vertices modulo $2k$, a reflection $i \mapsto c - i$ interchanges edge parities precisely when c is even. This is excluded by the hypothesis. The involution must therefore be the half-turn $i \mapsto i + k$, which interchanges the matchings only when k is odd. This proves (i).

Every perfect matching contained in the union uses all common edges and chooses one of the two alternating matchings on each difference circuit. A fixed circuit makes an invariant choice impossible. On a pair D, tD , invariance requires flipping exactly one of its two circuits relative to M , giving two choices independently for each pair. This proves (ii). Finally,

$$(M \triangle D) \triangle (tM \triangle tD) = (M \triangle tM) \triangle D \triangle tD,$$

and the two distinct circuits are disjoint components of the original difference, proving (iii). $\square$

Thus flipping a paired circuit makes the matching agree with its involution translate on a larger region. Neither this identity nor the existence of an invariant mixture guarantees fewer odd complementary circuits, or the exclusion of all- a pentagons. Moreover, Example 9.31 below shows that these local gains cannot in general be organised as monotone growth of the full matching stabiliser.

The local-validity constraint for *several* simultaneous flips has a particularly simple form. Call a perfect matching locally valid if it does not use all five a -edges at any x -pentagon.

Lemma 9.29 (implication-closed mixtures). *Let $s = 5$, and let M, N be locally valid perfect matchings of X . Write $D_0, \dots, D_{r-1}$ for the circuits of $M \triangle N$. The perfect matchings contained in $M \cup N$ are*

$$M_z = M \triangle \bigcup_{z_i=1} D_i, \quad z \in \{0, 1\}^r.$$

Construct a directed graph $\Gamma(M, N)$ on the circuit indices as follows. For each pentagon on which M and N each use exactly one x -edge, say $e \in M$ and $f \in N$, with disjoint endpoint sets, add an arc $i \rightarrow j$ when $e \in D_i$ and $f \in D_j$. Loops may be omitted. Then M_z is locally valid if and only if

$$z_i \leq z_j \quad \text{for every arc } i \rightarrow j.$$

Thus the allowable flip sets are exactly the successor-closed unions of strongly connected components of $\Gamma(M, N)$. They are closed under union and intersection, and their inclusion-minimal nonempty members are precisely the sink strongly connected components.

Proof. The description of perfect matchings in the union follows by choosing an alternating matching on each difference circuit. On a pentagon, a locally valid matching uses one or two x -edges, covering two or four vertices. An all- a mixture can occur only if the vertices covered by M 's x -edges and those covered by N 's x -edges are disjoint: a vertex covered by both has no incident a -edge in the union. Two nonempty even-sized subsets of a five-element set can be disjoint only when both have size two. Hence the only dangerous configuration consists of the two disjoint single x -edges in the statement.

In that configuration the mixture has no x -edge precisely when it removes e but does not insert f , namely when $z_i = 1$ and $z_j = 0$. This is excluded by exactly the displayed implication; if $i = j$, that configuration is impossible. All other pentagons are automatically safe. This proves necessity and sufficiency. Closure under union and intersection is immediate. A closed set contains every strongly connected component it meets and every component reachable from it. Every nonempty such set contains a sink component, proving the final assertion. $\square$

This lemma removes local invalidity by grouping dependent circuits; it does not make their effects on odd circuits additive. In particular, a union of two individually non-improving valid flips can improve, as the second certificate in Proposition 9.32 shows. The remaining question is about a suitably chosen *closed union*, not necessarily one circuit or even one minimal block.

We made a small diagnostic computation, reproducible with `code/cdc_palette_experiment.py`. On K_4 , five distinct full-rank flows gave 40 labelled outputs, all with palette chromatic number 4. For three distinct full-rank flows on the Petersen graph, one choice of local orders per flow led to affine solution spaces of dimensions 3, 3, 4; all 32 resulting palettes had chromatic number 5, as they must if Proposition 9.1 is to pass this negative control. For the colourable graph $\text{Cay}(S_5, \{(0\ 1), (0\ 1\ 2\ 3\ 4), (0\ 4\ 3\ 2\ 1)\})$, twenty distinct full-rank flows gave 600 affine solutions (counted as labelled outputs): 240 had palette chromatic number 7 and 360 had palette chromatic number 8, with none 4-colourable. A direct SAT search does find a rank-three flow with a tetrahedral palette on this graph, as Proposition 9.3 predicts, but that search simply solves an equivalent form of the Tait-colouring problem. A second SAT search finds a generator-separated rank-three flow and an output certified by a transvection matching, while forcing all four lower-coordinate differences to occur on x -edges; hence none of the four parallel matchings can certify that output. This confirms that the linearised target is strictly more flexible than Proposition 9.14, although the search again uses the desired colouring condition rather than proving it. The experiment shows that merely taking an arbitrary 8-flow and solving the paper's linear system need not reveal the desired colouring, even for a small Cayley graph. The new method therefore supplies sharp reformulations and a potentially useful family of augmented linear systems, but not at present a proof of the Alspach–Zhang conjecture.

For that separated transvection witness the compatibility system has nullity 8, so $\dim \mathcal{K}_f = 5$. Its quotient code $\mathcal{K}_\phi$ is zero, and hence all five dimensions are extendable layer potentials in $\mathcal{R}_f$. The unique layer potential of minimum support has weight 8 upstairs: its four selected a -edges form a monochromatic 4-cycle in Q , consecutive at every vertex, exactly as in Proposition 9.11. Of the graph's 120 x -edges, 40 are dangerous for the displayed matching. The restriction map $A|_{U_f}$ has rank 5, its global-translation part has rank 1, and hence the induced map $\bar{A}$ in

Proposition 9.13 has rank 4. Thus the forty collision equations reduce, after compatibility and translation are taken into account, to a four-dimensional effective syndrome which the witness hits. The outstanding proof problem is to force that hit without first knowing a Tait colouring.

As a check against selection bias, the script also constructs twelve distinct quotient $\mathbb{F}_2^2$ -flows, chooses lifts and local data independently of any colouring target, and tests all twelve transvection matchings after solving the original compatibility system. The resulting weighted cut-code dimensions were 0 once, 1 eight times, and 2 three times; their quotient codes were all zero, so every nonzero direction again came from $\mathcal{R}_f$. None of the 144 augmented systems was consistent. This small sample is not evidence against the conjecture—the graph is colourable and the preceding witness succeeds—but it shows that the successful dimension-five layer-potential space is special and that an arbitrary application of the cycle-double-cover solver is unlikely to find it.

A separate experiment, `code/layer_potential_experiment.py`, checks Proposition 9.8 against the original vertex equations on the Petersen graph and on this S_5 Cayley graph, including cases with several simultaneous layer potentials. It then fixes a quotient flow sampled without any colouring constraint and searches its lifts using only the rank of B_c . The subflow space has dimension 14, and the search finds a 14×14 matrix of rank 7. Its radical has dimension 7, so $\dim \mathcal{R}_{f_c} = 6$; the quotient cut code is zero, giving $\dim \mathcal{K}_{f_c} = 6$. Nevertheless all twelve transvection systems fail for this flow. Thus even a larger space than that of the successful dimension-five witness need not solve the colouring equations. This is a finite search result, not a claim that rank 7 is minimal for the matrix family. The local-order invariance proved after Proposition 9.4 is also checked exhaustively on all 252 ordered local-flow configurations by the palette script.

The experiment `code/transvection_switch_experiment.py` gives a stronger explanation of that failure. The same quotient flow has respectively 6, 4, 4 vertices where values 1, 2, 3 are repeated in nonconsecutive positions. Proposition 9.18 therefore excludes all twelve transvection targets for *every* one of its 2^{48} lift choices, not just the rank-seven choice found by the search. The script also checks the affine layer equation against the full compatibility system on this flow, the successful separated witness, and the Petersen graph, and verifies all 180 order-five word/transvection cases in the displayed switch table.

To test the new construction without assuming the colouring constraint, a SAT sampler chooses twelve distinct quotient flows subject only to the flow equations and the absence of the impossible local pattern for the transvection fixing 1. There are 50 allowed local words out of the 60 nonzero zero-sum words of length five. The sampler does not encode the switches or any Tait colouring. Exactly one of these twelve flows has a consistent switch system; its solution space has dimension one, and the resulting Tait colouring is constructed and verified directly at all 120 vertices. The other eleven show that the local filter alone does not suffice. As a negative control, twelve filtered quotient flows on the Petersen graph have inconsistent switch systems for all three transvections. These are diagnostic samples, not an exhaustive classification or a probability estimate. The useful reduction is an affine test for a *fixed* quotient flow; choosing a flow that is guaranteed to pass remains unproved.

Finally, `code/colour_support_experiment.py` applies Proposition 9.19 to the same twelve flows, keeping only their colour-1 supports. Eight tests now succeed, and all eight resulting colourings are checked directly; seven were excluded by the extra subflow requirement in the restricted switch system. The four failures each have two odd-sign circuits, giving explicit inconsistency certificates even though the input graph is Cayley. The twelve flows have eight distinct supports, five successful and three unsuccessful, so the counts over flows compare the two tests on the same inputs rather than estimate a success rate over supports. Each of the three unsuccessful supports is repaired by one alternating-circuit flip against a left translate of its forced matching; the flipped circuits have lengths 38, 42, 26, respectively. The repaired colourings are independently verified.

The same script then samples twelve distinct perfect matchings of Q , without a quotient-flow or colouring constraint, implementing the guaranteed starting choice of Proposition 9.24. Nine supports give a colouring directly. The remaining three each have two odd-sign circuits and are repaired by one translated alternating-circuit flip, of lengths 24, 42, 60. These computations test the proposed move but do not establish its availability beyond the displayed inputs. As independent checks, the script compares the signed circuits with the actual complementary 2-factor upstairs, verifies the local relations on all $3^5 = 243$ words, and checks all 32 supports of the Petersen quotient: just five are locally valid, and each has two sign-odd circuits of length two. Thus the negative control is exhaustive at the support level.

The stronger audit in `code/translated_repair_audit.py` tests strict decrease, not merely a one-step jump to a colouring. It uses canonically indexed Cayley graphs, checks every reported flip by a separate set-based traversal, and distinguishes exhausted enumerations from bounded samples. In the smallest simple-group case it gives a complete finite verification of the proposed repair rule.

Proposition 9.30 (finite A_5 descent check). *For every generating pair $G = A_5 = \langle a, x \rangle$ with $a^2 = 1$ and $\text{ord}(x) = 5$, every locally valid support P with an odd-sign circuit admits a strictly decreasing translated alternating-circuit flip. Every locally valid start on these graphs therefore reaches a Tait colouring in at most two such flips.*

Computer-assisted verification. Up to simultaneous conjugation in S_5 , take $x = (01234)$. Of the fifteen involutions of A_5 , ten generate A_5 together with x ; they form two orbits of size five under conjugation by $\langle x \rangle$. The script verifies this classification by explicit permutation multiplication. Representatives and complete support counts are as follows; the last three columns give the number of odd circuits in the complementary 2-factor.

a	Locally valid supports	0 odd	2 odd	4 odd
$(01)(23)$	5875	3475	2330	70
$(02)(13)$	4625	2935	1590	100

One enumeration uses only perfect-matching clauses and the exclusion of all- a pentagons, blocking each output until the SAT instance is unsatisfiable. A second enumeration, using no SAT solver, backtracks over the ten allowed local support patterns and checks the edge agreements. Their complete sets of 10 500 matchings agree exactly, not just their counts. For every one of the 4090 obstructed matchings the audit supplies a group element and a component of the symmetric difference whose flip remains locally valid and strictly decreases the odd-circuit count. The component, new matching, and decrease are checked independently, and the signed-circuit calculation is compared with the actual complementary cycles upstairs. Since all states have at most four odd circuits, at most two strictly decreasing flips suffice. The stabiliser divisibility in Proposition 9.27 is checked on all obstructed states as well. $\square$

This verifies an auxiliary repair statement on A_5 , not a new infinite case of the conjecture. Larger tests remain finite: on the A_6 graph with $a = (05)(12)$ and the S_6 graph with $a = (05)$, both with $x = (01234)$, the audit takes 1024 distinct locally valid matchings invariant under $\langle (012) \rangle$ in each graph. Respectively 470 and 625 starts have odd complementary circuits; all reach colourings by at most three successive decreasing flips. In a separate exhaustive run on the A_6 graph, all 1664 locally valid matchings invariant under $C_{A_6}(a)$ are enumerated. This centraliser has order eight: 1588 matchings have no odd circuits, 74 have eight, and two have sixteen, in agreement with Proposition 9.27. All 76 obstructed starts reach colourings in at most four flips, which need not preserve the initial centraliser invariance. These data do not establish descent on every matching of A_6 or S_6 .

For a control on the role of regularity, the audit also uses the vertex-transitive group of order twenty preserving the two pentagons of the Petersen graph. Each of the five locally valid

matchings has stabiliser of order four, yet its complement consists of two odd pentagons: the action is not regular, so Proposition 9.27 does not apply. Every nonempty difference with a translate is a single alternating circuit of length eight. The four distinct possible moves merely give the other four matchings and preserve the odd-circuit count. Thus the checker detects genuine stalls, and vertex transitivity alone cannot justify the proposed repair rule.

The experiment `code/involution_repair_experiment.py` strengthens the complete A_5 check: all 4090 obstructed supports admit a decreasing flip of a *paired* difference circuit for some involution, as in Lemma 9.28(iii). This uses the non-SAT local-pattern enumeration and checks every involution's circuit decomposition, as well as each successful flip independently. The same restricted rule passes independent samples of 512 locally valid matchings on each of the displayed S_5, A_6, S_6 graphs, with respectively 214, 298, 301 obstructed starts. These are bounded samples, not exhaustive larger-group results. Full symmetrisation is a different matter: in the first A_5 case there are ten obstructed supports for which every involution not already preserving the matching has a fixed difference circuit. The following explicit representative explains the limitation without any matching search.

Example 9.31 (a barrier to monotone symmetry growth). Take $G = A_5$, $a = (01)(23)$, $x = (01234)$ and $H = \text{Stab}_{A_5}(4) \cong A_4$. Use right-action notation i^g for permutations, so that the H -orbit coordinate of g is 4^g . On the five-vertex quotient by H , select the a -edge $\{0, 1\}$, the a -semi-edge at 4, and the x -edge $\{2, 3\}$. Their lift is a perfect matching M . Its selected a -positions are 4, 0, 1, consecutive on every pentagon, so it is locally valid. Its complement follows the quotient cycle with generator word x^2ax^2 , an element of order three; consequently $X - M$ consists of four circuits of length fifteen. The stabiliser of M is exactly H : it contains the maximal subgroup H , but cannot be all of G since some, but not all, a -edges are selected.

For $t = (04)(12)$, the difference has one fixed circuit of length ten and two pairs of lengths six and ten. Direct permutation multiplication gives the following five single-circuit outcomes; the listed lengths are those of $X - (M \triangle D)$.

Multiplicity	$ D $	Locally valid?	Complementary circuit lengths
1	6	yes	15, 45
1	6	no	5, 5, 5, 15, 15, 15
3	10	no	5, 15, 15, 25

The valid outcome has matching stabiliser of order three. The twelve involutions outside H form one H -conjugacy orbit, and their images of M comprise all four translates different from M . Hence the table, transported by H , covers *every* translate, not just this involution. These finite claims are checked by the script's explicit permutation certificate, without a SAT solver or a search for M ; all distinct flips give exactly four valid outcomes and sixteen invalid ones.

It follows that every locally valid decreasing single flip from M changes the odd-circuit count from four to two and the stabiliser order from twelve to three. Also, for every involution $u \notin H$, no u -invariant perfect matching is contained in $M \cup uM$, by Lemma 9.28(ii). Thus neither reaching a colouring in one flip nor preserving or increasing stabiliser order at every decreasing flip can be required of the general repair rule. This particular A_5 example does not obstruct strict descent: Proposition 9.30 supplies a second decreasing flip to a colouring.

As additional stress tests on already colourable solvable groups, the translated-repair audit includes $C_5 \wr C_2$ of order fifty and $\mathbb{F}_{16}^+ \rtimes C_5$ of order eighty. In the first, a swaps two five-point blocks and x cycles the first block. In the second, writing $\mathbb{F}_{16} = \mathbb{F}_2[z]/(z^4 + z + 1)$, take $a(v) = v + 1$ and $x(v) = z^3v$. The complete enumeration on the fifty-vertex graph has 1480 locally valid matchings: 1170 colour directly, and all 310 obstructed matchings have a decreasing translated flip. A bounded sample of 30 000 matchings on the eighty-vertex graph has 19 441 even complements, 10 191 complements with two odd circuits and 368 with four;

every obstructed sample admits a decrease. These tests challenge the stronger proposed repair rule, not the already established solvable-group theorem.

The next test disproves that single-circuit rule on a non-solvable group, and also rules out the requirement that a decreasing closed set be inclusion-minimal.

Proposition 9.32 (two barriers to restricted descent). *On $G = \text{PSL}(2, 11)$, acting on $\mathbb{F}_{11} \cup \{\infty\}$, take*

$$a(z) = \frac{2z + 7}{4z + 9}, \quad x(z) = \frac{8z}{z + 7}.$$

These generate G , with $\text{ord}(a) = 2$, $\text{ord}(x) = 5$ and $\text{ord}(ax) = 11$. In the resulting 660-vertex Cayley graph there are locally valid matchings M_1, M_2 , each preserved by $H = C_G(a)$ of order twelve, such that:

- (i) *$X - M_1$ has four odd circuits, but every single-circuit flip against every translate creates an all- a pentagon. A simultaneous flip of two difference circuits, of lengths 22 and 214, gives an even 2-factor with circuit lengths 182, 478.*
- (ii) *$X - M_2$ has four odd circuits, and no inclusion-minimal nonempty closed flip set for any translate reduces this count. Nevertheless two individually valid single-circuit flips, each preserving four odd circuits, jointly give an even 2-factor. Their circuits both have length 100.*

Computer-assisted certificates. Both matchings are specified by eleven local patterns each. The script `code/translated_block_repair.py` constructs them without a SAT search and verifies the two-generator group against an independently generated projective-line model of $\text{PSL}(2, 11)$. Both displayed fractional transformations have determinant one. The H -action on the 132 pentagons has eleven orbits of size twelve. In the script's canonical ordering their representative local support masks are

$$\begin{aligned} M_1 &: (1, 1, 19, 19, 2, 28, 28, 2, 25, 25, 25), \\ M_2 &: (2, 2, 16, 14, 8, 7, 1, 14, 25, 14, 25). \end{aligned}$$

Bit j selects position j , indexed from zero. Vertices are lexicographically ordered permutation tuples; pentagons start at their least vertex, take the smaller of the two x -neighbours first, and are ordered by their starting vertices. The first eleven pentagons represent the eleven H -orbits. Propagating the masks by H and filling in the forced x -matching constructs M_1, M_2 ; the script checks all edge agreements and local patterns. Their full matching stabilisers both have order twelve. The initial complementary circuit lengths are respectively four copies of 27, six of 38, and 120, 204; and four copies of 15, three of 44, and six of 78.

For M_1 , all 660 translations give 168 distinct single-circuit outcomes, every one invalid. This is independently verified by rebuilding the Cayley edges and translations as sets of vertex pairs, finding difference components by a separate graph traversal, and exhibiting an entire unmatched x -pentagon in each resulting complementary factor. The same traversal verifies the perfect-matching condition and complementary parities; it does not use the repair audit's stored edge actions, validity test, or factor traversal. There are 55 distinct translates of M_1 , including itself. The other 54 have the following implication systems; each has exactly one minimal nonempty closed block.

Distinct translates	Difference circuits	Minimal block size	Odd circuits after flip
18	2	2	4
12	3	3	4
12	4	4	4
12	4	2	0

Every subset of the difference circuits is checked against Lemma 9.29. For one successful order-five translate, the dependency graph is

$$D_0 \longleftrightarrow D_2 \longrightarrow D_3 \longleftrightarrow D_1.$$

The sink block $\{D_1, D_3\}$ has circuit lengths 22, 214, and its flip gives the two even circuits asserted in (i). Thus M_1 is genuinely isolated under locally valid *single*-circuit moves, but not under simultaneous moves.

For M_2 , the script enumerates every mixture for all 54 different translates and determines inclusion-minimal valid subsets directly, independently of the strongly connected component routine. There are 96 minimal-block occurrences: 84 leave four odd circuits and 12 give six. Hence none decreases. For an explicit order-five translate, two singleton sink blocks have circuit lengths 100, 100; flipping either alone leaves four odd circuits, whereas flipping both gives complementary lengths 28, 28, 36, 184, 192, 192. Set-based traversals verify these counts. Both final colourings are constructed and checked directly at every vertex.

As a local check on the general criterion, all 100 ordered pairs of pentagon matching patterns, with all 250 local component choices, are also verified. The counterexamples concern the rules quantified over *every* locally valid matching; they do not exclude a rule restricted to a carefully chosen class of starts. The displayed graph is colourable, not a Cayley snark. $\square$

The same script tests successive closed-set repairs on twelve specified order-five generating pairs of $\text{PSL}(2, 11)$, sampling 512 distinct locally valid matchings invariant under $C_G(a)$ for each pair, with seed 509 and no colouring constraint. Among the 6144 starts, 5583 already have even complements; all 561 obstructed starts reach colourings in at most nine strictly decreasing steps. The algorithm first tries sink blocks and, when those do not improve, enumerates larger closed sets. The second certificate above was found by the one sink-block failure in this run; a larger closed set repairs it. None of the twelve sampling runs exhausts its matching space. These successes do not prove that an improving closed set exists in general, and the implication criterion controls local validity rather than global oddness.

The smaller face of Proposition 9.25 gives a different test. The new script `code/quotient_face_experiment.py` compares SAT enumeration with an independent exact-cover enumeration on quotient vertices; for invariant matchings the latter uses whole subgroup edge orbits. On each of the same twelve specified $\text{PSL}(2, 11)$ pairs it exhausts the $C_G(a)$ -invariant members of $\mathcal{F}$. Their sizes are 268, 372, 444, 444, 372, 268, each occurring for both x and x^2 : 4336 starts in total. Of these 3906 colour directly, and all 430 obstructed starts colour by single-circuit descent in at most three moves (214 in one, 185 in two and 31 in three). Intermediate matchings stay in $\mathcal{F}$ but need not retain centralizer invariance. This is exhaustive only in the specified invariant classes, not on all quotient matchings or all groups.

For each of those twelve invariant spaces, every member having the minimum possible number of complementary circuits is already even. A separate bounded test samples 512 unrestricted quotient perfect matchings on each pair, with seed 1847. Among these 6144 starts, 2581 are even and each of the other 3563 has an immediate strictly decreasing translated single-circuit flip. These starts are not required to be centralizer-invariant, but the test is sampled rather than exhaustive; it supports strict descent within $\mathcal{F}$ without proving it.

The dual criterion also has an exact barrier in this test set. In the four pair cases with $\text{ord}(ax) = 11$, the dual Q^* has 60 vertices, valency eleven and chromatic number five: the script proves non-4-colourability by SAT and verifies a 5-colouring. Hence Corollary 9.16, even with its more general local four-colour hypothesis, cannot handle those graphs. They are nevertheless Tait-colourable. In flow language, restricting to face-potential flows discards essential solutions; a general proof must allow a non-trivial homology class or return to the matching/interlacement formulation.

On the two A_5 cases, the same two enumerators give 125 members of $\mathcal{F}$ each: 115 colour directly and ten colour after one flip. On $C_5 \wr C_2$ there are 120, of which sixty colour directly and sixty after one flip; on the order-eighty affine example there are 705, all colouring directly. The component-by-component lift is checked for all 15750 unordered pairs with repetition in the two A_5 spaces, and the local lift formula for all 285 ordered pairs of unmatched positions on cycles of lengths 3, 5, 7, 9, 11. The implementation `code/parity_interlacement.py` checks Equation (19) against direct factor traversal on all 1075 small-face matchings. The face-potential construction of Corollary 9.16 is also checked directly: the dual is 3-colourable in the first A_5 and the wreath-product cases, but not in the second A_5 and affine cases; the more general local four-colour condition succeeds in all four, and the induced Tait colouring is verified on X .

The same exhaustive data verify Proposition 9.20 in three forms: solving each affine circuit system, multiplying its signed-loop factors, and grouping its nonnegative contributions into G -orbits. In the order just used, the exact values of $Z_k(Q)$ are

$$540, \quad 540, \quad 440, \quad 8320.$$

An optional larger run exhausts all 26305 quotient perfect matchings in each of the two displayed S_5 maps and obtains $Z_k(Q) = 115960$ in both. These positive values are evidence for the minimum-face strengthening, not a non-vanishing proof: both A_5 and the wreath-product spaces contain obstructed matching orbits, and the orbit expansion has no presently known term that must survive for every regular map.

The separate checker `code/loop_congruence_experiment.py` evaluates the unsigned count by traversing the complementary components upstairs. It then enumerates every negative state (M, B) and its $G \times C_2$ orbit, checks the stabiliser restrictions in the proof of Proposition 9.21, and compares the signed vertex-subset expansion with the affine quotient-colouring count. The complete results are:

Map	$U(Q)$	$Z_k(Q)$	$\mu(G)$	$U(Q) \bmod \mu(G)$
A_5 , either case	860	540	16	12
S_5 , either case	191480	115960	32	24
$C_5 \wr C_2$	880	440	4	0
$\mathbb{F}_{16}^+ \rtimes C_5$	8320	8320	64	0

Each A_5 case has 160 negative states, in four enlarged-group orbits of size forty. Each S_5 case has 37760, in sixteen orbits of size eighty and 152 of size 240; all these stabilisers have odd order. The script also verifies both sharpness examples and the nonfree Petersen control from the proposition.

The nonzero residues certify the minimum-face colourings in the A_5 and S_5 tests using only unsigned component counts. In these particular tests the residue already comes entirely from Hamiltonian complements: all states with at least two complementary circuits contribute zero in aggregate modulo $\mu(G)$. The two zero residues show that this particular modular test does not detect every colourable face. There is still a useful refinement because any invariant subfamily is allowed. In the affine order-eighty example, each of the six matching orbits with two complementary circuits has forty members; restricting to one such orbit gives $U_B = 160 \equiv 32 \pmod{64}$ and hence an even complement, although none of its matchings has a Hamiltonian complement.

The script exhausts the matching orbits and verifies Corollary 9.22 on each. Each A_5 case has six certifying orbits, all Hamiltonian. Each S_5 case has 92, 55 and eight certifying orbits with respectively one, two and three complementary circuits, so the certificate also works without Hamiltonian complements in these non-solvable examples. The affine example has six, eight and six with respectively two, three and four circuits. The wreath-product face has no certifying orbit at all; consequently *every* invariant subfamily of that face has zero unsigned

residue, although sixty of its matchings have even complements. This rules out a universal nonzero-residue assertion for all order-five faces, not merely for the full-family count. A proof along this route must establish the circuit-count/stabiliser bound in the remaining non-solvable cases, or obtain information beyond this congruence. Neither step has been established.

An additional run of the same checker exhausts the twelve documented $C_G(a)$ -invariant faces on $\text{PSL}(2, 11)$. Here $|C_G(a)| = 12$ contains the full power of two dividing $|G|$, so at most three complementary circuits suffice. Among the 4336 matchings, 240 pass this certificate, including 224 with more than one complementary circuit; each of the twelve cases has at least one. In the six representative cases the respective counts are 4, 16, 40, 40, 16, 4, unchanged on replacing x by x^2 . This is exhaustive only in those fixed-subgroup classes.

The centraliser restriction does not always combine well with the small-circuit bound. For S_6 with $a = (0\ 5)$ and $x = (0\ 1\ 2\ 3\ 4)$, all 21 $C_G(a)$ -invariant minimum-face matchings have even complements, but their circuit counts are sixteen (four matchings), twenty-eight (sixteen) and forty (one). Since $|C_G(a)| = 48$ contains a Sylow 2-subgroup of S_6 , the bound in Corollary 9.22 is five for every one of these matchings. None passes it. This excludes that specified invariant class as a way to certify this graph by the congruence; it does not exclude other matchings in the full face.

The script `code/transposition_quotient_experiment.py` checks the monodromy construction of Proposition 4.9 on the two-subset quotient, without enumerating the factorial-sized group. A separate exact-cover enumeration agrees on the entire set of selected supports for each parameter. The complete counts are:

s	invariant matchings	matchings with an odd quotient circuit
5	21	5
7	275	50
9	5145	1029
11	124659	26244

Every odd quotient circuit is a pentagon with monodromy of order $s - 1$, and therefore lifts to even circuits. At $s = 5$, one matching has two such quotient pentagons; for $s \geq 7$ there is at most one. The script multiplies the actual traversal words and checks their permutation orders. For S_6 , it also lifts all 21 matchings to the full 720-vertex graph, checks their complementary circuits and colourings, and compares the entire matching set with the independent exact-cover enumeration. Thus the finite tests verify the construction at larger parameters and independently upstairs at $s = 5$; the assertion for all odd $s \geq 5$ rests on the displayed proof, not on extrapolation from the enumeration.

The companion `code/two_cycle_monodromy_experiment.py` implements Proposition 4.10. It verifies directly that the canonical quotient complement has no circuit for all 33 coprime odd pairs through total degree thirty. It also exhausts every one-dart-per-orbit quotient matching through degree ten. For $(m, n) = (3, 5)$ it finds 103 good and eight bad matchings, the latter having a quotient 9-cycle of monodromy order five; for $(3, 7)$ the corresponding counts are 1775 and 250. These bad cases independently check that the monodromy parity test is detecting a genuine selection issue. The universal even-degree conclusion is proved above rather than inferred from these finite tests.

The script `code/odd_transposition_monodromy_experiment.py` checks every case of Proposition 4.12 except the adjacent chord, which is already covered by the cited Hamiltonicity theorem. It constructs the displayed matching, multiplies every quotient-circuit word, and verifies all 246 nonadjacent coprime chord cases through degree fifty-one (among 270 coprime cases in total). All constructed quotient complements are circuit-free except at $(N, k) = (5, 2)$, where the unique circuit has even length two. Independent full lifts verify the three exceptional displayed matchings on S_5 and S_7 . The theorem rests on the symbolic attachment-set proof above, not this finite range.

Minimising the total circuit count alone is insufficient: on the wreath-product example its minimum within $\mathcal{F}$ is two, attained by ten even factors and ten factors with two odd circuits. There is no Hamiltonian complement anywhere in that face. Nevertheless, lexicographically minimising first the number of circuits and then oddness gives zero oddness in all four complete small spaces and all twelve exact invariant $\text{PSL}(2, 11)$ spaces. The positive-drift certificate below shows that a useful repair may increase the total circuit count, so this observation needs a global minimisation argument rather than the presently tested descent.

Even within this safe face, a uniform averaging argument fails.

Example 9.33 (positive mean drift in the quotient-matching face). Use the 660-vertex graph and canonical edge ordering of Proposition 9.32. Number its 330 a -edges from zero in lexicographic endpoint order. Select those indexed by the set bits of the integer whose hexadecimal expansion is the concatenation of the following two strings:

3844401993410402c00010004800424501022008a
15c9048880001002480003053203206140084082.

This support is a quotient perfect matching. Its forced matching M has trivial matching stabiliser and complementary lengths 15, 27, 618, hence $o(M) = 2$. Choose g uniformly from all 660 group elements. Consider either of two moves: choose a difference circuit uniformly and flip it (do nothing if there is none), or choose a subset of the difference circuits uniformly and flip their union. Every outcome lies in $\mathcal{F}$, but the respective expected changes in odd-circuit count are

$$\mathbb{E}[o(M') - o(M)] = \frac{197}{2475} > 0, \quad \mathbb{E}[o(M') - o(M)] = \frac{677}{21120} > 0.$$

These rational values are verified by exhaustive finite enumeration in the face script. Across all ordered pairs (g, D) there are 1518 single flips, counted with multiplicity: 215 give zero odd circuits, 1021 give two, 272 give four and ten give six. For the first expectation each translate's subtotal is divided by its own number of difference circuits before averaging over g ; simply averaging the displayed multiplicities is a different distribution. For the second expectation all 4349 pairs (g, S) are enumerated, where S ranges over component subsets (at most seven components for any g), again weighting each translate uniformly. A separate vertex-set traversal checks perfect matchings and complementary lengths, and quotient/lift difference components are compared independently. An explicit single flip to a colouring is also verified; this witness increases the total circuit count from three to five while removing the two odd circuits. Thus this example refutes these two negative-drift claims, not restricted descent or colourability.

10 Open problems

1. (Conjecture 1.1 for simple groups.) Is $X(G; a, x)$ 3-edge-colourable for every non-abelian simple group $G = \langle a, x \rangle$, $a^2 = 1$, $\text{ord}(x)$ odd? Even the case $G = \text{PSL}(2, q)$ or $G = A_n$ for all q, n is open, although Corollary 7.2 covers those generating pairs with $\text{ord}(ax) = 3$.
2. (The bi-Cayley case.) For a non-abelian simple group T with an outer involutory automorphism α and $x \in T$ of odd order at least 5 with $\langle x, \alpha(x) \rangle = T$, is $B(T; x, \alpha(x))$ always 3-edge-colourable? Same question for $T = S \times S$ with α the swap. The order-three case is Proposition 3.9; for larger odd orders, is the coset-incidence graph in Proposition 4.17 guaranteed to have a consecutive 2-factor? The case $T = A_n$, α conjugation by a transposition, is already interesting.
3. (Arc-regular cubic graphs.) Is every connected cubic graph admitting an arc-regular group of automorphisms 3-edge-colourable (Corollary 3.7)? By Corollary 3.8 the answer is yes up to 10 000 vertices, and by Corollary 3.10 it is enough to exclude counterexamples whose arc-regular group is non-abelian simple.

4. (Vertex-transitive graphs.) Are the Petersen graph and its truncation the only connected cubic vertex-transitive graphs that are not 3-edge-colourable? Corollary 8.1 says yes up to 1280 vertices, and Theorem 5.2 says yes for solvable vertex-transitive groups.
5. (Girth.) Does Conjecture 1.1 hold for cubic Cayley graphs of girth 3 and 5 (equivalently, by Proposition 3.4, for generators x of order 3 and 5), and of girth 6? In the girth-three case Corollary 3.10 reduces the problem to non-abelian simple groups. Does the conjecture hold for girth at least 7, i.e. is there a cyclically 7-edge-connected Cayley snark?
6. (Hamiltonicity.) Is every $(2, s, 3)$ -Cayley graph with $|G| \equiv 0 \pmod{4}$ and $s \equiv 2 \pmod{4}$ hamiltonian? (This is the remaining case of Theorem 7.1; it does not affect Conjecture 1.1.)
7. (Structure.) Is there an algebraic characterisation of the even 2-factors of $X(G; a, x)$, for instance in terms of a subgroup H of odd order such that the colouring is H -invariant? Our computations show that in all cases tried an H -invariant colouring exists for a subgroup H of index at most 9240 (Table 2), often a point stabiliser or an involution centraliser; is there always an H -invariant colouring with H of bounded index (say $|G : H| \leq c |G|^{1/2}$), or with H from a specific family such as the centraliser of an involution?
8. (Matching monodromy.) Can one construct a proper subgroup H avoiding the conjugacy class of x and a matching in $H \setminus X$ for which every odd complementary quotient circuit has an even-order traversal word, as in Lemma 4.8? Theorem 4.13 settles every symmetric-group generating pair with a a transposition, including matchings whose complementary colours cannot descend. Can the two-subset construction be extended to involutions with several transpositions, or can an analogous small coset action be found for other almost simple groups? The existence of an arbitrary matching alone does not control the circuit-word orders.
9. (Cycle-double-cover palette and quotient switches.) For the generator-separated flow of Proposition 9.5, can one choose the quotient flow, its lifts around the x -cycles, and one of the twelve transvection matchings so that the augmented linear system in Proposition 9.12 is consistent? Can the duality argument in the cycle-double-cover proof be extended to these extra equations? When $s = 5$, it suffices instead to choose a quotient flow whose affine switch system in Proposition 9.18 is consistent: this already constructs a colouring without the lifts. Equivalently, in the regular-map formulation, can symmetry or duality select a homology class h and a face potential z satisfying Proposition 9.17? The chromatic-number-five duals on $\text{PSL}(2, 11)$ show that the zero class is not enough. The local repeated-value filter is necessary but not sufficient. Can the Cayley action rule out a global inconsistency certificate for a suitable choice of quotient flow? Arbitrarily choosing an admissible quotient flow would merely assume the desired colouring, so the choice must have an independent existence proof.
10. (Weighted cut code.) Can one choose a member of the symmetric matrix family B_c in Proposition 9.8 whose radical supplies directions solving the transvection syndrome (14)? The affine layer assignments are exactly the characteristic vectors in Proposition 9.9, and lift changes preserving B_c preserve this whole affine set. For $s = 5$, its intersection with the affine switch system gives an additional necessary condition for a transvection palette. Making B_c have small rank alone is insufficient; the quotient flow of the dimension-six experiment is locally obstructed for every choice of lifts. After avoiding these obstructions, can the remaining lift freedom control the full syndrome?
11. (Colour support and signed circuits.) For $s = 5$, can the Cayley action be used to choose a support P satisfying Proposition 9.19 with every circuit of H_P sign-even? Proposition 9.24 guarantees a locally valid start. The universal single-circuit descent rule is false, even on

$\text{PSL}(2, 11)$, by Proposition 9.32; restricting to minimal valid blocks also fails. Does every obstructed locally valid matching admit a translate and a proper implication-closed union of difference circuits whose simultaneous flip reduces the odd-circuit count? Lemma 9.29 characterises local validity exactly, but does not force this global parity improvement. Such a theorem would prove the order-five case by descent. Alternatively, can neutral or temporarily non-local-valid intermediate moves be controlled? Example 9.31 also excludes monotone growth of the matching stabiliser as a requirement at every step.

12. (The quotient-matching face.) Does the smaller, nonempty class $\mathcal{F}$ of Proposition 9.25 always contain an even-complement matching? All circuit exchanges against translates remain in this class, so no local-validity implications are needed. Can the Cayley action force equality in the parity-refined nullity bound (19), perhaps after neutral exchanges? Equivalently, can the orbit expansion (17) or a character-theoretic evaluation of its three-state tensor prove $Z_k(Q) > 0$? A concrete sufficient target from Corollary 9.22 is a matching with at most $1 + \log_2 |\text{Stab}_G(M)|_2$ complementary circuits when $4 \mid |G|$; no parity test is needed for this certificate. Its existence in general is not established. A single decreasing flip suffices in the documented finite tests, but Example 9.33 excludes two natural uniform negative-drift arguments. Forcing a single complementary circuit is too strong within this face, as the wreath-product and affine examples show.

Acknowledgements

This survey was prepared with the help of AI assistants (Claude and OpenAI Codex); the bibliographic details were checked against the cited sources where these were accessible, and the computations of Section 8 can be reproduced with the accompanying code.

References

- [1] E. Aboomahigir, R. Nedela, Cubic Cayley graphs of girth at most six and their hamiltonicity, *Acta Math. Univ. Comenian.* 88 (2019) 351–359.
- [2] M. Ahanjideh, A. Iranmanesh, The validity of Tutte’s 3-flow conjecture for some Cayley graphs, *Ars Math. Contemp.* 16 (2019) 203–213.
- [3] M. Ahanjideh, I. Kovács, Nowhere-zero 3-flows in Cayley graphs on solvable groups of twice square-free order, arXiv:2603.24175 (2026).
- [4] B. Alspach, Y.-P. Liu, C.-Q. Zhang, Nowhere-zero 4-flows and Cayley graphs on solvable groups, *SIAM J. Discrete Math.* 9 (1996) 151–154.
- [5] F. Castagna, G. Prins, Every generalized Petersen graph has a Tait coloring, *Pacific J. Math.* 40 (1972) 53–58.
- [6] R. C. Compton, S. G. Williamson, Doubly adjacent Gray codes for the symmetric group, *Linear Multilinear Algebra* 35 (1993) 237–293.
- [7] A. Biere, M. Fleury, N. Froleyks, M. J. H. Heule, CaDiCaL 2.0, in: *Computer Aided Verification (CAV 2024)*, Lecture Notes in Comput. Sci. 14681, Springer, 2024, pp. 133–152.
- [8] R. A. Wilson et al., Atlas of Finite Group Representations, version 3, <https://brauer.maths.qmul.ac.uk/Atlas/v3/>.
- [9] M. Conder, Trivalent (cubic) symmetric graphs on up to 10000 vertices, 2011, <https://www.math.auckland.ac.nz/~conder/symmcubic10000list.txt>.

- [10] M. Conder, R. Nedela, A refined classification of symmetric cubic graphs, *J. Algebra* 322 (2009) 722–740.
- [11] S. J. Curran, J. A. Gallian, Hamiltonian cycles and paths in Cayley graphs and digraphs—a survey, *Discrete Math.* 156 (1996) 1–18.
- [12] D. Ž. Djoković, G. L. Miller, Regular groups of automorphisms of cubic graphs, *J. Combin. Theory Ser. B* 29 (1980) 195–230.
- [13] J. A. Ellis-Monaghan, I. Moffatt, A Penrose polynomial for embedded graphs, *European J. Combin.* 34 (2013) 424–445; <https://arxiv.org/abs/1106.5279>.
- [14] J. A. Ellis-Monaghan, L. H. Kauffman, I. Moffatt, Edge colourings and topological graph polynomials, *Australas. J. Combin.* 72 (2018) 290–305; <https://arxiv.org/abs/1807.07500>.
- [15] H. Glover, K. Kutnar, D. Marušič, Hamiltonian cycles in cubic Cayley graphs: the $\langle 2, 4k, 3 \rangle$ case, *J. Algebraic Combin.* 30 (2009) 447–475.
- [16] H. Glover, K. Kutnar, A. Malnič, D. Marušič, Hamilton cycles in $(2, \text{odd}, 3)$ -Cayley graphs, *Proc. London Math. Soc.* (3) 104 (2012) 1171–1197.
- [17] H. Glover, D. Marušič, Hamiltonicity of cubic Cayley graphs, *J. Eur. Math. Soc.* 9 (2007) 775–787.
- [18] J. Geelen, OpenAI’s proof of the Cycle Double Cover Theorem, arXiv:2607.15399 (2026).
- [19] A. Hujdurović, K. Kutnar, D. Marušič, On prime-valent symmetric bicirculants and Cayley snarks, in: *Geometric Science of Information*, Lecture Notes in Comput. Sci. 8085, Springer, 2013, pp. 196–203.
- [20] A. Hujdurović, K. Kutnar, D. Marušič, Cubic Cayley graphs and snarks, in: R. Connelly, A. Ivić Weiss, W. Whiteley (eds.), *Rigidity and Symmetry*, Fields Inst. Commun. 70, Springer, New York, 2014, pp. 27–40.
- [21] F. Jaeger, Flows and generalized coloring theorems in graphs, *J. Combin. Theory Ser. B* 26 (1979) 205–216.
- [22] F. Jaeger, A survey of the cycle double cover conjecture, in: *Cycles in Graphs*, Ann. Discrete Math. 27 (1985) 1–12.
- [23] F. Jaeger, T. Swart, Problem session, in: *Combinatorics 79*, Ann. Discrete Math. 9 (1980) 305.
- [24] M. Kochol, Snarks without small cycles, *J. Combin. Theory Ser. B* 67 (1996) 34–47.
- [25] J. Kratochvíl, S. Poljak, Compatible 2-factors, *Discrete Appl. Math.* 36 (1992) 253–266.
- [26] K. Kutnar, D. Marušič, D. W. Morris, J. Morris, P. Šparl, Hamiltonian cycles in Cayley graphs whose order has few prime factors, *Ars Math. Contemp.* 5 (2012) 27–71.
- [27] D. W. Morris, K. Wilk, Cayley graphs of order kp are hamiltonian for $k < 48$, *Art Discrete Appl. Math.* 3 (2020) #P2.02; arXiv:1805.00149.
- [28] K. Kutnar, D. Marušič, Hamilton cycles and paths in vertex-transitive graphs—current directions, *Discrete Math.* 309 (2009) 5491–5500.

- [29] X. Li, S. Zhou, Nowhere-zero 3-flows in graphs admitting solvable arc-transitive groups of automorphisms, *Ars Math. Contemp.* 10 (2016) 85–90.
- [30] M. W. Liebeck, A. Shalev, Classical groups, probabilistic methods and the $(2, 3)$ -generation problem, *Ann. of Math.* 144 (1996) 77–125.
- [31] W. Mader, Minimale n -fach kantenzusammenhängende Graphen, *Math. Ann.* 191 (1971) 21–28.
- [32] A. Malnič, D. Marušič, P. Potočnik, On cubic graphs admitting an edge-transitive solvable group, *J. Algebraic Combin.* 20 (2004) 99–113.
- [33] A. Malnič, P. Potočnik, Invariant subspaces, duality, and covers of the Petersen graph, *European J. Combin.* 27 (2006) 971–989.
- [34] M. Nánásiová, M. Škoviera, Nowhere-zero flows in Cayley graphs and Sylow 2-subgroups, *J. Algebraic Combin.* 30 (2009) 103–110.
- [35] R. Nedela, M. Škoviera, Atoms of cyclic connectivity in cubic graphs, *Math. Slovaca* 45 (1995) 481–499.
- [36] R. Nedela, M. Škoviera, Which generalized Petersen graphs are Cayley graphs?, *J. Graph Theory* 19 (1995) 1–11.
- [37] R. Nedela, M. Škoviera, Cayley snarks and almost simple groups, *Combinatorica* 21 (2001) 583–590.
- [38] OpenAI, A proof of the cycle double cover conjecture, 2026, https://cdn.openai.com/pdf/04d1d1e4-bc75-476a-97cf-49055cd98d31/cdc_proof.pdf.
- [39] S.-i. Oum, A proof of the cycle double cover conjecture by OpenAI: an exposition, arXiv:2607.16356 (2026).
- [40] I. Pak, R. Radoičić, Hamiltonian paths in Cayley graphs, *Discrete Math.* 309 (2009) 5501–5508.
- [41] M. A. Pellegrini, M. C. Tamburini Bellani, The simple classical groups of dimension less than 6 which are $(2, 3)$ -generated, *J. Algebra Appl.* 14 (2015) 1550148; arXiv:1405.3149.
- [42] P. Potočnik, Edge-colourings of cubic graphs admitting a solvable vertex-transitive group of automorphisms, *J. Combin. Theory Ser. B* 91 (2004) 289–300.
- [43] P. Potočnik, M. Škoviera, R. Škrekovski, Nowhere-zero 3-flows in abelian Cayley graphs, *Discrete Math.* 297 (2005) 119–127.
- [44] P. Potočnik, P. Spiga, G. Verret, Cubic vertex-transitive graphs on up to 1280 vertices, *J. Symbolic Comput.* 50 (2013) 465–477.
- [45] J. Sawada, A. Williams, Solving the sigma-tau problem, *ACM Trans. Algorithms* 16 (2020) Art. 11.
- [46] L. Traldi, Binary nullity, Euler circuits and interlace polynomials, arXiv:0903.4405v2 (2009), <https://arxiv.org/abs/0903.4405>.
- [47] W. T. Tutte, The factorization of linear graphs, *J. London Math. Soc.* 22 (1947) 107–111, <https://doi.org/10.1112/jlms/s1-22.2.107>.

- [48] W. T. Tutte, A contribution to the theory of chromatic polynomials, *Canad. J. Math.* 6 (1954) 80–91.
- [49] W. T. Tutte, On the algebraic theory of graph colorings, *J. Combin. Theory* 1 (1966) 15–50.
- [50] A. J. Woldar, On Hurwitz generation and genus actions of sporadic groups, *Illinois J. Math.* 33 (1989) 416–437.
- [51] F. Yang, X. Li, Nowhere-zero 3-flows in dihedral Cayley graphs, *Inform. Process. Lett.* 111 (2011) 416–419.
- [52] J. Zhang, S. Zhou, Nowhere-zero 3-flows in nilpotently vertex-transitive graphs, [arXiv:2203.13575](#) (2022).
- [53] J. Zhang, S. Zhou, Nowhere-zero 3-flows in Cayley graphs on supersolvable groups, *J. Combin. Theory Ser. A* 204 (2024) 105852; [arXiv:2203.02971](#).

group G	$ G $	inv. classes	pairs (a, x)	orders of x	$\max \bar{V} $	result
A_5	60	1	6	3, 5	15	all colourable
S_5	120	2	2	5	5	all colourable
$\text{PSL}(2, 7)$	168	1	4	3, 7	21	all colourable
$\text{PGL}(2, 7)$	336	2	5	3, 7	42	all colourable
A_6	360	1	4	5	10	all colourable
$\text{PSL}(2, 8)$	504	1	42	3, 7, 9	63	all colourable
$\text{PSL}(2, 11)$	660	1	14	3, 5, 11	66	all colourable
M_{10}	720	1	0	—	—	no such pairs
$\text{PGL}(2, 9)$	720	2	10	3, 5	45	all colourable
S_6	720	3	2	5	10	all colourable
$\text{PSL}(2, 13)$	1092	1	45	3, 7, 13	84	all colourable
$\text{PGL}(2, 11)$	1320	2	17	3, 5, 11	120	all colourable
$\text{PGL}(2, 13)$	2184	2	26	3, 7, 13	168	all colourable
$\text{PSL}(2, 17)$	2448	1	60	3, 9, 17	153	all colourable
A_7	2520	1	11	5, 7	35	all colourable
$\text{PSL}(2, 19)$	3420	1	76	3, 5, 9, 19	180	all colourable
$\text{PSL}(2, 16)$	4080	1	204	3, 5, 15, 17	255	all colourable
$\text{PGL}(2, 17)$	4896	2	36	3, 9, 17	288	all colourable
S_7	5040	3	19	3, 5, 7	42	all colourable
$\text{PSL}(3, 3)$	5616	1	20	3, 13	78	all colourable
$\text{PSU}(3, 3)$	6048	1	6	7	28	all colourable
$\text{PSL}(2, 23)$	6072	1	107	3, 11, 23	276	all colourable
$\text{PGL}(2, 19)$	6840	2	58	3, 5, 9, 19	360	all colourable
$A_5 \wr \mathbb{Z}_2$	7200	3	8	15	25	all colourable
$\text{PSL}(2, 25)$	7800	1	134	3, 5, 13	300	all colourable
M_{11}	7920	1	14	5, 11	55	all colourable
$\text{PSL}(2, 16)$	8160	2	20	15, 17	68	all colourable
$\text{PSL}(2, 27)$	9828	1	186	3, 7, 13	378	all colourable
$\text{PSL}(3, 3).2$	11232	2	31	3, 13	156	all colourable
$G_2(2) = \text{PSU}(3, 3).2$	12096	2	21	3, 7	378	all colourable
$\text{PGL}(2, 23)$	12144	2	72	3, 11, 23	528	all colourable
$\text{PSL}(2, 29)$	12180	1	219	3, 5, 7, 15, 29	435	all colourable
$\text{PSL}(2, 31)$	14880	1	160	3, 5, 15, 31	480	all colourable
$\text{PGL}(2, 25)$	15600	2	90	3, 5, 13	300	all colourable
$\text{PSL}(2, 25).2_3$	15600	1	0	—	—	no such pairs
$\text{PSL}(2, 25)$	15600	3	30	13	26	all colourable
$\text{PGL}(2, 27)$	19656	2	123	3, 7, 13	378	all colourable
A_8	20160	2	33	5, 7, 15	35	all colourable
$\text{PSL}(3, 4)$	20160	1	60	5, 7	105	all colourable
$\text{PGL}(2, 29)$	24360	2	146	3, 5, 7, 15, 29	840	all colourable
$\text{PSL}(2, 37)$	25308	1	378	3, 9, 19, 37	684	all colourable
$\text{PSU}(4, 2) = \text{PSp}(4, 3)$	25920	2	42	5, 9	120	all colourable
$Sz(8)$	29120	1	210	5, 7, 13	455	all colourable
$\text{PGL}(2, 31)$	29760	2	113	3, 5, 15, 31	960	all colourable
$\text{PSL}(2, 32)$	32736	1	930	3, 11, 31, 33	1023	all colourable
$\text{PSL}(2, 41)$	34440	1	380	3, 5, 7, 21, 41	861	all colourable
$\text{PSL}(2, 43)$	39732	1	490	3, 7, 11, 21, 43	924	all colourable
$\text{PSL}(3, 4).2$ (field)	40320	2	11	5, 7	105	all colourable
$\text{PSL}(3, 4).2$ (graph)	40320	2	68	3, 5, 7	315	all colourable
$\text{PSL}(3, 4).2$ (graph-field)	40320	2	45	5, 7	105	all colourable
S_8	40320	4	47	5, 7, 15	35	all colourable
$\text{PGL}(2, 37)$	50616	2	242	3, 9, 19, 37	1368	all colourable
$\text{PSU}(4, 2).2 = W(E_6)$	51840	4	52	5, 9	40	all colourable
$\text{PSL}(2, 47)$	51888	1	437	3, 23, 47	1128	all colourable
$\text{PSL}(2, 7) \wr \mathbb{Z}_2$	56448	3	6	21	42	all colourable
$\text{PSL}(2, 49)$	58800	1	480	3, 5, 7, 25	1200	all colourable
$\text{PSU}(3, 4)$	62400	1	64	3, 5, 13, 15	195	all colourable
$\text{PGL}(2, 41)$	68880	2	250	3, 5, 7, 21, 41	1680	all colourable
$\text{PSL}(2, 53)$	74412	1	61	3, 9, 13, 27, 53	1431	all colourable
$\text{PGL}(2, 43)$	79464	2	325	3, 7, 11, 21, 43	1848	all colourable
M_{12}	95040	2	66	3, 5, 11	396	all colourable
$\text{PSL}(2, 59)$	102660	1	926	3, 5, 15, 29, 59	1770	all colourable
$\text{PSL}(2, 47)$	103776	2	288	3, 23, 47	2208	all colourable